

# Supplementary Material

## Synthesizing Staged Updates between Verified Controllers

### Routes from claims to evidence

Choose a route below, or use the complete contents on the next page. Blue references are clickable; all page numbers refer to this integrated PDF.

#### Complete examples: component and requirement choices

Cell gives the full local input and both completing paths. Policy separates audit and role lifetimes. S1 adds observation and interleaving witnesses.

[TA B, p. 6](#)   [TA C, p. 7](#)   [S1, p. 33](#)

#### Formal guarantees: correspondence, completion and handover

Follow the source-history and synthesis proofs, then the execution and endpoint results. Execution requires A1-A4; a progress rank bounds events, not elapsed time.

[TA A, p. 3](#)   [TA F, p. 12](#)   [TA G, p. 13](#)   [S2, p. 35](#)

#### Requirement meaning: what holds after activation

Separate safe-prior-history inclusion from exact start-scoped invariants and entry-scoped interval rules. Actual keys, observations and fallback remain additional obligations.

[TA D, p. 9](#)   [TA E, p. 9](#)   [S3, p. 38](#)

#### RQ1: when granularity changes feasibility

Start with definitions and all constructed outcomes, then read the source-derived PC2 case. Inherited inputs show no decision change; PC2 merged synthesis times out, while a separate source argument proves impossibility.

[TA O.11, p. 29](#)   [S4.1, p. 44](#)   [TA J, p. 19](#)

#### RQ2-RQ3: checks, fixed-budget costs and policy diagnoses

Independent fixtures and source-monitor checks have separate populations. The fixed-budget comparison covers all 27 contracts; paired costs omit unresolved runs. Detailed outcomes retain all failures and faster alternatives.

[S4.2, p. 45](#)   [S4.3, p. 47](#)   [S4.4, p. 51](#)

#### Related work: supplied inputs and generated guarantees

Complete Cell encodings expose the obligations supplied to native DUCS and GR(1) updating. The expanded comparison records each approach's input, progress condition and scope.

[TA H, p. 16](#)   [TA I, p. 18](#)   [S5, p. 77](#)

Auxiliary and historical records are retained in S4.5 onward and TA N-O. They do not enlarge the fixed-budget population. Saved campaign names may differ from the current RQ numbers; S4 identifies those mappings.

# Supplementary Material

## Synthesizing Staged Updates between Verified Controllers

### Complete contents

Part I follows the paper's argument. Part II supplies additional witnesses, implementation detail, history proofs, evaluation records and comparisons.

|                                                                       | PDF page  |
|-----------------------------------------------------------------------|-----------|
| <b>Part I: Proofs and definitions (Technical Appendix)</b>            | <b>3</b>  |
| A Source Histories and Synthesis Correctness                          | 3         |
| B Complete Cell Contract and Its Two Completing Paths                 | 6         |
| C Policy Requirement-Boundary Argument                                | 7         |
| D NEW and UPD History-Scope Derivations                               | 9         |
| E Activation-Scoped Initialization                                    | 9         |
| F Proof of Conditional Trace Lifting                                  | 12        |
| G Endpoint Interfaces and Certificate Reuse                           | 13        |
| H Complete Native DUCS Construction for Cell                          | 16        |
| I Cell through a GR(1) Bridge Interface                               | 18        |
| J Saved PC2 Activation Histories and Their Boundary                   | 19        |
| K Exploration Order and Work Bound                                    | 22        |
| L Complete Benchmark Input Counts                                     | 24        |
| M All Completed Direct-Full Exploration Pairs                         | 25        |
| N Separate Extended-Budget Trials                                     | 25        |
| O Separate Evaluation Populations and All-Contract Grid               | 25        |
| P Implementation and AI-Assisted Research Details                     | 32        |
| <b>Part II: Validation and experimental detail (Supplement S1-S5)</b> | <b>33</b> |
| S1 Granularity and observation witnesses                              | 33        |
| S2 Finite games, search invariants, and certificates                  | 35        |
| S3 Monitor intervals and conditional trace lifting                    | 38        |
| S4 Supplementary validation and cost tables                           | 44        |
| S5 Expanded related-work comparison                                   | 77        |

# Technical Appendix: Synthesizing Staged Updates between Verified Controllers

Anonymous Author(s)

This technical appendix follows the paper’s argument: contract and policy proofs (A–C), requirement meaning and execution (D–G), comparisons with existing interfaces (H–I), the PC2 case (J), then exploration and evaluation records (K–P). The main paper gives the complete Cell contract, the Policy mechanism, update-game specification, and proofs of contract correspondence, continuation preservation, and synthesis correctness. This document supplies the complete Policy contract, source-history recurrences, expanded proofs and auxiliary lemmas, residual-soundness formalization, complete tables, and further interpretation, execution and endpoint-interface results. References to the main paper use its section and algorithm numbers. The submission’s supplementary volume also contains S1–S5 with experimental details.

## A Source Histories and Synthesis Correctness

**Source histories without a global game.** Fix the eligible local contract of Section 3. A history is  $\ell = z(a_1, o_1) \cdots (a_n, o_n)$ , starting at a reachable old closed-loop tuple  $z$ ;  $o_j$  records the realized physical successor tuple. For each prefix, reconstruct installed versions  $v_i(\ell)$  and local states  $s_i(\ell)$  by the recurrences below. Let  $D(\ell)$  be its executed update commands. Initially  $D = \emptyset$ , every version is old and local states are those in  $z$ . Old monitor memory is inherited from  $z$ , each interval monitor starts at its supplied initializer, and every new monitor is inactive. The old endpoint controller imposes no restrictions after entry.

**Enabled local events and outcomes.** For an ordinary label  $a$ , let  $J_a(\ell) = \{i : a \in \Sigma_{e_i(v_i(\ell))}\}$ . It is enabled exactly when  $J_a(\ell) \neq \emptyset$  and every  $i \in J_a(\ell)$  has a nonempty local  $a$ -successor set. Every tuple choosing one such successor per participant, and retaining all other local states, is an outcome. This test ignores monitors, safety and targets. The prefix is UC-quiescent exactly when no enabled ordinary label is uncontrollable, including labels with unsafe monitor outcomes. A command  $a$  may occur only at a UC-quiescent prefix, with  $a \notin D(\ell)$  and all its declared predecessors in  $D(\ell)$ .

An ordinary event preserves versions and  $D$ . A transfer  $\rho_i$  additionally requires  $v_i = \text{old}$  and chooses any  $s'_i \in g_i(s_i)$ , changes only that installed component to  $(\text{new}, s'_i)$ , and adds  $\rho_i$  to  $D$ . An empty transfer image permits no extension. A stop or start leaves all physical states unchanged and adds itself to  $D$ ; start additionally requires the current version-tagged tuple in its initializer domain. Every enabled outcome yields a source extension, even if it violates a monitor. Thus monitors test behavior rather than disabling physical events.

**Memory, boundaries and source objectives.** Monitor memory follows a deterministic fold on the history. Ordinary events and transfers advance every active monitor by its transition function. A stop makes its named old monitor inactive without letting it observe that stop, and advances all other active monitors once. A start advances previously active monitors once, then installs the named new monitor’s initializer value; the new monitor does not consume start again. Other inactive memories

remain inactive. The executed-command set makes every boundary one-shot. A prefix is safe when every active memory is outside its error set.

Handover is allowed precisely at safe UC-quiescent prefixes with every command executed and with the physical tuple and new-monitor memories matching a supplied target; the selected target also supplies its controller state. A source policy may depend on the entire history and may disable only controllable labels. Every enabled uncontrollable, and every outcome of every permitted label, remains possible. A policy wins if all its prefixes from every old entry are safe, no non-handover prefix is a dead end, and every maximal history reaches handover after finitely many events. Neither fairness nor a choice of local outcomes is available to the policy. These operational clauses precede the global-state construction and are the source obligations used in Proposition 3.1.

Source histories describe supplied monitor behavior. They do not infer the semantic activation-history sets  $H_r$  in RS or prove those sets complete for an application. The scope-check populations and their limits are unchanged. The main paper states Proposition 3.1 for these source operations.

### Proof of Proposition 3.1.

*Proof.* Write  $H$  for the source histories specified above and define

$$F(z(a_1, o_1) \cdots (a_n, o_n)) = (\xi_n, \mu_n, \Sigma_{\text{upd}} \setminus D_n),$$

where  $\xi_n, \mu_n, D_n$  are obtained by the local recurrences. At length zero this is exactly  $\text{Init}(z)$ , so the images of initial histories are exactly  $Q_0$ . Distinct old-controller states may have the same image; old control has been released, so this identification removes no subsequent source restriction.

We show the one-step property: a source history  $h$  permits label  $a$  with realized component outcome  $o$  iff the generated game at  $F(h)$  permits that label with the same component outcome, and the resulting source extension has the resulting game state as its image. For an ordinary label, the participating set  $J_a(h)$  is determined by the current versions. Both definitions require a nonempty participating set and every participant to have a transition and include their full Cartesian product, retaining nonparticipants. Both then apply the same deterministic transition to every active monitor. For  $\rho_i$ , both definitions require UC quiescence, an unexecuted command with all predecessors executed, an old component and a successor in  $g_i(s_i)$ . Both change only that component, advance active monitors and remove that command from the pending set. For a stop, both make the named monitor inactive without observing its own boundary and advance all other active monitors once. For a start, both require the initializer domain, advance previously active monitors once and install the named monitor's initializer without a second observation. The executed/pending sets are complements, so one-shot and precedence tests coincide.

UC quiescence in each case is computed from every enabled ordinary UC label before any monitor-safety filtering. Thus even unsafe UC outcomes are preserved. Monitor safety, completion of all commands and equality to each projected loadable target are read from identical coordinates. Consequently  $F$  preserves and reflects safety, handover eligibility and every compatible full target. Histories with the same image have the same extensions up to their realized physical outcomes and the same resulting images.

Map a source history to the entire game path of its prefix images, retaining its labels and component outcomes. Repeated application of the one-step property maps every finite source history to a game path. Conversely choose an old tuple witnessing any first state in  $Q_0$ , then lift every edge using its recorded component outcome and the one-step property. This produces a source history for every finite game path and, by the same recurrence, for every infinite path. Different initial witnesses can give different histories but admit the same extensions and objectives.

A game history-dependent policy lifts to a source policy by applying it to this image path. Conversely, for each initial game state choose one old-tuple witness and consistently lift each subsequent path from that witness; apply the source policy to the lifted history. This defines a game history-dependent policy. In either direction enabled labels, all uncontrollable choices and every permitted-label outcome are preserved. Every path of the translated game policy lifts to a source-policy path

from its chosen initial witness; every path of the lifted source policy projects to a game-policy path. A winning policy therefore acquires no unsafe prefix, non-goal deadlock or infinite avoiding path under either translation. Hence a winning source policy exists iff a winning game history-dependent policy exists. The finite-game memoryless-policy lemma below supplies a state-dependent winner with a strict rank, completing the proposition. This is an operational correspondence for the supplied contract, not a proof of frontend correctness or application-history completeness.  $\square$

**Lemma (memoryless policies suffice).** If any history-dependent policy wins this finite game, a state-based policy wins. *Proof.* At a safe non-goal, a proving set is the full nonempty UC successor union if UC is enabled, or one complete nonempty controllable bucket otherwise. Starting with goals, attractor layers add safe states whose proving sets lie in previous layers. Choosing those sets gives a state-based policy with decreasing layer index. Outside the fixed point, every controller choice has an outcome outside, or the state is unsafe or a non-goal deadlock. The environment can keep selecting such outcomes after every history, defeating any history-dependent policy.

**Lemma (monitor intervals).** Along every update prefix, each active monitor is its inherited or initialized state advanced over exactly the labels in its declared active interval. *Proof.* Entry inherits old states and initializes interval monitors. Ordinary events and transfers advance all active monitors once. Stop removes its monitor before observation, while others advance. Start advances existing monitors and installs the new one after its boundary. Pending commands enforce one-shot execution and consistent versions/activity. Induction gives the stated interval; safety makes it non-error, and goal matching preserves new-monitor states into the endpoint.

**Lemma A.1** (UC priority). *Disabling controllable events at UC-enabled states preserves the existence of a policy for finite safe completion.*

*Proof.* A winning policy must handle every UC outcome. Disabling optional controls removes runs and creates no deadlock while a UC event remains enabled. Every surviving maximal run was already a maximal run of that policy, so it still completes safely. Conversely, the reduced policy is admissible in the original game because controllers may disable those events.  $\square$

**Lemma A.2** (Search coverage). *At loop boundaries, every state in  $X$  has all UC candidates queried; every expanded unproved quiescent safe non-goal with candidates left is in Retry; and every discovered safe non-goal outside  $X$  is in Open.*

*Proof.* Initialization at entry states establishes coverage. Discovery queues each new safe non-goal; only expansion removes it, after querying all UC candidates. At quiescence, expansion and retries consume the cursor through a nonempty bucket or exhaustion, retaining every nonwinner with candidates left. Promotion removes only winners. These are the only changes to those sets; finite propagation drains before either return.  $\square$

### Proof of Theorem 6.1.

*Proof. Termination.* The finite input induces finitely many states and candidates. Each state is discovered and expanded at most once; each retry consumes a new candidate, and each promotion adds a state to  $W$  once. Reverse incidences are processed on those promotions, so every worklist drain is finite. Eventually all entries win or both exploration queues empty.

*Soundness.* Induction on promotions gives complete successor sets with smaller retained ranks: expansion supplies all uncontrollables, or the selected nonempty controllable bucket supplies all outcomes. A queried bucket never changes, and ranks are fixed on first promotion, so this evidence persists. If all entries win, their policy closure is safe and every non-goal has a successor. Strict decrease of the natural-number rank excludes infinite runs; nonempty successors exclude non-goal sinks. Thus

every maximal run reaches a goal within its initial rank many events. Goal recognition uses modeled uncontrollable quiescence and supplies each compatible  $\kappa(q)$ .

The monitor-interval lemma above supplies the interval invariant; goal matching preserves components and new-monitor states into the verified endpoint.

*Completeness.* At failure, promotion has reached its fixed point. An entry lies in  $L = V \setminus W$ , and  $L$  contains no goal. By Lemma A.2, empty Open ensures every safe non-goal in  $L$  was expanded. With uncontrollables, complete expansion and exhausted propagation imply an uncontrollable outcome in  $L$ ; otherwise promotion would apply. Without uncontrollables, empty Retry gives every enabled controllable bucket, each with an outcome in  $L$  for the same reason. Against any policy, the environment keeps choosing these outcomes; permitting no event produces a deadlock. The resulting run reaches an unsafe state, deadlocks outside a goal, or remains in  $L$  forever. Each contradicts winning. Hence a winning policy rules out failure; termination then forces success.

The losing-region strategy defeats history-dependent policies after every history; success supplies a ranked state-based policy. Lemma A.1 preserves winning in the original supplied game.  $\square$

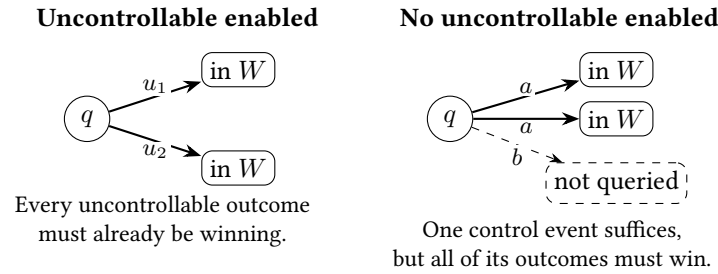


Figure A1: Safe promotion to  $W$ . The two  $a$  edges are nondeterministic outcomes of one control event, not two selectable actions. Other control buckets may be postponed.

**Promotion cases.** Figure A1 displays both promotion cases. Every UC outcome must win when UC is enabled. At quiescence one control bucket suffices, but all its nondeterministic outcomes must win; other control buckets may remain unqueried.

## B Complete Cell Contract and Its Two Completing Paths

Table 1 in the main paper gives all component, endpoint-controller and monitor transitions, including the conventions for absent edges.

Endpoint tuples list stations  $A, B$ , controller and monitor. States  $h/e$  mean holding/empty;  $k, c, d$  mean old-safe, inspection-clear and inspection-pending;  $c_o, c_0, c_1$  name controller states. Old reachable tuples are  $x_A = (h, e, c_o, k)$  and  $x_B = (e, h, c_o, k)$ ; new ones are  $z_A = (h, e, c_0, c)$ ,  $y_B = (e, h, c_1, d)$  and  $z_B = (e, h, c_0, c)$ . Initial tuples are  $x_A, z_A$  and loadable targets are  $\{z_A, z_B\}$ , excluding pending inspection at  $y_B$ . The interval initializer records the holder at either all-old one-workpiece tuple. The new initializer admits all eight version-tagged one-workpiece tuples, installing  $d$  when  $B$  holds and  $c$  otherwise. The paths starting with  $A/B$  holding map to  $z_B/z_A$  at handover.

With empty-only transfers  $g_A = g_B = \{(e, e)\}$  and exactly one workpiece, the two paths  $\rho_B; m_B; \text{start}; \text{stop}; j; \rho_A$  from  $A$  holding and  $\rho_A; m_A; \text{start}; \text{stop}; \rho_B$  from  $B$  holding have safe singleton outcomes: start installs inspection-pending if  $B$  holds and inspection-clear otherwise, stop removes the inspection ban, and both paths finish with new versions and clear inspection at matching supplied targets. Their remaining lengths  $6, \dots, 0$  and  $5, \dots, 0$  strictly decrease, proving finite completion from both reachable old states (13 policy states and 11 edges). Replacing the two transfer commands by one with relation  $g_A \times g_B$ , with endpoints and requirements fixed, requires both stations empty, but every ordinary move, inspection and requirement boundary preserves exactly one workpiece; that command can never execute, stays pending and makes every goal unreachable.

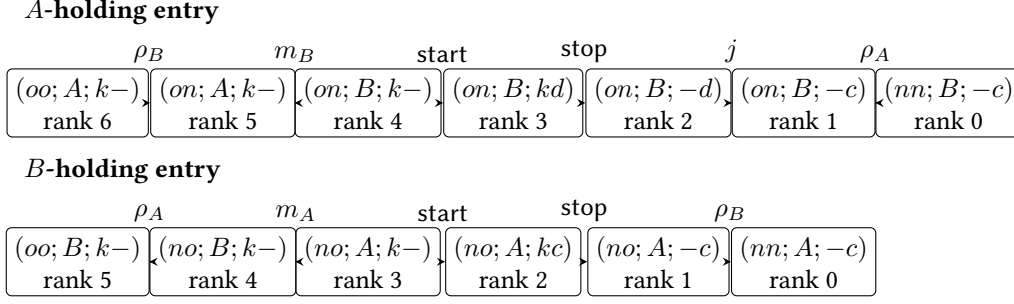


Figure A2: One policy from both entries. States list versions of  $A$ ,  $B$ , the workpiece holder, and old/new monitors; the holder also determines the interval monitor. A dash means inactive. Every edge decreases the remaining-step rank. The two rank-zero states match the respective fixed-controller targets.

### B.1 Residual Soundness and the Cell Reference

Residual soundness (RS) relates monitor memory to prior history. Let  $\text{SafePref}_r$  be requirement  $r$ 's prefix-closed language, and let  $H_r(\xi)$  contain all admissible activation histories consistent with observed component state  $\xi$ . A new requirement's history includes start once; its initialized monitor reads only later events. Residual-sound initialization means

$$\underbrace{\{w : \eta_r^*(\iota_r(\xi), w) \notin F_r\}}_{\text{continuations accepted by the initialized monitor}} \subseteq \underbrace{\bigcap_{h \in H_r(\xi)} \{w : hw \in \text{SafePref}_r\}}_{\text{continuations safe after every admitted history}} \quad (\text{RS})$$

for every  $\xi \in \text{dom } \iota_r$ . Equality is exact; strict inclusion is conservative and may exclude feasible updates. An empty history set makes the right side universal. Synthesis checks initializer states and domains, but neither infers histories nor establishes their completeness.

For Cell's declared start-only reference,  $H_{r_{\text{inspect}}}(\xi) = \{\text{start}\}$ . Initializing at  $c$  is exact for that reference; initializing at  $d$  when  $B$  holds is conservative, forbidding departure until inspection. The inclusion proof follows below. This distinction exposes a separate obligation: an exact history residual need not encode the intended duty at activation.

A Boolean event-history variable, or *fluent*, records setting/resetting labels. For an initially true fluent  $p$ , a history making  $p$  false then true and the empty history end at the same value but differ under  $\Box p$ .

**Deriving RS in the cell.** The declared pre-activation observation retains only start, so  $H_{r_{\text{inspect}}}(\xi) = \{\text{start}\}$  for every admissible cell tuple. The reference monitor starts at  $c$ , unchanged by start; the right-hand side of RS is therefore its non-error continuation language from  $c$ . If  $B$  is empty, initialization at  $c$  gives equality. If  $B$  holds, initialization at  $d$  is conservative: before the first  $j$  or  $m_B$ ,  $d$  additionally forbids  $m_A$ ; either  $j$  or  $m_B$  brings the two initial states to the same state. Thus every word safe from  $d$  is safe from  $c$ , while the single event  $m_A$  witnesses strict inclusion. This is a proof for the declared observation scope, not recovery of pre-activation history.

Initializing at  $c$  while  $B$  holds satisfies equality but omits the intended pending-inspection duty encoded by  $d$ . Declared histories and intended obligations are therefore separate input choices.

## C Policy Requirement-Boundary Argument

**Opposing requirement boundaries.** Policy(2) has two old/new requirement pairs: audit and role. It requires audit overlap and role nonoverlap. Table A1 gives its complete input. Write  $o, n$  for old/new,  $u, l$  for unlogged/logged,  $A, R$  for audit/role, and  $\ell_z, a_z$  for version- $z$  log/access.

Table A1: Complete Policy(2) input; all events controllable. Unlisted plant/controller edges are absent; unlisted monitor edges self-loop, including at error err. Monitors have alphabet  $\Sigma$ .

| Part              | Finite definition                                                                                                                                                                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Plant/transfer    | Each version has sole initial state $r$ with loops on $\Sigma_N = \{\ell_o, \ell_n, \text{service}, a_o, a_n\}$ . Transfer $\rho$ changes its version with $g = \{(r, r)\}$ .                                                                                                               |
| Controllers       | $C^z$ ( $z \in \{o, n\}$ ) has alphabet $\Sigma_N$ , initial $u$ , edges $u \xrightarrow{\ell_z} l \xrightarrow{\text{service}} u$ , and $a_z$ loops at $u, l$ . An unreachable third state has all $\Sigma_N$ loops.                                                                       |
| Endpoint monitors | $A_z$ : initial $u$ , states $u, l, \text{err}$ , edges $u \xrightarrow{\ell_z} l \xrightarrow{\text{service}} u$ and $u \xrightarrow{\text{service}} \text{err}$ . $R_z$ : initial ok, states ok, err, edge ok $\xrightarrow{a_z} \text{err}$ , where $\bar{z}$ is the other version.      |
| Interval monitor  | Initial 1010; states $\{b \in \{0, 1\}^4 : (b_1 \vee b_2) \wedge \neg(b_3 \wedge b_4)\} \cup \{\text{err}\}$ . Bits track active $(A_o, A_n, R_o, R_n)$ . Respective stop/start commands write $b_1 = 0, b_2 = 1, b_3 = 0, b_4 = 1$ ; unsafe results enter err. Other events preserve bits. |
| Initialization    | Four boundaries and $\rho$ , once each; no precedence. Old monitors are $A_o, R_o$ , new are $A_n, R_n$ . On either $(o, r)$ or $(n, r)$ , initialize $A_n = u, R_n = \text{ok}$ ; initialize the interval to 1010 only at $(o, r)$ . Domains are complete.                                 |
| Entries/targets   | For $x \in \{u, l\}$ , entry $q_x$ admits old tuple $((o, r), C^o = x, A_o = x, R_o = \text{ok})$ ; loadable targets are $z_x = ((n, r), C^n = x, A_n = x, R_n = \text{ok})$ . Admission, pending commands, activation and goal tests follow Section 3.                                     |
| Boundary merge    | One global start activates both new monitors; one global stop deactivates both old monitors. Compose member observations audit then role; other inputs stay fixed.                                                                                                                          |

From  $q_u$ , the saved policy executes  $\text{start}_A, \text{stop}_A, \text{stop}_R, \rho, \text{start}_R$ , reaching  $z_u$ . From  $q_l$ , service first reaches  $q_u$ . Ranks five/six decrease on every edge; other controllables are disabled. Figure A3 depicts this whole policy.

Both merged entries lose: the first global start overlaps roles; the first global stop removes audit coverage. Ordinary events and transfer preserve these bits, so every completing path encounters error. The saved checks confirm closure and both member orders. This compares boundary granularity, not DUCS expressiveness. Service continuity is not required. The interval monitor induces the opposite boundary orders without supplied precedence. Its role clause forbids the two role monitors from being active together; the access monitors alone only reject the other version's access event. The saved losing closure checks every alternative and both orders for observing merged member labels. Table A1 above gives the complete finite input. Figure A3 visualizes its saved policy and requirement lifetimes.

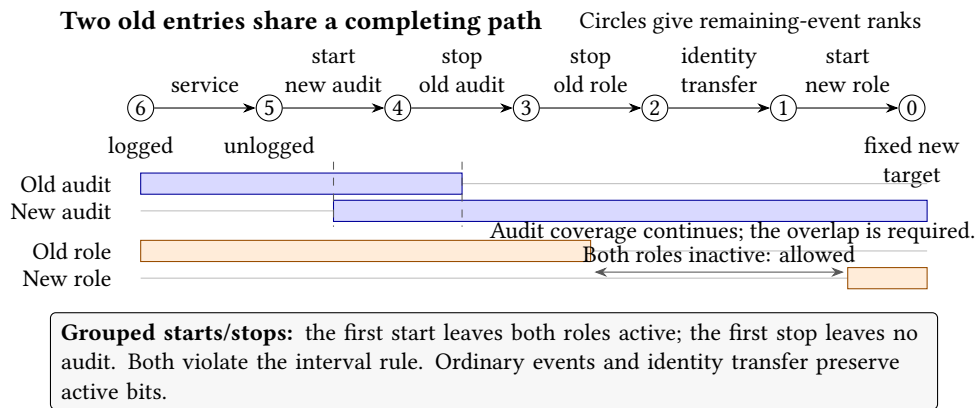


Figure A3: A saved Policy certificate with no supplied precedence; ranks need not be shortest. Logged/unlogged mark both entries; bars mark active requirements. New-audit start resets to unlogged; handover reaches new audit unlogged and new role valid. Arrows have one action and outcome. The role gap is allowed; ordinary events may be withheld, with no service-continuity guarantee.



## D NEW and UPD History-Scope Derivations

**History-scope checks.** Six target/quiescence fixtures match raw-LTS reconstruction. Across 27 Java exports, all 138 NEW lookup tables match 1,587 reconstructed entries (900 non-error, 687 ERROR). ERROR entries, missing signatures and unknown states are outside the partial initializer domain; only retained non-error entries enable initialization. These are functional checks, excluded from performance statistics.

For new requirements (NEW), independent monitor/observer exploration checks all 138 occurrences under the declared history scopes. Equal fluent vectors have equal continuation languages and match frontend lookup; every admitted safe history reaches such a pair, proving RS inclusion. Exactness also needs nonempty actual safe histories, still unverified.

For interval requirements (UPD), a fresh reference starts at entry, consumes its marker once and uses actual fluent values. The 89 R2 initializers use only update-stage fluents whose commands cannot precede entry. The 46 R1 formulas have  $\Box(G \Rightarrow \neg A)$ :  $G$  is the command-controlled interval guard and  $A$  a disjunction of ordinary-event predicates. The guard starts false and changes only on excluded pre-entry commands; the action predicates reset at entry. Static formula, label, reset and DFA checks therefore establish exact entry-scoped RS for all 135 UPD requirement occurrences (derivations in S3–S4). Execution assumptions remain unvalidated.

For R1, the general source/export condition checks 46 occurrences, 4,722 state–label pairs and 854 entry valuations; seven negative mutations are rejected (S3). The older  $E'$  condition, restricted to update-stage fluents, left these ordinary-action predicates unverified; those results are retained. GSM/R1’s concrete derivation is in Technical Appendix E.

## E Activation-Scoped Initialization

**Obligations that begin at activation.** *Activation-scoped* interpretation ACT starts a pure invariant  $\Box\varphi$  with the actual post-boundary observer vector, checking  $\varphi$  there and after each later label. It retains factual fluent values, not a past-violation bit. Start follows the declared observer rules and is not consumed twice. This differs from A’s safe-history restriction and H’s whole-past obligation.

**Proposition E.1** (Activation-scoped invariant initialization). *For such an invariant, let a safe reference history end at vector  $v$ . If the supplied initialized monitor has the same continuation language as that reference history’s residual, it is exact for ACT at every activation with vector  $v$ , including activations whose actual histories violated the invariant before its start.*

*Proof.* The safe reference prefix leaves only current/future invariant obligations. Deterministic observers give the same future valuations from  $v$  on every common suffix, proving equality with ACT. Earlier violations in the actual history do not affect those valuations.  $\square$

This requires an appropriate initializer, not merely A inclusion. Section 7.2.1 proves ACT directly for the thirteen invariants along the saved PC2 policy; the 138 A checks remain A checks.

### E.1 From a Safe Lookup Cell to an Activation-Scoped Invariant

Proposition 4.1 in the main paper gives the criterion and complete proof: a faithful invariant monitor reached safely with observer vector  $v$  accepts exactly  $\text{ACT}_\varphi(v)$ . The witness may come from the unrestricted monitor–observer product; physical executability of that reference history is unnecessary. We first check the constructed lookup, then compare its monitors with source formulas, and finally identify the conditions required at actual activations. Main-paper Tables 3 and 5 give the overall guarantee and validation maps.

**Reconstructing the frontend lookup.** The nine inherited inputs have 46 NEW requirement positions and three variants, yielding 138 occurrences. The saved independent reconstruction checks all 138 lookup tables and 1,587 entries, including 900 non-error entries, 687 error entries, 408 observer columns and 31,176 observer transitions. S4.2 gives the export protocol and reconstruction checks. A non-error entry is chosen from a pair reached by a finite safe word of the free monitor–observer product; equivalent-state alternatives must have the same error language. The reconstruction explores non-error successors, so induction on its search depth supplies that witness. The saved table records  $(v, m)$ , not the witness word itself.

**Matching monitors to source invariants.** The reconstruction establishes safe lookup reachability. To check the proposition’s separate faithfulness premise, we compare the saved monitors with the original source invariants independently of the Java formula compiler. All 46 NEW positions are pure invariants  $\Box\varphi$ : seven use declared fluents only, six use action predicates only, and 33 use both. Names such as Pending and SEQUENTIAL designate deterministic current-value observers, not additional temporal operators. A restricted parser reads propositional negation, conjunction, disjunction and implication inside  $\Box$ , assertion aliases, fluent initial values, event-set aliases/differences and finite indexed ranges. Unsupported syntax is rejected. Source-defined observer rules must match the saved rules, including action predicates’ last-event convention.

For each invariant, the finite check verifies  $\varphi(v_0)$  and a non-error initial monitor state, then explores every non-error monitor–observer pair. At each pair  $(m, v)$  and every checked label  $a$ , it verifies

$$\delta_M(m, a) \in \text{Error} \iff \neg\varphi(d(v, a)).$$

The checked alphabet contains every saved observer label and every source-referenced event, plus a fresh representative for all unmentioned events. The latter class has the same source-observer effect (declared fluents stutter and action predicates reset); the saved DFA uses its default transition. From the initial pair, induction on word length now proves that a prefix is non-error exactly when all its source valuations satisfy  $\varphi$ . Thus this finite condition establishes the monitor-faithfulness premise used above on these alphabets; it is not an assumption that a passing table check alone validates the compiler.

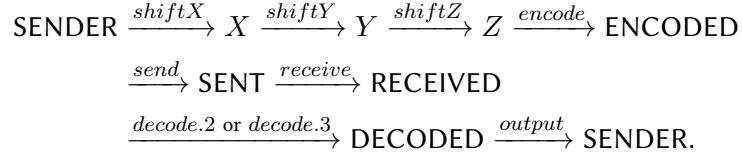
All 138 occurrences pass: 900 non-error product pairs, 35,283 pair/label checks, and all 900 non-error lookup cells contained in the safe product. There are no unsupported formulas or mismatches. The three variants agree on source formula, observers and monitor for each of the 46 positions; the counts are occurrences, not 138 independent formulas. This establishes ACT exactness for those lookup cells under their reference observation semantics.

**What this yields for a winning policy.** At a real activation, the adapter must use this lookup entry for the actual post-start observer vector and must subsequently follow the same observation and update rules. Start is accounted for once, before initialization, rather than observed twice. Every active suffix of a winning policy is accepted by the installed monitor; the proposition therefore gives the invariant at activation and after every observed prefix. A previous actual violation is irrelevant to this explicitly start-scoped requirement. This is not the history-inclusive H interpretation, nor does it turn an arbitrary A inclusion into ACT exactness. An undefined/rejected lookup or a separate fallback initializer is outside this argument. Conversely,  $\varphi(v)$  alone does not ensure that the table has a non-error entry for  $v$ .

**Remaining activation conditions.** Physical-state decoding, actual key selection and absence of fallback at every benchmark activation remain unchecked. In particular, the adapter preserves physical observers across commands, whereas an implicit action predicate can reset under the reference alphabet convention; matching future observations must be checked rather than inferred from table equality. The discrepancy is not shown to be conservative: the saved checks do not determine whether it rejects or admits additional actual activations. These results do not report 900 observed activation points or claim that all 25 WIN policies have newly validated source semantics.

The existing objects are `results/supplement-checks/frontend_lookup_demo/verification_details.json` (per-requirement normalized actual and independently reconstructed tables) and `lookup_check_summary.csv`; `verify_saved.py` contains the reconstruction and exact table comparison. The additional source analysis is supplied in `results/supplement-checks/source_invariant_lookup/`. These checks reuse saved objects and introduce no new synthesis or performance result.

**A source-derived policy with active ordinary behavior.** This is an analytical construction from the supplied `GSM_FG.lts` and saved frontend lookup, not an exported solver policy or an additional experiment. The old plant is the cycle



The new plant inserts  $\text{shift}T$  and state  $T$  between  $Z$  and  $\text{encode}$ ; the supplied transfer maps identically named old states to new ones. All ordinary events are controllable. The old safety requirement forbids  $\text{decode.3}$ ; the new one forbids  $\text{decode.2}$ . The R1 interval forbids ten ordinary labels between old-stop and new-start. Its initializer and source interpretation are derived below.

The new requirement's physical observer is initially zero, becomes one on  $\text{decode.2}$ , and becomes zero on every other ordinary label. The saved lookup maps observer zero to non-error monitor state zero and observer one to error (outside the safe initializer domain). In monitor state zero,  $\text{decode.2}$  enters error; every other label preserves zero. Update commands and transfer preserve the physical observer. In particular, start reads its current value and does not reset it.

**Proposition E.2** (GSM activation with ordinary events). *The supplied GSM/R1 contract admits a state-based completing policy from every old entry whose new-decode requirement is active during ordinary  $\text{decode.3}$  and  $\text{output}$  events. At these activations, the supplied new-requirement initializer is exact for the activation-scoped requirement  $\Box \neg \text{decode.2}$ ; the policy reaches a compatible supplied new-controller state within twelve update events.*

*Proof.* While old-stop is pending, follow the old cycle only until the first RECEIVED state, then select old-stop and new-start consecutively. Select  $\text{decode.3}$ , then  $\text{output}$ , then the identity transfer at SENDER; disable other controllables. No UC event occurs and each selected label has a singleton outcome.

Every old plant state reaches the first RECEIVED in zero to seven edges without taking a decode edge. This prefix preserves the old requirement and the initially false R1 gap guard. At RECEIVED, the previous  $\text{receive}$  has set the observer to zero; the same holds for an old entry at RECEIVED. Hence the new initializer is defined. Stop and start preserve that observer, and no ordinary event occurs while the R1 gap guard is true. After start, both  $\text{decode.3}$  and  $\text{output}$  preserve monitor state zero. A choice of  $\text{decode.2}$  at that same plant state would enter error, so the active requirement constrains an actual ordinary choice.

Following  $\text{output}$ , the physical state is old SENDER with observer zero. Transfer yields new SENDER with observer zero and new monitor zero, exactly the supplied new endpoint's initial projection; all three commands are consumed. The remaining safe R1 monitor is of interval type and is not matched against an endpoint requirement. The fixed new controller subsequently excludes  $\text{decode.2}$ . Its monitor therefore recognizes precisely the safe active suffixes; violations before activation have no bearing on this ACT claim.

Let  $d$  be distance along the old cycle to the first RECEIVED. Give pre-stop states rank  $d+5$ ; give the states after stop, start,  $\text{decode.3}$ ,  $\text{output}$  and transfer ranks 4, 3, 2, 1, 0, respectively. Every edge strictly decreases rank, with maximum twelve and rank zero at a matching goal. The pending-command set and active monitors distinguish these phases, so no extra history memory is needed. This proves all-entry

safety and finite completion. The same event sequence also places both boundaries before transfer, as required by the separately supplied R2 order; no R2 solver result is inferred here.  $\square$

An old execution can already have performed *decode.2* before the update request. This construction thus does not establish H or safe full histories for A. It establishes a non-vacuous ACT obligation over ordinary update actions and the fixed endpoint continuation, within this supplied model. It does not validate the compiler, application intent or runtime adapter.

**From the source interval to an entry guarantee.** GSM/R1 requires  $\square(G \Rightarrow \neg A)$ :  $G$  starts at old-decode stop and ends at new-decode start;  $A$  denotes ten decode/encode, communication and shift events. Every old entry has  $G = \text{false}$ , since no command has occurred. Entry preserves this guard and resets action predicates. Its monitor has guard-false, guard-true and error states: stop sets the guard, start clears it, and  $A$  enters error while the guard holds. Installing guard-false therefore gives exactly the fresh E obligation at every old entry.

The source/export check covers this formula, initializer, all 45 state-label pairs and both action-predicate entry values. This establishes declared E meaning; application intent and runtime conformance remain unvalidated. Technical Appendix D retains the full check population and older unverified results.

## F Proof of Conditional Trace Lifting

**Linking the phases.** The linked model connects disjoint old/update/new phases by entry  $z \mapsto \text{Init}(z)$ , policy edges  $\text{Succ}_\sigma$  and handover  $q \mapsto \kappa(q)$ . Here  $\text{Init}(z)$  is the admitted update state copied and initialized from old state  $z$ ,  $\text{Succ}_\sigma$  gives the policy successors, and  $\text{CL}^v$  is the monitored closed loop of endpoint version  $v$ . Entry and handover add no game event. Let  $\alpha$  be a phase-disjoint observation map.

**A1 (endpoints):** actual endpoint observations map to same-labeled edges of  $\text{CL}^v$  and preserve its component and monitor states.

**A2 (update observations and progress):** at a non-goal, filtering the complete environment event set disables every controllable event outside  $\sigma(q)$  and disables no uncontrollable event; every remaining event/outcome maps into  $\text{Post}(q, a)$  and is reported with its successor identity. The resulting set is nonempty and a next event is eventually observed. This progress premise assumes no fairness among event choices.

**A3 (commands and observation):** each command starts at uncontrollable quiescence, finishes atomically and failure-free in finite time, and reports one modeled transfer or monitor-lifecycle transition with its successor identity. Internal  $\tau$  steps, unobservable outcomes and divergence are outside the model.

**A4 (boundaries):** entry atomically observes the actual reachable old state  $z$  and switches to  $\text{Init}(z)$ ; at a quiescent goal, ordinary controllables are disabled and handover installs its selected controller state and switches control. Each boundary is a finite, failure-free linearizable operation preserving the specified component and monitor states [17].

For the history-based interpretation below, let  $H_r^{\text{exec}}(\xi)$  contain the actual activation histories at  $\xi$ , using the declared reference origin and event projection of  $r$  (and consuming a new requirement's start exactly once). *History completeness* is the explicit premise  $H_r^{\text{exec}}(\xi) \subseteq H_r(\xi)$  at every reached activation. It is not established by A1–A4 or by synthesis.

**Theorem F.1** (Conditional trace lifting). *For a returned WIN certificate  $(\sigma, \text{rk}, \kappa)$  of an eligible contract, every execution satisfying A1–A4 maps through  $\alpha$  to the linked model. From an observed entry  $\alpha(x_0) = \text{Init}(z) \in Q_0$ , the update reaches quiescent handover within  $\text{rk}(\text{Init}(z))$  observed game events and then hands control to the compatible endpoint. With history completeness and RS for a supplied initializer, its active interval satisfies the declared residual interpretation. Agreement with application intent remains an input obligation.*

*Proof.* Fix an execution satisfying A1–A4 and a returned certificate  $(\sigma, \text{rk}, \kappa)$ . We use the phase-disjoint observation map  $\alpha$  from the theorem; the assumptions require it to preserve the specified component and monitor states, not merely event names. A1 maps the old prefix to same-labeled edges in the monitored old closed loop. At entry, A4 supplies one linearization point at which the actual reachable old state  $z$  is observed and control changes to  $\text{Init}(z) \in Q_0$ . Entry contributes the linked model’s boundary edge and no game event. Thus the update observation starts in the certificate domain.

Suppose the current observation is a non-goal certificate state  $q$ . A2 filters controls according to  $\sigma(q)$  while retaining every enabled UC event. Its next reported event and successor identity therefore give an edge  $q \xrightarrow{a} q'$  in the complete policy successor relation. For a command, A3 additionally supplies the required UC quiescence, finite atomic completion and exact modeled successor. Ordinary reports satisfy A2 directly. The checked WIN certificate includes every such successor, so  $q'$  remains safe and in its domain, with  $\text{rk}(q') < \text{rk}(q)$ . This establishes the update-prefix simulation and rank invariant by induction, for every outcome rather than a selected favorable one.

A2 makes a next event eventually observable at every non-goal; A3 rules out a nonterminating command between observations. Consequently the execution cannot stop or stall at a non-goal. An infinite sequence of update events would give an infinite strictly descending sequence of natural ranks, which is impossible. At most  $\text{rk}(\text{Init}(z))$  game events therefore reach a goal. This is an event bound: the assumptions provide no uniform bound on the time between observations. They also impose no fairness on which enabled event or outcome occurs.

The goal predicate supplies UC quiescence and a compatible reachable new-endpoint state  $\kappa(q)$ . A4 disables ordinary controls, preserves the specified component and monitor states, and completes handover at a finite linearization point. Representing this operation by the linked boundary edge adds no game event. A1 then maps every new-phase observation to the selected monitored endpoint’s same-labeled continuation. Thus the complete observed execution maps into the linked model. The argument assumes the two boundary implementations have the A4 properties; it does not prove their linearizability or atomicity from implementation code.

Finally, certificate safety preserves each supplied monitor while it is active. Fix a reached activation at  $\xi$ , and an actual history  $h \in H_r^{\text{exec}}(\xi)$  measured from the declared reference origin using the requirement’s event projection. The explicit completeness premise  $H_r^{\text{exec}}(\xi) \subseteq H_r(\xi)$  gives  $h \in H_r(\xi)$ . Monitor safety gives  $w \in L_r(\iota_r(\xi))$  for every finite observed active suffix  $w$ . RS then gives  $hw \in S_r$ . The same argument applies at every activation; prefix closure covers every finite prefix of an infinite active interval. It does not establish completeness of the supplied histories or their agreement with intent. If a history-inclusive interpretation contains an earlier violation, an activation-safe scope that omits it does not satisfy this premise. Cell’s activation duty and the stated ACT cases instead have their separate direct interpretation arguments in Section 4.1. Unobservable transitions, divergence, failed commands and implementations violating A1–A4 are outside the theorem.  $\square$

## G Endpoint Interfaces and Certificate Reuse

Let  $T$  project loadable targets onto component and new-monitor states, omitting controller state. The update-game construction uses endpoint controllers only through entries  $Q_0$  and  $T$ .

**Theorem G.1** (Endpoint-interface reuse). *Keep components, transfers, monitors, initializers, event control and precedence fixed. Eligible contracts with equal entries and target projections have identical update games. Every WIN policy/rank and LOSS certificate transfers between them; selecting a compatible target of the second contract preserves that endpoint’s continuations. A winning policy also transfers to fewer entries and more projected targets, stopping at newly admitted goals.*

The following proof establishes each clause. This interface result does not claim an implemented cache or a measured reuse speedup. Let  $K$  and  $K'$  satisfy Theorem G.1. Their local data define the same finite ambient state space of version-tagged component tuples, monitor memories and pending-command sets. Let  $I, I'$  be their entry sets, and let  $T, T'$  be their target projections. A target projection

retains all component and new-monitor coordinates and removes only the endpoint-controller coordinate. All targets retain their eligibility premises, including reachability and endpoint safety, deadlock freedom and preservation of uncontrollable events.

**Identical interfaces.** Assume  $I = I'$  and  $T = T'$ . On every ambient state, enabled ordinary events and their complete outcome buckets are equal because the component models and controllability agree. Transfer guards and outcomes agree because the transfer relations, pending commands and precedence agree. Boundary guards and memory updates agree because monitors and initializers agree. UC quiescence and safety therefore agree as well. Induction on path length from  $I$  gives equal reachable state sets and label/outcome transitions. No rule inspects either endpoint-controller state after entry. Old states collapsed by  $\text{Init}$  may have different controller coordinates, but such coordinates impose no source restriction once old control is released.

A reached state is a goal precisely when it is safe, UC-quiescent, has no pending command, and its component/new-monitor projection belongs to  $T$ . Hence goals agree. The identity map preserves the policy successor relation, safety, nonblocking, all maximal paths and every strict rank inequality. It also preserves the losing-certificate obligations: goals are absent from  $L$ , a losing entry remains in  $L$ , and at each safe member the same UC escape witness or all-control counter-outcomes remain in  $L$ . Thus WIN and LOSS certificates transfer, including their universal quantification over all enabled outcomes.

For every reached goal  $q$ , membership of its projection in  $T'$  supplies at least one full target  $z' \in Z'_{\text{load}}$ . Choose one and define  $\kappa'(q) = z'$ . The selected endpoint controller is eligible and the handover copies precisely its component, monitor and controller coordinates. Proposition 3.2 yields an isomorphism to that endpoint rooted at  $z'$ . This preserves that endpoint's complete continuation language; it does not claim equivalence between the two endpoint controllers' continuation languages. Nondeterministic local outcomes are unchanged and remain adversarial. Several targets with one projection cause no difficulty: selecting a compatible controller state is part of handover, not selection among an event's nondeterministic outcomes.

**A Cell replacement with different continuation behavior.** Delete only  $c_0 \xrightarrow{j} c_0$  from the new controller in Table 1. Its alphabet still contains  $j$  through  $c_1 \xrightarrow{j} c_0$ . The reachable endpoint tuples remain exactly  $z_A = (h, e, c_0, c)$ ,  $y_B = (e, h, c_1, d)$  and  $z_B = (e, h, c_0, c)$ : the cycle  $z_A \xrightarrow{m_B} y_B \xrightarrow{j} z_B \xrightarrow{m_A} z_A$  remains, and the deleted edge removes only the inspection self-loop at  $z_B$ . This closed loop remains safe and deadlock-free; all events are controllable. Keeping  $Z_{\text{load}} = \{z_A, z_B\}$  preserves its target projection, and the old endpoint is unchanged. Thus Figure 1's whole update certificate and its handover choices transfer. The endpoint languages differ: from  $z_B$ , the original admits  $j^\omega$ , whereas the replacement's first event must be  $m_A$ . Reuse therefore does not require continuation-language equivalence between endpoints. This finite construction adds no solver measurement.

**Smaller entry interface and larger target interface.** Assume instead  $I' \subseteq I$  and  $T \subseteq T'$ , and let  $(\sigma, \text{rk}, \kappa)$  win for  $K$ . Starting from  $I'$ , use  $\sigma$  until the first goal of  $K'$  and stop there. Every resulting nonterminal prefix is also a prefix of the original policy from  $I$ , because all local transitions agree. Safety and nonblocking are inherited. An infinite resulting path would be an infinite original-policy path, contradicting its finite completion. Every original goal has its projection in  $T'$  and is also a goal when reached from  $I'$ , so a maximal resulting path cannot pass an old goal without stopping. A new goal may be reached earlier, which only truncates a path.

Restrict  $\text{rk}$  to this truncated closure and assign zero at its goals. Edges between remaining non-goals keep their original strict inequalities. An edge entering a goal has a positive-rank non-goal source and zero-rank target, so also decreases. Choose a compatible target in  $Z'_{\text{load}}$  at each such goal. This supplies a complete WIN certificate for  $K'$ . The converse is not claimed: fewer entries or additional targets can make a formerly losing contract win. Nor does the theorem preserve the particular policy returned by a new solver run, which may use a different endpoint-dependent exploration order.

## G.1 Exactly When a Fixed Returned Policy Can Be Reused

The following result is relative to an already returned WIN certificate; it does not require equal or nested target interfaces.

**Theorem G.2** (Policy reuse). *For a returned WIN certificate  $(\sigma, \text{rk}, \kappa)$ , write  $D = \text{Reach}_\sigma(Q_0)$  and  $G_D = D \cap Q_{\text{goal}}$ , and extend  $\sigma$  by the empty set on  $G_D$ . Fix components, transfers, monitors, initializers, event control and precedence. For another eligible endpoint pair with entries  $Q'_0 \subseteq D$  and goals  $G'$ , let  $D'$  be the closure under the extended policy, stopping at  $G'$ . That policy wins exactly when  $D' \cap G_D \subseteq G'$ .*

Here is the complete construction and proof.

Let  $K$  have certificate  $(\sigma, \text{rk}, \kappa)$ , finite support  $D = \text{Reach}_\sigma(Q_0)$  and terminal set  $G_D = D \cap Q_{\text{goal}}$ . Define  $\bar{\sigma}$  on  $D$  by retaining  $\sigma$  off  $G_D$  and choosing no controllable at  $G_D$ . The latter states have no UC event by the original goal condition, so they have no successor under  $\bar{\sigma}$ . Every state of  $D$  is safe. Each original nonterminal has at least one successor, retains every UC and every selected-label outcome within  $D$ , and has strictly larger rank than each of those successors.

Let  $K'$  be eligible with the same components, transfers, monitors, initializers, controllability and precedence, but possibly different endpoint controllers and loadable targets. Assume  $Q'_0 \subseteq D$ . After admission the successor rules inspect only the fixed local data, so they agree on  $D$  in both contracts. Evaluate the new goal test  $G'$  on these states using the new projected targets. Starting from  $Q'_0$ , form the least closure  $D'$  under  $\bar{\sigma}$ , expanding only states outside  $G'$ . Closure of the old certificate gives  $D' \subseteq D$ . In particular, constructing  $D'$  requires only the stored policy successor graph and the new entry/goal information; it does not explore unselected update actions. Obtaining and validating that endpoint information remains a separate task.

Suppose  $D' \cap G_D \subseteq G'$ . At a reached non-goal, the state cannot be in  $G_D$ , so it retains the original nonempty successor set, with all uncontrollables and selected-event outcomes. Every such successor is in  $D'$ . Safety is inherited from  $D$ . Set  $\text{rk}'(q) = 0$  on  $D' \cap G'$  and  $\text{rk}'(q) = \text{rk}(q)$  elsewhere. An original nonterminal has positive rank: otherwise its nonempty successor set would require a negative natural-number rank. Thus every edge entering a new goal strictly decreases, and every remaining edge retains its original strict inequality. No infinite non-goal run exists, and no finite maximal run can end at a non-goal. The policy therefore wins from every new entry. For each reached new goal, choose any compatible full target in  $Z'_{\text{load}}$  to define  $\kappa'$ . Its existence follows from the new goal test, and endpoint eligibility plus Proposition 3.2 preserves that endpoint's own continuation language.

Conversely suppose  $q \in D' \cap G_D$  but  $q \notin G'$ . By the definition of  $D'$ , a finite run from a new entry reaches  $q$  without having stopped at a new goal. As an original terminal,  $q$  is UC-quiescent, and  $\bar{\sigma}(q) = \emptyset$ . Hence this is a finite maximal run ending at a non-goal deadlock in  $K'$ , so the fixed policy is not winning. This proves both directions. It says nothing about another policy or about extending the old policy beyond its former terminals; either may solve  $K'$ .

**A simple sufficient target condition.** If the new target projection contains the projections of every member of  $G_D$ , every original terminal remains a new goal. The criterion then holds for any eligible  $Q'_0 \subseteq D$ . Unlike whole-interface equality or target-set enlargement, it permits removing old loadable targets never used by the certificate. It is only sufficient: new goals can also stop every path before a removed old terminal.

**Application to the saved PC2 certificate.** The PC2-Rolling certificate used in Technical Appendices J and J.1 has 126 entries, 381 states and 522 edges, with maximum rank 37. Its only all-commands-completed state is its single terminal, whose handover selects new-00000009. Keeping the old endpoint and narrowing the loadable set to this one reachable target leaves that same certificate valid, even if the original loadable set contained other targets. This is a direct specialization of the sufficient condition, using the already saved certificate; it is not a new synthesis or timing result. The eligibility of a different supplied new controller, and the presence of a target with the required component

and all-new-monitor coordinates, must still be checked. No equivalence between different endpoint controllers' continuation languages is assumed.

## H Complete Native DUCS Construction for Cell

Table A2 specializes main-paper Table 2 to the analytical Cell inputs. This section gives the DUCS construction; Appendix I gives the GR(1) construction. Table A3 below maps each Cell obligation to its encoding.

Table A2: Analytical Cell constructions: supplied and generated structures. No general translation, minimum-size, effort or performance result is claimed.

| Interface                                 | Supplied Cell structure                                                                                                                                                                 | Generated or selected structure                                                                                                                                         |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DUCS [34, Defs. 4.1–4.3, 5.1]             | Holder-refined old controller; environment transfer $R$ ; eight version/holder and three endpoint states in $E'$ ; guarded transfers, fluent goals $G, G', T$ and eventual handover $h$ | Protocol/interrupt composition, update environment/goal, update-and-subsequent controller with map $H$ ; legal, deadlock-free completion from all old-controller states |
| GR(1) updating [2, Def. 3.1, Secs. 4–4.1] | Phase-extended specifications with version/lifetime variables, transfer guards, $p' = [h = B]$ at start, old inspection ban released at stop, switching condition and endpoint edges    | New controller $C_2$ , bridge and switching bookkeeping; bounded switch from winning states, then $C_2$ unchanged                                                       |
| FG-DUCS                                   | Local components, $g_A = g_B = \{(e, e)\}$ , typed monitors, initializers, start-before-stop, endpoint controllers and $Z_{\text{load}} = \{z_A, z_B\}$                                 | Mixed-state and boundary transitions, target tests; policy, rank and handover map preserving the supplied continuation                                                  |

The eight stages record versions and holder, excluding protocol and monitors. FG-DUCS's new-monitor initializer also covers all eight tuples. All three interfaces require justified input meanings.

We construct a correct Cell update using native DUCS environments, relation  $R$ , transition requirement  $T$  and boundary protocol  $O$  [34, Defs. 4.1–4.3 and 5.1].

DUCS's `hotSwap` event requests the update and interrupts old control. Its `reconfigure` milestone switches environments; our extra  $h$  connects mixed stages to a fixed endpoint.

**Environments and entry.** Refine the old controller to its reachable closed loop  $c_A \xrightarrow{m_B} c_B \xrightarrow{m_A} c_A$ , preserving ordinary traces and recording the holder. The old environment has these same moves. Let  $E'$  contain eight stages  $(v_A, v_B, L) \in \{o, n\}^2 \times \{A, B\}$  and three post-handover states. In a stage,  $m_B$  moves  $A$  to  $B$ ,  $m_A$  moves  $B$  to  $A$ , and  $j$  self-loops only if  $v_B = n, L = B$ . Transfer  $\rho_i$  changes  $v_i = o$  to  $n$  only while station  $i$  is empty. A controllable  $h$  maps  $(n, n, L)$  to  $z_L$ . Post-handover edges copy the fixed new closed loop:

$$z_A \xrightarrow{m_B} y_B \xrightarrow{j} z_B, \quad z_B \xrightarrow{j} z_B, \quad z_B \xrightarrow{m_A} z_A.$$

Here  $y_B$  is inspection-pending;  $z_A, z_B$  are clear targets. The standalone initial state is  $z_A$ , but interrupt entry uses  $R : L \mapsto (o, o, L)$ . The 11-state count excludes protocol and monitors and is not a minimality result.

**Requirements and memory.** Write  $s = \text{startNewSpec}$  and  $t = \text{stopOldSpec}$ . Protocol  $O$  accepts these physical stutters. Let  $\dot{a}$  record last event  $a$ , resetting on every other label. Fluents  $B = \langle m_B, m_A, \text{false} \rangle$  and  $P = \langle m_B, j, \text{false} \rangle$  retain holder and uninspected-arrival history. Take  $G = \neg \dot{j}$  and  $G' = \neg(P \wedge \dot{m}_A)$  under native old-until-stop and new-after-start scopes.



For each event  $a$ , let  $S_a$  be its initially false, never-reset occurrence flag. Take  $T$  to be the conjunction

$$\begin{aligned} & \Box(\dot{s} \Rightarrow S_{\text{reconfigure}}), & \Box(t \Rightarrow S_s), \\ & \Box(\dot{h} \Rightarrow S_{\rho_A} \wedge S_{\rho_B} \wedge S_s \wedge S_t \wedge \neg(B \wedge P)), & \Box(\text{reconfigure} \Rightarrow \Diamond \dot{h}). \end{aligned}$$

Interrupts,  $O$ , version guards and the post-handover region ensure at most one occurrence of each update event. This analytical  $T$  includes eventual  $h$ , beyond the published safety- $T$  implementation [34, Sec. 5.3]; no solver experiment is claimed. Native DUCS still requires eventual stopOldSpec, reconfigure and startNewSpec, including in its safety- $T$  implementation. Here reconfigure enters the mixed stages;  $h$  is an additional fixed-target connection, so those three milestones do not by themselves require  $h$ . We do not rule out another encoding within safety- $T$ .

**Proposition H.1** (Cell witness preservation). *This staged native DUCS instance admits a correct update from both reachable old entries. On the segment from hotSwap through  $h$ , erasing these two events and reconfigure and renaming  $s, t$  to Cell's boundaries gives the displayed FG policy paths. After  $h$ , its ordinary traces equal those of the supplied new closed loop rooted at the selected target.*

*Proof.* Use DUCS's assembled environment, including  $O$ , its old-event renaming/fluent extension, and controller extraction [34, Sec. 5, Theorem 5.1]. Preserve old behavior before hotSwap. Map  $H(c_L)$  to the corresponding pre-reconfiguration controller state; then choose reconfigure, the six-/five-event path of Figure 1,  $h$ , and every fixed-endpoint edge thereafter. All choices are enabled in the assembled environment; no ordinary UC event remains after initiation, ensuring legality and deadlock freedom.

Old  $j$  is disabled until  $t$ . On safe active prefixes the correspondence is: the FG monitor is pending iff  $B \wedge P$ . At  $s$ , a  $B$  holder has an earlier  $m_B$  and no  $j$ , so both are pending. At an  $A$  holder,  $B$  is false and FG is clear, although  $P$  may remain true from an earlier visit to  $B$ . This is equivalence of permitted continuations on the plant, not equality between  $P$  and the monitor state:  $m_A$  and  $j$  are disabled at  $A$ , and the next  $m_B$  makes both representations pending. Thereafter  $j$  clears  $P$  and the monitor. An enabled  $m_A$  comes from  $B$ ; it leaves  $P$  unchanged while setting  $\dot{m}_A$ , so it violates  $G'$  exactly when FG was pending. Otherwise it leaves  $B \wedge P$  and FG clear. Transfers and boundaries preserve the correspondence. Thus every ordinary extension has the same safety outcome in both representations. The finite paths satisfy the boundary order and reach all-new, clear targets, fulfilling  $T$  and the native update milestones. After  $h$ , the synchronized environment/controller copies retain every fixed-endpoint edge. Their reachable product is diagonal, giving the claimed ordinary-trace equality.  $\square$

The native encoding explicitly supplies stages, guards, holder refinement, history fluents, target gate and post-handover graph. FG-DUCS generates the update-game counterparts from local transfers, lifecycle rules, initializers and fixed targets, returning rank and handover certificates. This deterministic, fully controllable example establishes one-direction witness preservation, not a general translation, solution-set equivalence, measured effort saving or performance advantage.

Table A3: The same Cell obligations through the two interfaces. Native entries refer to this explicit construction; they are not lower bounds on other encodings. FG-DUCS supplies the generated structure in addition to the modeler’s inputs.

| Obligation                | Explicit native Cell modeling                                                                         | FG-DUCS input and generated structure                                                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Empty-station replacement | Eight mixed stages in $E'$ , holder moves and version/empty-station guards for $\rho_A, \rho_B$ .     | Old/new models for $A, B$ and $g_A = g_B = \{(e, e)\}$ generate mixed versions and transfer edges with lifecycle/quiescence guards.                       |
| Inspection lifetime       | Holder-refined old control; $B, P, \dot{a}, S_a$ fluents and $G, G', T$ encode memory and boundaries. | Typed monitors, initializers and declared precedence generate boundary events, active-memory updates and initialization under the type-defined lifetimes. |
| Fixed continuation        | Post-handover graph in $E'$ , a guarded $h$ and eventual- $h$ in $T$ .                                | Fixed controllers and load targets generate matching/goal conditions; synthesis selects a load state and handover map.                                    |

Native DUCS also constructs its standard protocol  $O$  and interrupt composition; these are not additional modeler inputs. The native witness retains every fixed-endpoint edge after  $h$ .

## I Cell through a GR(1) Bridge Interface

Amram et al.’s update problem takes old/new GR(1) specifications and a switching condition. Before switching, an update strategy satisfies the old system safety relation; afterwards it satisfies the new relation and new justice, under the new environment safety assumptions [2, Def. 3.1]. Their construction generates the bridge and its *switch/allowed* bookkeeping [2, Sec. 4.1]. We describe concrete predicates for Cell to identify the additional mixed-version and lifetime structure supplied by this Cell construction. This is an analytical input construction, not a Spectra run or a general translation.

**A phase extension of the endpoint specifications.** Cell is deterministic and all its events are controllable. Take no environment variables and the environment safety relation to be true. System variables are the holder  $h \in \{A, B\}$ , versions  $v_A, v_B \in \{0, 1\}$ , new-start and old-stop flags  $s, t$ , inspection debt  $p$ , a phase  $d \in \{\text{bridge}, \text{post}\}$ , an event selector  $a$ , and the post-phase controller coordinate  $c \in \{c_0, c_1\}$ . Before start,  $p = 0$ . The old initial valuation has  $h = A$ , all five bits zero,  $d = \text{bridge}$ ,  $c = c_0$  and an unused initial selector value; its other old entry has  $h = B$ . The two-state old closed loop only alternates  $m_B, m_A$  and preserves all added bits.

Let  $G = v_A \wedge v_B \wedge s \wedge t \wedge \neg p$ . Define  $R_B$  on bridge-phase states with  $\neg G$  by the following guarded assignments. A row sets  $a'$  to its label and performs its assignments; every other unlisted coordinate is unchanged. There is no additional waiting round.

| Label    | Guard                               | Assignment             |
|----------|-------------------------------------|------------------------|
| $m_A$    | $h = B \wedge (\neg s \vee \neg p)$ | $h' = A$               |
| $m_B$    | $h = A$                             | $h' = B, p' = s$       |
| $j$      | $h = B \wedge v_B \wedge t$         | $p' = 0$               |
| $\rho_A$ | $\neg v_A \wedge h = B$             | $v'_A = 1$             |
| $\rho_B$ | $\neg v_B \wedge h = A$             | $v'_B = 1$             |
| start    | $\neg s$                            | $s' = 1, p' = [h = B]$ |
| stop     | $s \wedge \neg t$                   | $t' = 1$               |

The holder type preserves exactly one workpiece; transfer guards express empty-station replacement;  $s \wedge \neg t$  enforces start-before-stop. The  $j$  guard releases the old inspection prohibition only after local stop. Thus  $R_B$  is a *phase extension* of the old specification, not the unmodified old requirement. Keeping the original global prohibition until the final specification switch would exclude the displayed path that inspects before its last transfer. The supplied old controller nevertheless satisfies this extension because it never changes the added bits or inspections.

Take the old system safety relation to be  $R_B$  and the switching condition to be  $d = \text{bridge} \wedge G$ . The new system relation permits one auxiliary handover from that condition, preserving  $h, v_A, v_B, s, t, p$  and setting  $d' = \text{post}$ ,  $c' = c_0$ ,  $a' = \text{handover}$ . In the post phase it permits exactly the endpoint edges

$$z_A \xrightarrow{m_B} y_B \xrightarrow{j} z_B, \quad z_B \xrightarrow{m_A} z_A, \quad z_B \xrightarrow{j} z_B.$$

Here  $p = 0, 1, 0$  and  $c = c_0, c_1, c_0$  at  $z_A, y_B, z_B$ , respectively, and both versions and lifetime flags are one. The new initial valuation is post-phase  $z_A$ ; both justice conditions are true. The two loadable bridge projections reach this new winning region through the auxiliary handover.

**What this construction establishes.** Each edge of Figure 1 satisfies its row: start records the actual holder, inspection clears debt only after old-stop, and transfer preserves debt. Both displayed paths reach  $G$  and then take the auxiliary handover. With FG rank  $r$ , assign  $r + 1$  before handover and zero afterwards; every bridge round, including the auxiliary handover, strictly decreases rank. The six-/five-event paths therefore give seven-/six-round bridge witnesses. Erasing the auxiliary handover preserves their local events, initialized memories and selected physical targets. Unsafe controllable choices are excluded by the guards; we assert witness preservation, not equality of unsafe game buckets.

The post relation permits every supplied endpoint edge, although synthesizing a new GR(1) controller may select fewer of them. Once that  $C_2$  is selected, Amram et al. execute it unchanged after the bridge [2, Sec. 4.2, Alg. 2]. FG-DUCS instead takes the replacement controller and loadable targets as inputs; its continuation guarantee concerns that supplied controller. GR(1) predicates also avoid enumerating every mixed state: the comparison concerns who supplies the version, lifetime, initialization and target equations, not a state-count lower bound. FG-DUCS derives those equations' operational counterparts from the local contract. No modeling-effort or runtime reduction is inferred.

The construction does not extend unchanged to contracts with UC or nondeterministic outcomes. Their environment/system move order and observation of realized successors must be aligned: replacing a control choice followed by all its outcomes with an environment-first round can change  $\exists a \forall o$  into  $\forall o \exists a$ . The complete deterministic Cell comparison avoids that issue; the main all-outcome guarantee remains defined by FG-DUCS's own game.

## J Saved PC2 Activation Histories and Their Boundary

The supplied old controller and pre-synthesized new controller are fixed. Baseline PC2's old requirements and 12 new requirements remain; added new-endpoint calibration availability yields 13 new starts, separate from the interval duty. Transfers map old destinations to calibration counterparts, retaining ordinary observer memory. Work, calibration requests and updates are controllable; arrivals, tool outcomes, resets and calibration completion are UC. Entry inherits old memory and initializes the interval monitor; new-start looks up the current ordinary observer tuple. No precedence is supplied. Targets match components and new-monitor memory at reachable new-endpoint states.

Interval  $R_{CAL\_READY}$  observes transfers as calibration entry; endpoint availability observes ordinary requests. Completion restores both. Handover matches component, observer and new-monitor memory; real-history and runtime obligations remain unvalidated.

**Generation responsibilities beyond Cell.** PC2 separates an application obligation from its generated update protocol. The designer supplies the two old/new arm models, their local transfer maps, twelve old and thirteen new requirements, the interval duty and fixed endpoints. For interval availability, each readiness bit starts true, becomes false on its arm's transfer or calibration request, and becomes true on completion; their disjunction must hold. This obligation is supplied, not inferred from the plant.

From these inputs, FG-DUCS constructs the twenty-five independent requirement boundaries and two local transfers, their active-monitor updates, mixed-version successors, UC-quiescence guards and

target tests. The frontend prepares current-observer lookup maps and physical closure; the lazy solver generates the update product on demand. The saved result covers 126 entries with 381 policy states, 522 edges and maximum rank 37. Thus neither the sequence nor those policy states are part of the input. These counts describe one saved result, not a bound on the full mixed product.

For the published DUCS interface, a representation of these independently changing arms and requirement memories would place their staged behavior and admissibility in the supplied  $E'$  and transition requirements. DUCS itself still generates its global update protocol; that work is not assigned to the designer [34, Secs. 4.2, 5]. A GR(1) representation would supply corresponding version, local-state, active-memory and observer variables with transition constraints; its existing construction generates switching bookkeeping and the bridge [2, Def. 3.1, Sec. 4]. Unlike the deterministic Cell construction, PC2 has UC outcomes, so this responsibility comparison does not establish an equivalent GR(1) game or a general translation. It identifies which local operations FG-DUCS generates directly, without measuring modeling effort, minimum encoding size or external-solver performance.

Table A4: One saved 31-event PC2-Rolling path. O/N: old/new; ARM: unprocessed workpiece; CAL: its calibration counterpart. Ordinary observers stay zero.

| Reached after                  | Arm 1     | Arm 2     | Available | Rank |
|--------------------------------|-----------|-----------|-----------|------|
| Update entry                   | O/initial | O/initial | 2         | 31   |
| Two arrivals, twelve old stops | O/ARM     | O/ARM     | 2         | 17   |
| Transfer arm 1                 | N/CAL     | O/ARM     | 1         | 16   |
| Calibration completion 1       | N/ARM     | O/ARM     | 2         | 15   |
| Transfer arm 2                 | N/ARM     | N/CAL     | 1         | 14   |
| Calibration completion 2       | N/ARM     | N/ARM     | 2         | 13   |
| Thirteen new starts: handover  | N/ARM     | N/ARM     | 2         | 0    |

At handover, all 13 new monitors are 0 and no command remains; the matched endpoint has both arms operational. Certificate identifiers are in Technical Appendix J. This path supplies no timing and does not replace all-entry/all-outcome checking.

This analysis traverses existing PC2-Rolling exports; it runs no solver and adds no timing trial. The selected requirement is  $\Box(out_1 \Rightarrow Drilled_1 \wedge Cleaned_1 \wedge Painted_1)$ . The three fluents start false, successful tool events set them, and *reset*<sub>1</sub> clears them. An *out*<sub>1</sub> with any false fluent violates the requirement. Other update commands stutter for these fluents.

The saved old closed loop contains 126 reachable states and 348 edges; the policy contains 381 states and 522 edges. Its maximum state and entry rank is 37; all 522 edges decrease rank, and the longest entry-to-goal path has 37 events. The stored certificate therefore covers all 126 entries; the rank-31 path below is one representative. Matching component states, old-monitor memory and ordinary observers identifies each policy entry with one old state. Every maximal policy run crosses the single selected start edge  $q_{13} \rightarrow q_{12}$ , which installs monitor state 0 at two NEW/ARM components with all 24 ordinary observers zero. Ranks strictly decrease; removing that start edge makes the unique goal unreachable from every entry. The path in Table A4 starts at  $q_{126}$  and ends at  $q_7$ , whose handover target is new-00000009. All 13 new monitors are 0 there and no command remains. Its prefix to  $q_{12}$  has two arrivals, twelve stops, four transfer/completion events and eight starts, with no *out*<sub>1</sub> or successful tool result.

For each entry, find an old prefix from the declared old initial state and a policy prefix ending at this start. Keep an edge only if the source fluent values and its label do not violate the selected new requirement. Forward reachability in the old graph and reverse reachability from the start in the policy graph yield the table. Separate per-entry breadth-first traversals construct positive witnesses or record failure of each portion; set-based fixed points reproduce the same partition.

| Safe old prefix exists | Safe policy prefix exists | Entry count |
|------------------------|---------------------------|-------------|
| Yes                    | Yes                       | 39          |
| No                     | Yes                       | 21          |
| Yes                    | No                        | 66          |

All 126 entries reach the start through some unrestricted saved-model history. The 39 positive rows establish existential nonemptiness at this one reached tuple, not safety of all histories. An old cycle

$in_1; drill_1; drillOk_1; polish_1; polishOk_1; clean_1; cleanOk_1; out_1; reset_1$

returns to the identical old initial tuple. It violates the new requirement at  $out_1$  because  $Painted_1$  is false. Prepending this cycle to any entry's history preserves its final activation tuple while retaining a past violation. The new monitor is absent in every old state and all 375 policy ancestors of its start, so none of these negative histories refutes its active-lifetime safety. They show why activation-safe and history-inclusive scopes cannot be conflated.

Replay of source-declared plant moves and all 24 observer updates agrees with all 348 old and 522 policy edges. This establishes consistency of these saved histories, not independent frontend correctness or completeness of exported outcomes. The artifact retains all 126 rows, positive paths, negative reasons and 18 boundary edges separating the safe reachability sets. Only one reached tuple of one requirement is covered; the reported 38,288 initializer guard tuples, the other 12 PC2 new requirements and the 138-occurrence population are not checked for nonemptiness by this analysis.

**Direct ACT argument after the selected start.** ACT begins the invariant with the actual observer values at activation. For the selected requirement these values are all zero. The saved policy graph's complete suffix is:

| Source (rank) | New requirement started | Target (rank) |
|---------------|-------------------------|---------------|
| $q_{12}$ (5)  | OUT_IF_FINISHED_2       | $q_{11}$ (4)  |
| $q_{11}$ (4)  | TOOL_ORDER_1            | $q_{10}$ (3)  |
| $q_{10}$ (3)  | TOOL_ORDER_2            | $q_9$ (2)     |
| $q_9$ (2)     | PAINT_ONCE_1            | $q_8$ (1)     |
| $q_8$ (1)     | PAINT_ONCE_2            | $q_7$ (0)     |

All six states have two NEW/ARM components, 24 zero observers and selected-monitor state 0. No ordinary event, transfer or uncontrollable edge occurs in this suffix. The unique goal matches new-00000009. Consequently the source invariant holds at activation and on every remaining update step; matching the verified new endpoint supplies its post-handover guarantee.

The saved graph is closed over 381 entry-reachable states; all 522 edges decrease a natural rank, and no non-goal deadlocks. Removing the unique selected start leaves 375 reachable states and no goal from any of the 126 entries. Thus every maximal saved-policy run crosses the start and follows this suffix. The static checker `check_saved_suffix.py` reproduces these facts from the existing certificate without invoking synthesis. Completeness of modeled outcomes relies on the stored certificate checker; frontend/compiler correctness and real execution are not independently proved here. The argument concerns one requirement and one reached tuple, not all 138 NEW occurrences. The A/H diagnostics above remain unchanged.

## J.1 All New Requirements of the Saved PC2 Policy

The previous history analysis fixes one requirement to distinguish A from H. A separate inspection of the same saved policy establishes the active-lifetime obligation for all thirteen new requirements. It introduces no solver run, input change or timing trial.

The only new-start edges are the chain  $q_{20} \rightarrow q_{19} \rightarrow \dots \rightarrow q_7$ , in this order: stamping avoidance for arms 1 and 2; calibration availability; clean-once for arms 1 and 2; drill-once for arms 1 and 2;

output-if-finished for arms 1 and 2; new tool order for arms 1 and 2; and paint-once for arms 1 and 2. The selected start above is the eighth edge. Every state in this chain has two NEW/ARM components, all 24 ordinary observers zero and rank  $q - 7$  at  $q$ . Each newly installed monitor is 0 and stays 0. There is no ordinary event, transfer or UC edge after the first new-start.

**Proposition J.1** (All PC2 active obligations). *Every maximal run of the saved PC2-Rolling policy from any of its 126 entries activates all thirteen new requirements and satisfies each source invariant from its own activation through handover. The matching verified new endpoint preserves these obligations afterward.*

*Proof.* The saved domain is entry-reachable and closed, every edge strictly decreases a natural rank, and no non-goal deadlocks. Hence every maximal run reaches the unique goal  $q_7$ . Every  $q_j$ ,  $7 \leq j \leq 19$ , has only its preceding chain edge as an incoming policy edge, and no entry lies on the chain. Thus every run traverses all thirteen starts.

For each arm, the source invariants forbid stamping; forbid cleaning, drilling or painting an already completed workpiece; require all three completed-tool fluents at output; and constrain tool-pending fluents by the new tool order. At activation all completed-tool and pending fluents are false, and no ordinary action occurs on a start label. Each invariant therefore holds. The calibration invariant is  $\Box(\neg \text{Calibrating}_1 \vee \neg \text{Calibrating}_2)$ . Its source fluents start false and are set only by ordinary calibration requests. The old plant has no such request, so every old entry retains false values; none occurs in this policy either, while completion clears them. They too are false on the start chain. Start commands preserve these fluents and produce no ordinary action, so all active source invariants continue to hold until  $q_7$ . This goal matches new-00000009, with both components, all ordinary observers and all thirteen new monitors aligned. The supplied verified endpoint and continuation preservation give the post-handover part.  $\square$

This covers all new requirements of this one saved policy at their reached activation tuples. It does not prove initializer exactness over every permitted tuple, safe pre-activation histories, all 138 NEW occurrences, compiler correctness or runtime conformance. The 39/87 history partition above is unchanged. PC2 completes calibration before activating any new requirement; its active update suffix contains only starts. The GSM construction in Proposition E.2 complements it with ordinary behavior under an active new requirement.

The earlier violations have a concrete source distinction: old PC2 requires drilling, polishing and cleaning, whereas the new output rule requires drilling, cleaning and painting. The update contract preserves its interval calibration obligation and whichever old requirements remain active; it begins each new duty at its start. This defines the specified obligation, without establishing that an operator would choose that scope.

## K Exploration Order and Work Bound

The RQ3 LTS configurations disable the preliminary guided search (both guided limits are zero). The complete algorithm queries every uncontrollable candidate first. If any such bucket is nonempty, it omits controllable queries by the UC-priority reduction. The underlying Post relation still contains any enabled ordinary controllable edges, which Direct-Full materializes. Otherwise lazy and update-first consume their sorted controllable cursor through the first nonempty bucket, deferring the rest; eager consumes every controllable candidate. A deferred candidate is never removed merely because it appears unpromising.

The default lazy/eager priority depends on the pending operations and endpoint projection, as follows. Entries in each row are tried from left to right; all listed types remain candidates if the game enables them.

| Condition                             | Priority classes (first to last)       |
|---------------------------------------|----------------------------------------|
| Some old stop pending                 | stop; transfer; ordinary; other update |
| No stop, some transfer pending        | transfer; ordinary; other update       |
| No stop or transfer, endpoint aligned | start; all other events                |
| No stop or transfer, not aligned      | events other than start; start         |
| Update-first ablation                 | every update; every ordinary event     |

Within a class, the implementation prefers updates if update-first is configured or the endpoint projection is aligned, and ordinary events otherwise; remaining ties use type order start, transfer, stop, ordinary, then Java string order on the event label. Endpoint mismatch is the minimum over load signatures of the number of disagreeing component/new-monitor coordinates. An old coordinate agrees when its transfer contains the target; an inactive new monitor agrees when its defined activation map produces the target state.

Open is a LIFO stack. Initial roots and each newly discovered batch are sorted by increasing endpoint mismatch, then pending-set size, then canonical state order, and inserted in reverse so the least element is popped first. Existing stack entries keep their positions; this is not a global priority queue. Canonical ties compare component count, then each component's version and state, sorted active-monitor identifiers and states, and sorted pending labels; the problem's comparators define local state order. Already queued/expanded states are not inserted again. Retry is FIFO: append a state when it first has deferred candidates, remove from the head, skip winners/exhausted cursors, consume through the next nonempty bucket, and append again if candidates remain. Drain propagation after each expansion/resumption, and always serve Open before Retry. These rules are implemented by `FineGrainedSuccessorOracle` and `OfDucSynthesizer`; the artifact retains their exact source. Eager changes bucket materialization and update-first changes candidate priority, providing the reported sensitivity checks without selecting an order after observing results.

**Scope of the work bound.** Let  $d$  count discovered states,  $c$  candidate queries (including empty results), and  $o$  returned outcome incidences over those queries. With constant-time indexed state predicates,  $O(1 + |\text{Post}(q, a)|)$  successor enumeration and constant-time bookkeeping, Algorithm 1 admits an implementation with  $O(d + c + o)$  work and storage. With preprocessing costs  $T_{\text{end}}, S_{\text{end}}$ , total bounds are  $O(T_{\text{end}} + d + c + o)$  time and  $O(S_{\text{end}} + d + c + o)$  storage. Entry sets can themselves be exponential in compact component descriptions. The search term counts the finite explored portion and excludes representation-dependent predicate costs. Java sorts and scans active transitions for quiescence and repeatedly checks roots; its timings do not measure constant-cost semantic queries.

**Counters and charges.** Intern each state once and queue its expansion once. A cursor advances monotonically through each state's finite candidates; thus each query and each cursor/Retry transition is charged to a candidate or state. For each queried nonempty bucket, retain an outcome counter and the maximum rank of already proved outcomes. An outcome pointing to a winner updates these immediately; otherwise store one reverse incidence at its target. When a state first enters the winning set, process each of its reverse incidences once. Hence all counter decrements and maximum updates total  $O(o)$ .

At a state with enabled uncontrollables, combine their returned outcomes into the required universal proving set; promotion is enabled only after every UC candidate has been queried. At a quiescent state, any nonempty control bucket whose counter reaches zero can certify promotion. Recording the first such bucket needs constant work per completed bucket. Nonempty flags prevent an empty bucket from certifying progress. A state is promoted at most once; its rank is one above its proving set's accumulated maximum, and the retained witness supports extraction.

Maintain Open, Retry and the promotion worklist by indexed membership and removable entries, so enqueue/dequeue/removal is constant time and exhaustion needs no global scan. A remaining-root counter avoids repeated all-root scans. Expected-amortized constant-time state dictionaries suffice. Goal initialization and unsafe-state exclusion cost  $O(d)$ . The final policy traversal or losing-region

extraction scans only stored states and incidences. Summing these charges proves the stated bound. The worklist computes the same monotone promotion fixed point as the abstract algorithm; this argument does not assert a linear running time for every implementation of its instruction to exhaust promotion.

Full-expansion containment and the complete independent-family argument are given in S2.7. Direct-Full constructs the reachable game with unsafe/goal terminals; Lazy may reach that whole graph in the worst case.

## L Complete Benchmark Input Counts

Inputs have 1–5 components, 4–131 entries, 1–12 old/new requirements each and 3–26 commands, using PC2/R2’s input log.

Table A5: The nine adapted inputs. Old/new requirements are shared across variants; interval counts are Base/R1/R2.

| Input        | Comp. | Entries | Old/New | Interval B/R1/R2 | $T$ | $U$ |
|--------------|-------|---------|---------|------------------|-----|-----|
| GSM          | 1     | 8       | 1/1     | 0/1/2            | 8   | 3   |
| Industry     | 3     | 131     | 3/3     | 0/3/6            | 16  | 9   |
| MetaSocket   | 2     | 4       | 1/1     | 0/1/2            | 3   | 4   |
| PowerPlant   | 2     | 10      | 4/3     | 0/3/7            | 4   | 9   |
| PC Arms=1    | 1     | 9       | 6/6     | 0/6/12           | 7   | 13  |
| PC Arms=2    | 2     | 126     | 12/12   | 0/12/24          | 14  | 26  |
| Railcab      | 4     | 22      | 5/7     | 0/7/11           | 11  | 16  |
| Surveillance | 2     | 112     | 7/7     | 0/7/14           | 16  | 16  |
| Workflow     | 5     | 28      | 5/6     | 0/6/11           | 12  | 16  |

$T$  counts declared transfer pairs before fluent observers;  $U$  counts transfer, stop and start commands. Entries agree across completed variants; PC Arms=2/R2 times out and its requirement counts come from its input log. Only Railcab has branching transfers. These nine inputs declare no explicit precedence; the separate cell witness requires start  $\prec$  stop; R2 uses interval monitors.



## M All Completed Direct-Full Exploration Pairs

The all-contract outcome summary is Table 6 in the main paper. The table below reports every completed pair.

Table A6: Absolute exploration costs for all 14 completed Direct-Full pairs. Figure A4 displays every fixed-budget contract for all four methods.

| Input / variant   | States (L/DF)    | Queries (L/DF)      | Solver s (L/DF) |
|-------------------|------------------|---------------------|-----------------|
| GSM / Base        | 31 / 75          | 27 / 658            | 0.0224 / 0.0416 |
| GSM / R1          | 34 / 115         | 44 / 706            | 0.0228 / 0.0450 |
| GSM / R2          | 34 / 68          | 78 / 352            | 0.0201 / 0.0327 |
| Industry / Base   | 272 / 282,709    | 3,336 / 5,058,452   | 0.190 / 154     |
| Industry / R2     | 7,717 / 51,096   | 52,919 / 725,391    | 1.15 / 28.0     |
| MetaSocket / Base | 8 / 82           | 12 / 289            | 0.0149 / 0.0408 |
| MetaSocket / R1   | 8 / 95           | 12 / 257            | 0.0136 / 0.0426 |
| MetaSocket / R2   | 16 / 34          | 24 / 79             | 0.0152 / 0.0240 |
| PowerPlant / Base | 73 / 8,168       | 165 / 47,618        | 0.0463 / 1.85   |
| PowerPlant / R1   | 759 / 23,686     | 1,927 / 66,394      | 0.125 / 5.51    |
| PowerPlant / R2   | 1,330 / 3,594    | 3,118 / 13,512      | 0.180 / 1.01    |
| PC Arms=1 / Base  | 83 / 407,019     | 840 / 3,842,690     | 0.0741 / 157    |
| PC Arms=1 / R2    | 56,443 / 158,218 | 354,259 / 1,363,292 | 6.16 / 77.8     |
| Railcab / R2      | 97,522 / 317,094 | 437,383 / 2,842,593 | 15.5 / 283      |

L: Lazy; DF: Direct-Full. All 14 WIN/WIN pairs are shown. States and successor-oracle queries are exact first-trial counts. Solver seconds are medians of five valid runs, including internal checking and excluding JVM/frontend/endpoint time, rounded to three significant digits.

## N Separate Extended-Budget Trials

**Extended-budget trials.** Table A7 separates single-trial populations on the same Xeon/JAR: 5 comparable Direct-Full decisions match fixed-budget Lazy; 3 Travel (4, 8) WIN decisions resolve prior timeout. Direct-Full timeouts, Surveillance/R1 heap exhaustion and Lazy PC2/R2 timeout persist at expanded heaps. The shared Travel (6, 1) timeout triggers the prespecified (6, 4) skip (records in S4).

Table A7: Separate enlarged-budget single trials (7,200 s cap). S4 retain every run, failure and skip.

| Population             | Heap (GiB) | Planned | WIN | LOSS | TO | OOM | Skip |
|------------------------|------------|---------|-----|------|----|-----|------|
| Earlier (four methods) | 64/200     | 27      | 7   | 1    | 16 | 1   | 2    |
| Eager                  | 200        | 10      | 3   | 1    | 5  | 1   | —    |

On the same Xeon/JAR, Eager’s 200 GiB/7,200 s single trials return 3 WIN where Direct-Full remained unresolved at that heap, discovering about  $(1.08\text{--}2.66) \times 10^7$  states with UC pruning (60–68,000 times Lazy); Industry/R1 returns LOSS with 94,046 states (2.4 times). All 4 decisions match fixed-budget Lazy; ratios concern decided pairs only. The other 6 trials yield 5 TO/1 OOM at about 209–212 GiB peak RSS, a process-memory measure (S4).

## O Separate Evaluation Populations and All-Contract Grid

The following descriptions preserve the separate secondary populations reported in the main paper. They are not pooled with the fixed-budget comparisons. Figure A4 displays all 108 fixed-budget outcomes. Figure 2 in the main paper gives absolute costs for every completed Lazy/Direct-Full pair.

|               | Lazy |    |    | Update-first |    |    | Eager |    |    | Direct-Full |    |    |
|---------------|------|----|----|--------------|----|----|-------|----|----|-------------|----|----|
| Adapted input | B    | R1 | R2 | B            | R1 | R2 | B     | R1 | R2 | B           | R1 | R2 |
| GSM           | W    | W  | W  | W            | W  | W  | W     | W  | W  | W           | W  | W  |
| Industry      | W    | L  | W  | W            | L  | W  | W     | T  | W  | W           | T  | W  |
| MetaSocket    | W    | W  | W  | W            | W  | W  | W     | W  | W  | W           | W  | W  |
| PowerPlant    | W    | W  | W  | W            | W  | W  | W     | W  | W  | W           | W  | W  |
| PC1           | W    | W  | W  | W            | W  | W  | W     | W  | W  | W           | T  | W  |
| PC2           | W    | W  | T  | T            | T  | T  | T     | T  | T  | T           | T  | T  |
| Railcab       | W    | W  | W  | W            | W  | W  | T     | T  | W  | T           | T  | W  |
| Surveillance  | W    | W  | W  | W            | W  | W  | T     | T  | T  | T           | T  | T  |
| Workflow      | W    | W  | W  | W            | W  | W  | W     | T  | W  | T           | T  | T  |

W: WIN    L: LOSS    T: timeout    B: Base; R1/R2: interval variants

Figure A4: Every fixed-budget contract–method outcome. Each tile is one of the same 9 inputs  $\times$  three requirement variants, under a 1,200 s cap and 64 GiB heap. WIN and LOSS are completed decisions; timeout leaves feasibility unresolved. The methods change exploration, keeping entries, successors and goals fixed. The comparison evaluates complete exploration procedures. Workflow/R2’s Direct-Full tile uses the recorded same-setting timeout after interruption; both attempts are retained.

Base/R1/R2’s 5/3/6 pairs give median state ratios 111.89/11.88/2.75 and solver ratios 40.02/3.12/9.14. The largest state savings are in Base, but different completed populations do not isolate interval requirements’ effect.

**Order sensitivity in R2.** Eight R2 contracts complete all five repetitions with both Lazy and Update-first; PC2/R2 times out with both. Update-first discovers fewer states and has a lower solver median on Industry, PowerPlant, PC1, Railcab, Surveillance and Workflow; Lazy is lower on both measures for GSM and MetaSocket. For example, PC1/R2 has 56,443 versus 9,304 discovered states and 6.155 versus 1.166 seconds for Lazy versus Update-first. These saved results expose a substantial order dependence, but do not isolate a causal interaction between the order and interval monitors. The performance contribution concerns the full lazy procedure relative to construction alternatives, not a uniformly best exploration order.

## O.1 Full+UC observations

Full+UC yields 16 WIN/1 LOSS/10 TO on Mac, 15 WIN/12 TO on Xeon; Industry/R1 times out on Xeon despite Mac’s LOSS. State counts equal Eager in 14/16 cells, differing on GSM/R1 and PowerPlant/R1. On PC1/Base, Direct-Full/Full+UC/Eager each discover 407019 states versus Lazy’s 83. Full+UC, Eager and Lazy all use UC pruning. Builds, hosts and repetitions differ, precluding single-factor timing claims (all rows/conditions in S4).

## O.2 Legacy reference

**Separate Legacy reference.** Legacy uses our MTSA fork’s traditional DUCS/GR backend with a subsequent-control objective, differing from FG’s fixed endpoints. It yields 9 matching WIN, 13 OOM and 5 unmeasured snapshot-cap cases. At 200 GiB/3600 s, 18 single trials give 11 OOM and 7 instrumentation stops, no decisions. S4 retains build identities, source/fork discrepancies and enlarged-heap records; these failures prove no impossibility.

## O.3 Controlled, Travel and extended-budget populations

Controlled families vary fanout, transfer outcomes, UC events/outcomes, requirements or precedence. All methods agree on 31 WIN/two LOSS across 33 inputs (132 single-run jobs); multiple graph measures may vary together (definitions/counts in S4).

Travel couples an agency with  $N$  amenities,  $K$  UC choices and  $0-K-1$  selection steps: size, branching and length covary. Adaptations change purchasing, admission, transfers and monitors without claimed trace equivalence (S4). Across 192 first trials on 48 settings, each OTF variant yields 20 WIN/26 TO/2 OOM, Direct-Full 18 WIN/28 TO/2 OOM. Only  $(N, K) = (4, 4), (4, 6)$  extend OTF's decided range. Both approaches still time out on  $(6, 1)$  at 200 GiB/7,200 s, so the original heap cap is not the sole limit. S4 retains every setting and the 18 completed-pair state/time comparisons.

**Extended-budget limits.** At 7,200 s and heaps up to 200 GiB, 27 earlier jobs include 16 TO/1 OOM/2 skips; 10 Eager jobs include 5 TO/1 OOM. Comparable decisions agree with fixed-budget Lazy. The separate extended-budget OTF decisions on Travel  $(4, 8)$  agree with one another, but fixed-budget Lazy timed out there, so they are not baseline decision matches. These single trials are excluded from its medians; all outcomes remain in the artifact appendix and S4.

## O.4 Additional granularity witnesses

The boundary fixture requires alignment after old-stop and before new-start: the requirement-block restriction forbids that intervening ordinary event and loses, whereas boundary merging merely renames the single stop and single start and remains WIN.  $\text{Cell}_n$  has one workpiece circulating through  $n$  stations with empty-only transfers; its single old stop and new start make boundary merging a renaming. It remains WIN with merged boundaries and LOSS with both merges. Restricting transfers to initial local states yields 0 fine WIN/merged LOSS, 6 both-LOSS and 2 both-WIN pairs, with 10/18 incomplete; enabled UC requests can obstruct transfer quiescence.

Eight original finite files produce ten WIN/four LOSS across fourteen jobs covering cell transfer, observed branches, requirement interleaving and neutral controls (parameters in S1). S4 retains excluded populations and earlier failed inputs. The alignment sequence states a required boundary order, not the complete saved Java execution trace.

## O.5 Application policy and losing-contract details

**Following an application policy.** Railcab Base retains both transfer outcomes and their distinct continuations, memories and targets. R1's braking ban causes earlier braking or completion without it; crossing requires approach without a last emergency brake. Both all-entry exports pass certificate/Link checks, with no timing/runtime validation (complete tuples/branches in S4).

**What a losing contract means.** Industry has three validation components: gate-form GF1, TOR and DSD1. Industry/R1 loses at 42 of 131 entries by independent exhaustive checking. A reachable safe quiescent state has no supplied target: validating GF1 or TOR violates the new requirements; the remaining safe alternative, DSD1 validation, admits a rejection cycle, while withholding that validation deadlocks. The full losing fixed point, not this one path, excludes every update order. R1 forbids validation between old stop and new start. DUCS instead synthesizes continued new-specification control [34, Defs. 4.1–4.3]; its original build and aligned fork win on the source model. FG LOSS depends on supplied initializers, requirements and targets. Exempting GF1 from R1 makes all 131 entries WIN via earlier paths before the DSD1/TOR alternatives are activated, not escape from the displayed trap; this separate Mac contract variant supplies no timing data.

## O.6 Input adaptation details

Nine inputs manually adapt eight DUCS sources; both ProductionCell sizes share a parent. Base/R1/R2 vary interval requirements, fixing components, transfers, endpoints and old/new goals (Technical Appendix L). Adaptations make PC tool results UC, revise Surveillance battery behavior/safety, order GSM/R2's decode boundaries before ENV transfer, shorten Railcab transfer paths while keeping both

outcomes, and omit PowerPlant’s initial ENV transfer pair. Recorded provenance does not establish whole-problem equivalence.

PC1/PC2 retain their DUCS parent’s repeatable arrival cycle: UC  $in_i$  enters an occupied arm; controllable  $out_i$  and UC  $reset_i$  permit its next arrival. With work and output withheld, pending tool results or reset/arrival chains finish in quiescent arm states. PC2-Rolling adds one UC calibration-completion step. This is finite drainage of pending physical events, not a total arrival budget or a safety proof. DUCS’s production-cell study likewise distinguishes an empty-line condition from delaying newly arrived raw work [34, Sec. 6.1]. FG commands require no enabled UC event; a temporal gap between arrivals alone does not ensure this.

## O.7 Independent-update construction

For  $n \geq 1$ , take  $n$  components, each with one old and one new state and a singleton transfer. Every version and its endpoint controller admit a controllable *tick* self-loop, making both endpoints deadlock-free. There are no monitors, dependencies or uncontrollables; the unique new endpoint is available for handover. These inputs are eligible. Each update state is the subset of transferred components, and all  $2^n$  subsets are reachable; Direct-Full discovers them all.

Under Lazy’s default order, pending transfers precede ordinary events in this family, so expansion follows one  $n$ -step chain. Because Open precedes Retry, it reaches the sole goal after discovering exactly  $n + 1$  states. Backward propagation then assigns ranks  $0, \dots, n$  and proves the entry state before any alternative query. The ordinary self-loops are losing choices under finite completion; both methods use the same terminal goal. Thus the separation concerns exploration of one supplied game.

## O.8 Memory and returned policies

Lazy’s five-run median peak working sets span 0.102–7.36 GiB (median 0.837), including JVM/native overhead; Eager/Update-first sometimes use less. Its 25 WIN certificates contain 8–1,071 update states (median 55) and maximum ranks 4–34 (median 19). Ranks bound events, not downtime or optimal length. Counts match all five repeats; LOSS/TO has no winning policy.

The rank domain is recorded as `certificate_states` in saved `rq3/stage1.csv`. S4’s “Output policy states” instead uses `output_controller_states`: the linked old–update–new controller, with 15–1,260 states for Lazy. Linking includes reachable old and new closed-loop states and non-goal update states, identifying each update goal with its selected new state. These columns describe the same successful runs at different construction stages; neither counts explored game states.

## O.9 Lifetime gaps in two saved policies

The separate qualitative Railcab Base/R1 exports in S4 retain their full policy graphs. Define a gap on a completed path as the labels strictly after its last old-stop edge and strictly before its first new-start edge, excluding both boundaries. A normal label is outside the complete command set pending at entry. All 22 entries of each export have every old stop before the first new start on every possible policy path. There are 34 Base and 41 R1 completed paths when paths from different old entries are counted separately.

| Saved policy | States / edges | Entries | Gap labels | Ordinary gap labels |
|--------------|----------------|---------|------------|---------------------|
| Railcab Base | 115 / 118      | 22      | 5–8        | 1–4                 |
| Railcab R1   | 44 / 46        | 22      | 3          | 0                   |

The Base policy exploits a period without old or new requirements; R1 keeps its seven interval monitors active even when those requirements are inactive. The observed zero ordinary events in R1 describes this selected policy, not every winning policy. These counts quantify events, not elapsed time or suppressed alternatives.

The revised source's `scripts/analyze_saved_railcab_gap.py` traverses every saved outcome from every entry. It verifies entry coverage, strict rank decrease, one-shot command consumption, active-monitor membership, full certificate reachability and all terminal paths. Its inputs are the two `handoff-bundle.json` files under `results/supplement-checks/railcab_policy_demo/` and `results/supplement-checks/railcab_r1_policy_demo/`; it returns per-entry ranges and extremal path segments. Counting internal edges of the region with no active old/new monitor independently gives the same ranges for all 44 entries. The input bundles and this analysis script are included in the matching public snapshot.

This is a static analysis of the already saved qualitative exports, with no new synthesis or timing trial. Lazy's 125 successful fixed-budget repetitions instead retain summary-only controller output, certificate sizes and maximum ranks. Their unrecorded policy edges cannot establish corresponding gap ranges for all 25 WIN contracts. Neither these two exports nor their analysis reconstructs enabled but unselected events or an optimal completion rank.

## O.10 Separate Performance Populations

The 12 capped Direct-Full attempts on Lazy-decided contracts give total-JVM lower bounds against Lazy medians, excluding PC2/R2 (both unresolved). These are single attempts, not solver ratios or medians; S4 retains all 27 absolute times/outcomes.

Full+UC matches Eager's state counts in 14/16 cells; state-set equality is unchecked. OTF without pruning is unimplemented. Different builds, hosts and repetitions prevent single-factor timing claims. Technical Appendix O also retains Legacy's subsequent-control reference and unresolved outcomes.

Controlled families agree on 31 WIN/two LOSS over 33 inputs, without full factor isolation. Travel covaries agency/amenity size, branching and length: OTF gives 20 WIN/26 TO/2 OOM per method; Direct-Full gives 18 WIN/28 TO/2 OOM. Both time out on (6, 1) at 200 GiB/7,200 s. Technical Appendix O retains extended TO/OOM/skips and the extended WIN versus fixed timeout at (4, 8); extended trials never enter medians.

## O.11 Finite contracts for the constructed families

These definitions specify the inputs behind Table 4 and the Rolling grid. They describe finite models, not implementations of deployment platforms. Component versions have distinct tags even when their transition graphs coincide. Unlisted plant and endpoint-controller edges are absent; unlisted monitor labels stutter, and errors are absorbing. Ordinary events are controllable unless explicitly called UC. There is no explicit command precedence. OLD/NEW monitors are empty except in Rolling+Audit; the stated interval monitors apply until handover. The fixed endpoint graphs determine every reachable old entry and the listed loadable new targets. The main paper defines quiescence, command execution and the compared merges.

**Rolling( $n, m$ ) and block size  $k$ .** For  $2 \leq n \leq 6$ ,  $1 \leq m < n$ , each old component has one ready state with a  $serve_i$  loop. Its new version has booting and ready states, with UC  $ready_i$  from booting to ready and  $serve_i$  only at ready. A local transfer maps old ready to new booting. The interval monitor starts at  $r = n$ , decrements  $r$  by the transferred block size, increments it on  $ready_i$  (capped at  $n$ ), and errors below  $m$ . Both fixed endpoints are all-ready tuples whose one-state controllers admit all  $serve_i$  loops; these supply one entry and one target. Consecutive blocks of at most  $k$ , including the final remainder, are transferred atomically. The 15  $(n, m)$  pairs compare  $k = 1$  with  $k = n$ ; the separate 70-cell grid varies  $1 \leq k \leq n$ . Proposition 5.2 proves the threshold for these inputs.

**Canary( $n, m$ ).** The same 15  $(n, m)$  values use one old healthy state  $H$  and new states  $H, H_P, B_P, B$ . Only  $H$  serves. A transfer has two adversarial outcomes,  $H_P$  and  $B_P$ ; UC reports take  $H_P$  to  $H$  and  $B_P$  to  $B$ . A controllable restart takes  $B$  to  $H$ . The interval monitor is a bit vector initially zero: a broken report sets bit  $i$ , a restart clears it, and more than  $n - m$  set bits cause error. Other labels stutter. This

is a *reported-broken* budget; pending physical states are not identified with already reported failures. The saved certificate also checks physical health separately. The all-healthy old/new tuples and one-state serve-only endpoint controllers give one entry and one target. Merging transfers synchronizes their commands and retains every Cartesian-product outcome. The healthy-outcomes-only and no-recovery variants restrict transfer images or remove restart, respectively; their unchanged both-WIN and both-LOSS results delimit the recovery mechanism.

**Canary certificate scope.** The existing separate check passed on every state of all 30 returned WIN certificates: the 15 parameter pairs under Lazy and Direct-Full. It counts replicas in  $H$  or  $H_P$  against the threshold  $m$ ;  $B_P$  and  $B$  do not count. Thus it checks model-state health, while  $H_P$  still cannot serve. It does not check continuous service, every explored state or runtime behavior. The other 30 saved checks concern LOSS mechanisms, not physical-health success. Fine and merged transfers share capacity reasoning with Rolling, but their reported-health contracts are different; no safety-preserving reduction to the Rolling game is claimed.

**DB-Rolling**(3,  $m$ ). Here  $m \in \{2, 3\}$ . Each old component has primary  $P$ , secondary  $S$ , and election-pending  $E$  states; the new version adds booting  $B$ . Serving loops exist at  $P, S, E$ . A controllable  $stepdown_{i,j}$  synchronously changes  $P_i$  to  $S_i$  and  $S_j$  to  $E_j$ ; UC  $elect_j$  changes  $E_j$  to  $P_j$ . A transfer maps only  $S$  to  $B$ , and UC  $ready_i$  changes  $B$  to  $S$ . The interval budget starts at three, decreases by the transferred block size, increases by one on readiness (capped at three), and errors below  $m$ ; election does not consume this nonbooting budget. The old endpoint is  $(P, S, S)$ , the new target  $(S, P, S)$ , both with serve-only controllers; unreachable padding retains the remaining ordinary alphabet. Thus the contract specifies a change of primary. With  $m = 2$ , a fine policy can change the primary and transfer secondary components successively. The merged command requires all-secondary states, which cannot be quiescent while an election remains pending. With  $m = 3$ , even the first fine transfer violates the budget. These models do not assert uninterrupted write service.

**Rolling+Audit.** This is Rolling(2, 1) with controllable  $log_{old}$  and  $log_{new}$  loops in every local plant state. Each version's audit monitor has unlogged, logged and error states. Its own log sets logged; either serve changes logged to unlogged but errors from unlogged. The other log and all remaining labels stutter. The new initializer selects unlogged on all nine tagged physical tuples. The interval state is  $(r, o, a)$ , initially  $(2, 1, 0)$ : transfers/readiness change  $r$  as above, old-stop clears  $o$ , and new-start sets  $a$ . Safety requires  $r \geq 1$  and  $o \vee a$ , enforcing overlap of the requirements. Each fixed all-ready endpoint controller alternates its own log with either serve; both its unlogged and logged states are entries/targets, and unreachable controller padding retains the remaining ordinary alphabet. These specify the memories, initializer and coverage obligation used in the comparison. Merging the two transfers loses; the single old-stop and new-start cannot be combined further, so merging requirement groups alone is only a renaming.

**Threads**( $n, B$ ). For  $2 \leq n \leq 6$  and queue capacity  $B \in \{1, 2\}$ , each worker is idle or busy; component 1 additionally stores queue length  $0 \leq q \leq B$ . UC arrival increments  $q$  below capacity. At capacity, backpressure disables arrival whereas saturated arrival is a self-loop. Controllable  $dispatch_i$  requires  $q > 0$  and worker  $i$  idle, decrements  $q$  and makes that worker busy; for  $i > 1$  it synchronizes with component 1. UC  $finish_i$  returns a busy worker to idle. Old/new plant graphs coincide. Each local transfer is identity on idle states and preserves component 1's queue. There are no monitors. One-state endpoint controllers admit all ordinary labels; their reachable products from the empty, all-idle initial tuple give all old entries and all new targets. These two arrival modes yield 20 pairs. Under backpressure, disabling dispatch lets the finite queue fill and busy workers finish, permitting globally quiescent transfer. Saturated arrival can persist forever and prevents even fine transfer under the contract's quiescence rule. The negative control therefore distinguishes a genuine transfer opportunity from an implicit fairness assumption; it is not a reproduction of a multithreaded runtime.

## O.12 Granularity need not preserve feasibility

The comparisons above do not assert that finer commands always retain every merged solution. The following analytical contract gives the reverse separation; it is not an additional measured instance or part of the 55-pair population.

Both components have one state per version and the shared ordinary alphabet  $\{x, y, tick\}$ , with  $x, y$  UC and  $tick$  controllable. Every version has a  $tick$  self-loop. The only other loops are:  $y$  in old component 1,  $x$  in new component 1,  $x$  in old component 2, and  $y$  in new component 2. All other transitions are absent. Shared labels synchronize, so neither  $x$  nor  $y$  is enabled in the all-old or all-new product;  $x$  loops in new/old and  $y$  loops in old/new. Each local transfer maps its unique old state to its unique new state. There are no requirements, initializers or precedence edges. The one-state old and new controllers loop on all ordinary labels. Their closed loops each have one reachable state and a  $tick$  loop, hence are safe and deadlock-free; these states supply the entry and target.

With separate transfers, either first transfer reaches a mixed state with an enabled UC self-loop. The remaining transfer is then permanently blocked by the quiescence rule; the environment can repeat the loop forever, so every fine policy loses. Merging both transfers into their single Cartesian-product command instead moves directly from the quiescent entry to the quiescent all-new target, with rank one. Thus fine LOSS/merged WIN is possible. Atomic merging can remove a harmful mixed state, whereas Cell and Rolling show that it can remove a necessary intermediate state. The reported obstructions establish when a specified merge loses, not a universal refinement order between granularities.

## P Implementation and AI-Assisted Research Details

Algorithm 1 returns and internally checks the losing region  $L$ . The finite-JSON runner serializes its members in `certificate.json`; an independent checker reconstructs successors from the input and checks closure. The FSP/application benchmark driver instead records LOSS, region size and phase-mask statistics, without serializing solver membership. Its saved full-game exports support independent region reconstruction, but do not compare the solver’s member list. These output interfaces therefore provide different saved certificate evidence. The fixed RQ3 runs use Windows 10, an Intel Xeon W-2265, 256 GB RAM and Temurin 17.0.20.1. The main paper states the heap, time cap and repetition policy; separate Mac structural checks and extended trials retain their own provenance.

Codex (`gpt-5.5`, `gpt-5.6-sol`, `gpt-6-astra`; September 30–October 3: GPT-6, variant unavailable) assisted the research in the following roles. For design and theory, it proposed contract definitions, examples, counterexamples and proof drafts, which were checked by explicit derivations and finite-model checks where provided. For implementation, it edited the Java frontend, successor generation, solver, certificate checks and Python reconstruction tools; regression fixtures, exhaustive comparisons and certificate checks supplied validation. For experiments, it helped build and run drivers and aggregate saved outputs; the tables retain capped, failed and adverse outcomes, and reproduction scripts check their correspondence to saved results. For analysis, it compared source formulas, monitor automata, initializer tables and saved policies within each stated check’s scope. For writing, it drafted and edited text, diagrams and references; cited claims were checked against primary sources. These checks are evidence about their specified objects, not independent proof of all AI-produced code or reasoning.

Claude Fable 5.1 assisted planning, drivers, source checks and drafts. September 30–October 3 editing used Opus 5.5 and ChatGPT Pro/Latest reviews (response model unavailable); Claude Code inherited project context. These reviews were editorial advice and were not counted as experimental evidence or as human evaluation. Authors retain responsibility for the manuscript. Shared authorship, generated code and AI assistance across tools can produce correlated errors even when implementations or checks are separate.



# Synthesizing Staged Updates between Verified Controllers Supplementary Proofs and Tables

Anonymous Authors

Read S1 for additional granularity and observation witnesses, S2 for search invariants and certificates, S3 for requirement interpretation, S4 for the current evaluation and separately identified auxiliary records, and S5 for the expanded comparison. The Technical Appendix supplies the primary formal definitions and case constructions; this part supplies their detailed validation and experimental evidence. Complete Cell definitions and the full synthesis proof are in Technical Appendix B and A, respectively. Its proofs use the confirmed non-preemptive rule: update events have no successor whenever a modeled uncontrollable event is enabled. Goals are safe, uncontrollable-quiescent states with all update events completed and a compatible supplied endpoint. No fairness assumption is added. Section S4 follows RQ1–RQ3: constructed mechanisms, independent validation, then the fixed 27-contract comparison. Diagnostic, historical and extended-budget records follow in separate subsections; unresolved outcomes and trial provenance are retained.

## S1 Granularity and observation witnesses

The main paper gives the two completing paths and the local-versus-bulk argument. Here we tabulate its full closure, define the additional observation and requirement-boundary witnesses, and give their complete proofs. We also explain how the smaller executable fixtures relate to these contracts. In the two additional witnesses  $\mathcal{W}_b$  and  $\mathcal{W}_r$ , unlisted controller transitions are empty, the four successor rules supply transfer and requirement operations, and every endpoint admits *idle* and is deadlock-free.

### S1.1 Cell closure and the global-transfer embedding

The complete Cell input is in main-paper Table 1 and Technical Appendix B; the latter also supplies its residual interpretation. Main-paper Figure 1 gives the entire policy, and Technical Appendix B lists both paths. From the two old entries, the paths have lengths six and five, respectively, with 13 policy states and 11 edges in total. Their final states match the two loadable targets. The policy is safe and rank-decreasing. Joint transfer requires both stations empty, whereas every reachable physical tuple contains exactly one workpiece, so the merged contract has no goal. These arguments use the same endpoints and requirements.

For a global-transfer input with a single component  $\star$ , identify its old/new component states with the corresponding old/new product states. Carry monitor states and pending requirement operations unchanged. The map on update states is a bijection. The ordinary rule synchronizes the same transitions, the transfer rule applies the same global relation, and requirement stop/start applies the same monitor update and initialization. The sets of modeled uncontrollables coincide, so the non-preemption guard also coincides. Safety, roots, target matching, and quiescence are preserved. This establishes the game isomorphism for the global-transfer fragment.

The following exhaustive census retains unsafe successors to expose the full contract relation. It is distinct from the terminalized solver work and from the policy’s retained states.

Local counts are 30 for  $m_A$ , 26 for  $m_B$ , 20 for  $j$ , 15 for  $\rho_A$ , 10 for  $\rho_B$ , 12 for  $s$ , and 26 for  $t$ . Twelve safe-to-unsafe buckets remain visible: eight  $m_A$  edges violate inspection and four  $j$  edges violate the

Table S1: Full reachable closure of the paper’s illustrative two-station fixture, including successors beyond unsafe states. This census is distinct from terminalized solver work.

| Quantity                       | Local transfer | Bulk transfer |
|--------------------------------|----------------|---------------|
| Reachable states               | 56             | 10            |
| Safe / unsafe states           | 26 / 30        | 6 / 4         |
| Goals                          | 2              | 0             |
| Candidate buckets (all labels) | 392            | 60            |
| Singleton / empty buckets      | 139 / 253      | 16 / 44       |
| Retained policy states / edges | 13 / 11        | –             |

still-active old requirement. The holder monitor never reaches error. Bulk counts are 5 for each move, 2 for  $s$ , 4 for  $t$ , and zero for inspection and transfer. The primitives and omitted-transition conventions determine this census; these counts are not additional rules.

## S1.2 Observation-dependent choice

**Complete input.**  $\Sigma_N = \{idle, u, a, b, d_A, d_B\}$ ; only  $u$  is uncontrollable. The old plant has states  $\{w, a_0, b_0, a_1, b_1\}$ , initially  $w$ , and exactly the non-idle transitions  $w \xrightarrow{u} a_0$ ,  $w \xrightarrow{u} b_0$ ,  $a_0 \xrightarrow{a} a_1$ ,  $b_0 \xrightarrow{b} b_1$ ,  $a_1 \xrightarrow{d_A} w$ , and  $b_1 \xrightarrow{d_B} w$ . The new plant has states  $\{x, y\}$ , initially  $x$ , with  $x \xrightarrow{d_A} y$  and  $y \xrightarrow{d_B} x$ . Every old and new plant state has an *idle* self-loop; there are no other plant transitions.  $g = \{(a_1, x), (b_1, y)\}$ ,  $\Sigma_{\text{upd}} = \{\rho\}$ ,  $\prec = \emptyset$ . One-state  $C^o/C^n$  loop on all six ordinary labels/ $\{idle, d_A, d_B\}$ , respectively. All requirement, monitor and  $\iota$  families are empty.  $Q_0 = \{((o, s), (), \{\rho\}) : s \in \{w, a_0, b_0, a_1, b_1\}\}$ ;  $Z_{\text{load}} = \{(x, c_n), (y, c_n)\}$ . Goals are the two new states with empty  $P$ ;  $\kappa$  appends  $c_n$ . In  $\mathcal{W}_b$ , both branch paths are admitted by  $C^o$ , and  $d_A$  reaches  $y$  from the new initial  $x$ , establishing the stated entry and load sets.

**Policy and fixed-list obstruction.** In  $\mathcal{W}_b$ , all five old component states are reachable under the permissive endpoint controller. The policy waits at  $w$ , permits  $a$  at  $a_0$  and  $b$  at  $b_0$ , and transfers at  $a_1, b_1$ . Both outcomes of  $u$  are retained. Ranks 3, 2, 1, 0 by distance to the compatible new endpoint certify all five roots.

Consider any fixed finite list of controllable events, with uncontrollables accepted without advancing the list position. The environment can take  $u$  at  $w$  before the next listed controllable. Its two outcomes leave the same list position at  $a_0$  and  $b_0$ . Any finite sequence of *idle* heads changes neither component state; an infinite sequence would itself fail completion. The next non-idle head is one of  $a, b, d_A, d_B, \rho$ . At  $a_0$ , only  $a$  among these is enabled; at  $b_0$ , only  $b$  is enabled. Thus no common head can complete both outcomes. An exhausted list also fails at these non-goals. This proves failure for every such fixed list from the root  $w$ , and therefore for all-root admission.

The executable branching fixture tests the unrestricted policy and two all-time safety restrictions,  $\Box \neg b$  and  $\Box \neg a$ . Each restriction has a losing branch and hence gives one of the two observed LOSS jobs. These restrictions are concrete obstruction checks; the universal statement about fixed lists follows from the head argument above. In the neutral fixture, both resolving events are available after each branch, so the unrestricted and both restricted tests win.

## S1.3 Requirement interleaving

**Complete input.** All events controllable;  $\Sigma_N = \{idle, align\}$ ,  $g = \{(r, n)\}$ ,  $\Sigma_{\text{upd}} = \{\rho, t, s\}$ ,  $t = \text{stop}(r_o)$ ,  $s = \text{start}(r_n)$ ,  $\prec = \emptyset$ . The old plant has states  $\{w, r\}$ , initially  $w$ , with  $w \xrightarrow{align} r$  and an *idle* self-loop at each state. The new plant has the single initial state  $n$  and an *idle* self-loop. These are all plant transitions. Each monitor has initial safe state  $k$  and absorbing error state  $E$ ; *align* takes  $k$  to  $E$ , and every other label self-loops at  $k$ . One-state  $C^o$  has alphabet  $\{idle, align\}$  but only an *idle*

loop;  $C^n$  has only an *idle* loop.  $R^{old/new/upd} = \{r_o\}/\{r_n\}/\emptyset$ ;  $\text{dom } \iota_{r_n} = \{(o, r), (n, n)\}$ ,  $\iota_{r_n} = k$ . Its supplied residual forbids *align* after activation and erases ordinary events before activation (exact RS).  $\text{Reach}(\text{CL}^{\text{old}}) = \{(w, c_o, k)\}$ ,  $Q_0 = \{((o, w), (k, \perp), \{\rho, t, s\})\}$ ;  $Z_{load} = \text{Reach}(\text{CL}^{\text{new}}) = \{(n, c_n, k)\}$ . The unique goal is  $((n, n), (\perp, k), \emptyset)$ ;  $\kappa$  appends  $c_n$ . In  $\mathcal{W}_r$ , the old controller disables *align*, establishing its singleton entry set.

**Policy and interleaving obstruction.** The old endpoint of  $\mathcal{W}_r$  stays at  $w$  using *idle*; its controller disables the controllable *align* to satisfy the old monitor. The new endpoint at  $n$  is likewise safe and deadlock-free. Update admission drops the old controller while retaining the old monitor. The sequence

$$\text{stop}(r_o); \text{align}; \text{start}(r_n); \rho$$

first deactivates the old prohibition, reaches the transfer-ready state, starts the new prohibition from its supplied safe residual, and transfers. Every retained state is safe, and the final state matches the only load target.

Every completion must execute  $\rho$ , which requires the old ready state. Only *align* reaches that state. While  $r_o$  is active, *align* reaches its error; once  $r_n$  is active, *align* reaches its error. Therefore a safe completion must place *align* strictly between stop and start. A policy requiring the requirement operations to form a contiguous block excludes this necessary event. Simultaneous stop/start excludes the same gap. Starting first cannot help: it is disallowed at  $w$  by  $\text{dom } \iota_{r_n}$ , and overlap of the prohibitions would still forbid *align*.

The executable separation fixture is a related test input, not the same transition system as  $\mathcal{W}_r$ . It requires  $\text{startNewSpec} \Rightarrow \text{Aligned}$  with persistent *Aligned* set by *align* and reset at entry. Its NEW monitor instead uses the frontend's current-fluent lookup: the forbidden-event observer  $\text{align}_a$  must be false at activation. An ordinary *align* sets both *Aligned* and  $\text{align}_a$ ; *idle* clears  $\text{align}_a$  while preserving *Aligned*, and transfer/lifecycle events preserve these observer values. Consequently  $\text{stop}; \text{align}; \text{idle}; \rho; \text{start}$  is a source-derived five-step winning witness. The saved separation WIN has six certificate states and maximum rank five; the logs contain only transition summaries, so they do not establish the returned policy's exact placement of transfer. The four-step witness above belongs only to the abstract  $\mathcal{W}_r$ .

Both executable inputs retain the interval requirement that persistent alignment hold when new requirement starts. Neutral is not removal of this requirement. Both actual separation endpoints have one state and only *idle* loops. In the actual neutral model, the old endpoint has one *idle* state; the new endpoint has two states,  $\text{NEW\_WAIT} \xrightarrow{\text{align}} \text{NEW\_READY}$ , with *idle* at both. Its OLD/NEW monitors forbid *bad*, and NEW activation requires  $\text{bad}_a = \text{false}$ . It therefore permits transfer and alignment before the requirement block. The neutral full/block certificates have maximum ranks five/four; the full policy's extra step is not identified from the summary. The executable block forbids transfer as well as ordinary events between requirement operations. Its separation obstruction still follows because old requirement prevents alignment before stop and the interval requirement prevents starting before alignment. No trace equivalence or monitor-state isomorphism between the abstract and executable encodings is claimed.

## S2 Finite games, search invariants, and certificates

Technical Appendix A gives the complete synthesis proof and its UC-priority, memorylessness and search-coverage lemmas. Here we expand eligibility, invariants, checking, costs and implementation order. Let  $R = \text{Reach}_\sigma(Q_0)$ , computed without expanding goals. A maximal run is either infinite, or finite and ending at a goal or at a state with no policy successor. All statements below concern reachable states; a cycle outside  $R$  is irrelevant to a particular policy.

## S2.1 Complete eligibility checks

Before synthesis, we check consistency of event ownership and controllability; monitor totality, determinism, and absorbing errors; the types of  $g_i$  and  $\iota_r$ ; for each  $r \in R^{\text{upd}}$ , membership in  $\text{dom } \iota_r$  of the pre-update-tagged component state of every  $z \in \text{Reach}(\text{CL}^{\text{old}})$ ; that  $\prec$  is a strict partial order; requirement satisfaction and deadlock freedom of the pre-update and post-update closed loops; inclusion of every environment event in the event set of the corresponding controller; non-suppression of uncontrollable events by the pre-update and post-update controllers; and reachability of  $Z_{\text{load}}$ . RS remains external evidence. Synthesis checks  $\iota_r$ 's type, domain and non-error property; completeness of its activation histories is supplied, rather than inferred.

## S2.2 Maximal runs and ranks

Assume  $R$  is finite, all its states are safe, goals are terminal, and every non-goal has a successor. If the reachable non-goal subgraph has a cycle, first follow a path from a root to that cycle and then repeat it. Every edge is policy-permitted, including its selected nondeterministic outcome, so this is an infinite maximal run avoiding goals. Universal finite completion excludes it without invoking fairness.

Conversely, if that subgraph is acyclic, no path can visit a non-goal twice. Every path therefore has at most  $|R \setminus Q_{\text{goal}}|$  non-goal edges before ending. A non-goal endpoint is impossible by the successor premise. Thus every maximal run ends at a goal. Define  $\text{rk}(q)$  as the maximum length of a goal-ending path from  $q$ , and set goal ranks to zero. Such a path exists because the graph is finite, acyclic off goals, and has no non-goal sinks. For each edge  $q \rightarrow q'$  from a non-goal, prefixing a longest path from  $q'$  proves  $\text{rk}(q) \geq 1 + \text{rk}(q')$ . This gives a positive natural-number rank at non-goals. Finally, a strict decrease around a cycle would give  $\text{rk}(q) > \text{rk}(q)$ , a contradiction. These implications establish all three equivalences. Without the successor premise, a non-goal sink is an acyclic counterexample.

## S2.3 The four exploration invariants

For the partial table  $\mathcal{D}$ , queried-empty buckets are recorded by candidate consumption even though they are absent from  $\text{dom } \mathcal{D}$ . Each stored nonempty bucket is exact and contains all modeled outcomes. We state invariants at loop boundaries, after discovery queues and the finite reverse-edge worklist have been updated.

1. **Uncontrollable completeness.** On first expansion of  $q$ , all uncontrollable candidates are consumed before promotion can use the state. No candidate is added later: the finite event set is determined by  $q$ . Therefore every  $q \in X$  has its complete uncontrollable successor set known. The transient insertion into  $X$  inside the expansion block is not a return point.
2. **Controllable exhaustion at failure.** A safe non-goal state with no uncontrollable has a cursor over all controllable candidates. Expansion consumes it to the next nonempty bucket or exhaustion. If a nonwinner has candidates remaining, it is in Retry. Each retry consumes further candidates. The failure return is reached only after both queues and the propagation worklist are empty. Hence every such state in  $V \setminus W$  has exhausted its cursor. Removing an entry when its state becomes winning cannot remove a losing candidate from this argument.
3. **Monotone winning evidence.** A promoted state either has all uncontrollable outcomes at smaller retained ranks, or has a complete selected controllable bucket at smaller ranks and no uncontrollable. All those buckets are exact; growing  $\mathcal{D}$  adds other controllable choices rather than new outcomes to an already stored choice. The policy disables those additional controllables. Existing rank-decreasing witness edges consequently remain a valid certificate. Goals are recognized using modeled uncontrollable enabledness before terminalization, so an unqueried uncontrollable cannot invalidate rank zero.

4. **Open-state coverage.** Each newly discovered safe non-goal is inserted once in *Open*. It is removed only for first expansion, when it enters  $X$ . Thus  $V \cap \text{Safe} \setminus (X \cup Q_{\text{goal}}) \subseteq \text{Open}$ . Unsafe states and goals are discovered terminals and are handled separately.

The initial root insertion establishes these properties. Expansion, successor discovery, candidate consumption, and promotion preserve them as shown above; no other operation changes their relevant sets.

## S2.4 Attractor, termination, soundness, and completeness

Let  $W_0 = V \cap Q_{\text{goal}}$ . At the next layer a safe expanded state joins if its nonempty uncontrollable union is contained in the preceding layers, or if it has no uncontrollable and one complete nonempty controllable bucket contained there. Induction on these layers proves the existence of a winning policy for the current table. These minimum layer indices are analytical, not the implementation's retained certificate ranks. On first promotion the implementation assigns one plus the largest rank of the complete chosen successor set (all uncontrollable outcomes, or one winning controllable choice); goals have rank zero. It never lowers that assigned rank. Every stored choice used for promotion remains complete and rank-decreasing. Extraction may choose another stored controllable set only if all its successors also have smaller retained ranks. Goals satisfy the target and quiescence checks. Invariant I3 therefore preserves a valid certificate as exploration continues without claiming rank minimality.

The complete termination, soundness and completeness proof is in Technical Appendix A. The invariants above give its search-coverage and rank premises for the supplied successor and goal rules. A winning policy can be normalized by removing controls at UC-enabled states and retaining one winning control elsewhere; all UC outcomes and non-goal successors remain.

## S2.5 Checking the returned evidence

A WIN checker reconstructs the reachable policy domain from all roots and complete Post buckets. It verifies safety, enabledness of every retained action, nonempty successors outside goals, closure in the rank domain, and strict rank decrease for every outcome. At each goal it checks rank zero, physical uncontrollable quiescence, membership of  $\kappa(q)$  in  $Z_{\text{load}}$ , and component/new-monitor equality. Omitted policy entries mean the empty set. The maximal-run lemma then proves the model-level claim without reproducing the synthesizer's exploration order.

A LOSS checker verifies a nonempty intersection with the roots, no goal in  $L$ , and the obstruction conditions of that proof. Unsafe states are terminal losses. At a safe uncontrollable-free deadlock the universal condition on enabled controllables is vacuous, correctly certifying a loss. A safe uncontrollable-free state with choices requires inspecting every enabled candidate, so checking LOSS may cost as much as the corresponding exhaustive search. These checks trust the shared parser, successor, monitor, transfer, and endpoint/goal semantics; the separately implemented explicit and raw-LTS checks in the fixed correctness campaign (main-paper RQ2) provide additional, scoped cross-checks.

## S2.6 Work bound and static encoding

Technical Appendix K gives the complete counter-based  $O(d + c + o)$  work and storage argument, where  $d$  counts discovered states,  $c$  candidate queries and  $o$  returned outcome incidences. The bound assumes indexed predicates and constant-time bookkeeping; preprocessing and Java sorting/scans are additional costs.

For the following full-game encoding, let  $N$  count states,  $A$  candidate labels summed over states and  $B$  outcome incidences. Use a fresh environment root with edges to every  $Q_0$  state. At a safe non-goal with uncontrollables, an environment vertex chooses any uncontrollable event and then any outcome. At an uncontrollable-free state, a controller vertex selects a nonempty bucket and an environment vertex chooses its outcome. Unsafe states and deadlocks lose, and goals win. At most one intermediate vertex per nonempty bucket and one edge per outcome are required, giving  $O(N + A + B)$  construction.

Normalization above proves equivalence to the supplied game; a compatible  $\kappa$  remains additional typed evidence.

For any root/safety/goal/controllability-preserving isomorphism preserving all buckets, including empty ones, the strong-predecessor operation commutes with the state bijection. Induction preserves every attractor layer and its rank; complementing gives corresponding losing sets. Selected actions transport through the event bijection. Transport of load targets and  $\kappa$  additionally requires matching typed endpoint projections; an untyped graph isomorphism alone supplies no such map.

## S2.7 Full-expansion containment and the independent family

Direct-Full and OTF start from identical roots and use identical terminal tests, candidates, and successor rules. Induct on discovery by OTF. A non-root discovered state is an outcome queried at a previously discovered safe non-goal predecessor. Full expansion also discovers that predecessor by induction; since its terminal test is identical it expands the predecessor and queries the candidate, discovering every outcome. An unsafe or goal successor is recorded by both but expanded by neither. Applying this to the source of each queried bucket proves query containment as well as state containment. This reasoning covers arbitrary order and nondeterminism.

For the exact  $n + 1$  versus  $2^n$  family, component  $i$  has one old and one new state, each with a controllable *tick* loop, and its singleton transfer maps old to new. Both endpoint controllers have one state admitting *tick*; there are no requirements, uncontrollables, or dependencies. The new endpoint is the unique load target. The sole root has every transfer pending. Each update state is determined by a transferred subset  $S \subseteq \{1, \dots, n\}$ . It has its *tick* self-loop and, for each  $i \notin S$ , the transition  $S \xrightarrow{\rho_i} S \cup \{i\}$ . The full subset is the unique goal.

Every subset is reached by performing exactly its transfers in some order. Full expansion discovers all  $2^n$  subsets, even though it does not expand the final goal. Under pending-transfer-first order, OTF's first nonempty bucket at each non-goal discovers one fresh subset on a chain. The open queue is served before retries, so the chain reaches the goal after  $n$  transfers. Propagation then promotes every predecessor on the chain and covers the root before any other candidate is retried. Exactly  $n + 1$  states have been discovered. The *tick* self-loop at a non-goal cannot be a winning choice under universal finite completion. This is an existence construction for exploration cost, with no claim that every family or order has the same saving.

## S2.8 Fixed exploration order in the measured implementation

Technical Appendix K gives the complete measured order: disabled preliminary guided search, conditional event classes, endpoint mismatch and state ties, LIFO discovery, FIFO retries and promotion timing. It also derives the indexed work bound. The rules order queries, not policy execution; Eager and Update-first provide the reported materialization and order checks.

# S3 Monitor intervals and conditional trace lifting

The main paper proves the activation-invariant criterion, supported by the source checks in Technical Appendix E.1. Here we develop residual-history meanings and frontend premises. Technical Appendix F proves conditional trace lifting under A1–A4 and history completeness: runtime conformance, eventual observation, atomic boundaries and initializer meaning remain premises, while the linked-LTS checker verifies abstract composition.

## S3.1 Activation histories and residual meanings

For prefix-closed requirement language  $\text{SafePref}_r$ , let  $H_r(\xi)$  be the declared activation histories projected by  $\text{obs}_r$ . NEW checks use safe prior histories (A); UPD checks begin their reference at entry (E). Whole-history meaning  $H$  is a separate diagnostic.

NEW activation histories include start once: initialization supplies the post-start residual and online monitoring consumes later events. Game admission initializes monitors without an update-game event; the FSP boundary interpretation below accounts for `hotSwapIn` once in the supplied UPD boundary state. Define  $\mathcal{C}_r(\xi) = \bigcap_{h \in H_r(\xi)} \{w : hw \in \text{SafePref}_r\}$  and  $\mathcal{L}_r(m) = \{w : \eta_r^*(m, w) \notin F_r\}$ . Residual-sound initialization requires

$$\mathcal{L}_r(\iota_r(\xi)) \subseteq \mathcal{C}_r(\xi) \quad (\xi \in \text{dom } \iota_r) \quad (\text{RS})$$

with equality for an exact residual and strict inclusion for a conservative one. Distinct activation histories may require finite history bits in components and transfers. RS is supplied external interpretation evidence relative to the given requirement and observation meanings. For  $r \in R^{\text{upd}}$ ,  $\text{dom } \iota_r$  includes every reachable old component state.

### S3.2 What the frontend initializes

Cell is an explicit finite contract. The  $\text{Cell}_n$  measurements load its parameterized JSON representation through `CellRingDriver`, supplying the holder-dependent UPD initializer directly. The inherited FSP benchmarks use the distinct formula-compilation path below; its constant UPD initializer is not a restriction of the abstract contract. Java names entry `hotSwapIn` and the completion-side atomic quotient `hotSwapOut`; the paper calls the latter load.

The measured frontend has two distinct paths. For each NEW safety formula, it constructs the referenced fluent observers, including implicit action fluents. Observers track their initiating/terminating events, and unrelated events stutter. A product traversal associates each reachable observer vector with a compiled safety-monitor state. Multiple non-error monitor states for one vector are accepted only if their safe-continuation languages agree; the least state identifier is retained. An error association takes precedence. Crucially, the table-building traversal does not expand error monitor states, so it does not check how observer values evolve after a violation.

`MtsaRevisedOfDucsAdapter.newActivation` selects the observer indices belonging to that NEW formula (identity first, then a unique-name match), extracts their states from the augmented component tuple, and looks up the signature. A missing signature, an unknown monitor state or an error excludes the tuple. If observer indices are empty or the mapping is absent, it falls back to the monitor's initial state. Candidate tuples are the physical closure of all old endpoint roots under ordinary events and component transfers, allowing mixed versions.  $\text{dom } \iota_r$  retains exactly the non-error lookup entries; this does not prove past-history safety. The observer vector is stored with component zero, follows ordinary events in either version even while the NEW monitor is inactive, and is preserved by component transfers. At start, existing active monitors advance and the NEW monitor is installed from the lookup; it does not consume that start again.

For UPD, `compileTesters` instead computes a constant boundary state by taking the raw monitor from state zero through `hotSwapIn`, or stuttering if that label is absent. `updateActivation` assigns this same non-error state to every reachable old physical tuple; an error boundary is rejected. UPD has no fluent-based activation lookup. The symbolic contract allows both forms as supplied  $\iota_r$  maps, but the fluent-determination lemma below applies to an actual initializer only after its state agrees with the history-derived state. Our coverage CSV records the initializer mode separately. Source locations (including `UpdatingControllersDefinition`, which builds the table) and extraction commands are in the artifact's `experiments/rs_coverage/README.md`.

A lookup initializer uses only the current referenced fluent-observer vector  $f(\xi)$ , writing  $\iota_r(\xi) = \sigma_r(f(\xi))$ . It can forget prior violations or pending obligations. The actual new- and update-requirement frontend paths are distinguished above.

### S3.3 Safe prior histories for NEW requirements (A)

A new requirement becomes effective at its own activation. Its supplied meaning retains the progress or obligations represented by the current fluent vector, while excluding prior violating histories from

the reference intersection. With at most one occurrence of each update event, define

$$H_r^{\text{safe}}(\xi) = \{h \in H_r(\xi) : \eta_r^*(m_r^0, h) \notin F_r\}, \quad C_r^{\text{safe}}(\xi) = \bigcap_{h \in H_r^{\text{safe}}(\xi)} \mathcal{L}_r(\eta_r^*(m_r^0, h)).$$

Then  $RS_A$  is  $\mathcal{L}_r(\iota_r(\xi)) \subseteq C_r^{\text{safe}}(\xi)$ . This is a choice of requirement meaning, not a proof that a past violating trace satisfied the original invariant. In particular, conditioning on safe histories does not validate application intent. If  $H_r^{\text{safe}}(\xi)$  is empty, the intersection is the universal continuation language: inclusion is immediate, but equality generally fails. Exactness below is therefore explicitly restricted to nonempty safe-history domains.

**Lemma S3.1** (A: language-determined safe-prefix initialization). *Explore the freely reachable monitor/observer product over one-shot update histories, recording but not expanding ERROR pairs. Suppose all non-error monitor states reached at the same fluent vector have equal continuation languages. On the activation domain, let the actual non-error lookup  $\sigma_r(v)$  have that common language. Then  $\iota_r(\xi) = \sigma_r(f(\xi))$  satisfies  $RS_A$ , with equality whenever  $H_r^{\text{safe}}(\xi) \neq \emptyset$ .*

*Proof.* Every admissible safe history reaches a non-error product pair with observer vector  $f(\xi)$ . By language equivalence and lookup compatibility, its residual language equals  $\mathcal{L}_r(\iota_r(\xi))$ . A nonempty intersection of these equal languages is that language. For an empty history set, the universal intersection contains the initialized language, proving the stated inclusion.  $\square$

Language equivalence, rather than equality of state identifiers, is also the frontend's acceptance condition for multiple non-error states at a lookup vector. Our independent checker compares deterministic state pairs until an ERROR/non-error distinction or pair closure, and separately reconstructs the frontend's error-priority, equivalent-state/minimum-ID rule. This checks the earlier saved monitor/observer exports against the source rule; those exports do not contain a materialized lookup table. A separate fresh frontend export directly checks all 138 tables (S4), while leaving physical-state decoding and actual activation histories unverified. The check establishes  $RS_A$  inclusion for all 138/138 NEW initializers without a nonemptiness premise. Equality additionally requires nonempty safe histories at the actual activation; their plant-level nonemptiness remains unverified.

### S3.4 From update entry: UPD requirements (E)

An update-interval requirement becomes effective at entry. Its reference monitor starts afresh with the actual entry fluent vector  $v$ , consumes `hotSwapIn` once, and observes the subsequent update interval. Let  $\mathcal{L}_r^E(v)$  denote the resulting continuation language and  $b_r$  the frontend's constant post-entry monitor state. For the entry vectors  $V_r(\xi)$  compatible with  $\xi$ , define

$$C_r^E(\xi) = \bigcap_{v \in V_r(\xi)} \mathcal{L}_r^E(v), \quad RS_E : \mathcal{L}_r(b_r) \subseteq C_r^E(\xi).$$

The reference language must account for the effect of  $v$  on monitor initialization; simply attaching an unused observer vector to the fixed compiled DFA does not establish that relation. Before and after the one entry occurrence, each other update label is permitted at most once in the sufficient-condition overapproximation. This overapproximation may admit update labels before entry that the actual game excludes.

**Lemma S3.2** (E: entry-value-independent boundary language). *Let  $V_r$  overapproximate all reachable entry observer vectors under the one-shot event constraint. If  $\mathcal{L}_r^E(v) = \mathcal{L}_r(b_r)$  for every  $v \in V_r$ , then the constant initializer satisfies  $RS_E$  exactly at every actual entry with nonempty  $V_r(\xi) \subseteq V_r$ .*

*Proof.* Every actual entry vector belongs to  $V_r$ , so its reference post-entry language equals the boundary language by hypothesis. Thus the nonempty intersection over  $V_r(\xi)$  consists of copies of  $\mathcal{L}_r(b_r)$  and equals that language. Constant initialization uses  $b_r$ , giving equality in  $RS_E$ .  $\square$



**Lemma S3.3** (E': update-stage fluents suffice). *Suppose an UPD monitor references only declared fluents whose initiating and terminating labels are update commands: `stopOldSpec`, `startNewSpec`, `reconfigure` (including their indexed forms), or `hotSwapIn`. No such label occurs before actual entry. If the constant initializer is the non-error boundary state  $b_r$  obtained from compiled state zero by consuming `hotSwapIn` once, then it satisfies  $RS_E$  with equality at every actual entry.*

*Proof.* Before entry, no defining update label occurs, so all referenced fluents retain their declared initial values. The fresh E monitor therefore starts at state zero and consumes `hotSwapIn` once, reaching  $b_r$ . Every actual entry has reference language  $\mathcal{L}_r(b_r)$ , so the reference intersection equals that language.  $\square$

The E' check reads the expanded initiating/terminating sets of each referenced fluent from the saved model exports, checks the constant initializer and its non-error boundary, and certifies 89/135 UPD occurrences. It uses the actual game's exclusion of pre-entry update commands; the older free-history diagnostic admits such commands and is retained unchanged. A name such as `stopPump` is an ordinary plant event, not an update command.

The historical E' condition leaves 46 R1 occurrences unverified because they reference ordinary-action predicates. Attaching different observer vectors to a fixed DFA does not establish their reference languages. The additional source-specific check below proves their entry interpretation without changing those historical records.

**Lemma S3.4** (E: guarded-invariant entry initialization). *For a fresh entry-scoped requirement  $\square(G \Rightarrow \neg A)$ , suppose  $G$  is initially false, changes only on actual update commands excluded from old-endpoint histories, and is unchanged by entry. Let  $A$  be a disjunction of ordinary-event predicates, each reset by entry. If the saved DFA has the two-state invariant transitions described below and its constant initializer is the non-error false-guard state obtained by consuming entry once from its default initial state, then the initializer satisfies  $RS_E$  with equality at every actual entry.*

*Proof.* No update command occurs before entry, so  $G$  is false there. Every possible value of the action predicates satisfies the initial implication. Entry leaves  $G$  false and resets all action predicates; the fresh pure invariant has no pending obligation from an earlier position. The resulting continuation language is thus identical for every entry valuation and equals that of the supplied boundary state. The compatible entry set is nonempty at an actual modeled entry, so intersecting these equal languages establishes exactness.  $\square$

The added checker reads the original source formulas, expands action aliases, and checks their correspondence to saved monitor/observer exports. For each R1 occurrence, it requires one default-false declared guard, nonempty disjoint initiating/terminating sets consisting only of actual update commands, and ordinary-event predicates reset by entry. It checks source/export equality of these sets, a constant non-error boundary initializer, and all DFA transitions against the two-state invariant: initiating commands set the guard, terminating commands clear it, a forbidden ordinary event when the guard holds enters ERROR, and other events stutter. ERROR absorbs all labels; absent local-alphabet edges become ERROR and labels outside that alphabet stutter, matching the adapter. All 46 monitors omit the entry label from their local alphabets, so consuming entry stutters at the false-guard state.

Inspection of namespace validation and endpoint construction, together with the saved successful input-validation results, establishes that old histories contain only ordinary events. The adapter computes the boundary value once and installs it directly at every old root; initialization does not consume entry a second time. Seven negative mutations test rejection of a wrong guard value, non-update guard event, guard-changing entry, missing declared edge/label, non-rejecting forbidden event and missing entry reset. The checker verifies all 46 occurrences across 4,722 state-label pairs and 854 possible action-predicate entry valuations. These are static validation counts, not extra synthesis trials or enumerated physical histories. S4 preserves the older free-history and E' results, and the new CSV labels its distinct sufficient condition. The new proof uses the existing E meaning; it does not establish the history-inclusive H interpretation or application-level conformance.

### S3.5 Whole-history obligations: a stronger diagnostic

For this coverage audit,  $\text{obs}_r$  retains the events defining the referenced fluents, independently of the active component version. Run the requirement monitor from  $m_r^0$  on the whole projected history before activation and its continuation. With absorbing error states this gives

$$\mathcal{C}_r(\xi) = \bigcap_{h \in H_r(\xi)} \mathcal{L}_r(\eta_r^*(m_r^0, h)).$$

H counts violations before activation. If an admitted history reaches ERROR, its continuation language is empty; a non-error initializer accepts the empty suffix and cannot satisfy RS at that activation. We retain the H check as a diagnostic and assert no benchmark RS guarantee under H. The A and E interpretations above have different scopes; none changes the compiled game or any saved synthesis decision.

**Lemma S3.5** (Fluent-determined exact initialization). *Let the freely reachable product of the deterministic requirement monitor (including absorbing errors) and all its referenced fluent observers start at their specified initial states. Suppose equal fluent vectors in that product imply equal monitor states. Let  $\sigma_r(v)$  return that unique monitor state for each reachable vector  $v$ . For any activation state with  $H_r(\xi) \neq \emptyset$  whose histories are represented in this product and whose supplied initializer is  $\iota_r(\xi) = \sigma_r(f(\xi))$ , initialization is an exact residual for the history-inclusive interpretation.*

*Proof.* Each projected activation history reaches a product pair  $(\eta_r^*(m_r^0, h), f(\xi))$ . Fluent determination makes its first coordinate equal to  $\sigma_r(f(\xi))$ , independently of  $h$ . Thus every language in the intersection defining  $\mathcal{C}_r(\xi)$  is the same language  $\mathcal{L}_r(\iota_r(\xi))$ . Because the history family is nonempty, its intersection is that language, proving equality in RS. For an empty history family, the intersection is  $\Sigma^*$  and this argument gives inclusion only, not equality in general.  $\square$

A single non-error state is insufficient for this lemma. For  $\Box p$  with  $p$  initially true, let  $a$  make  $p$  false and  $b$  restore it. The empty history and  $ab$  have the same current fluent value but reach the safe and absorbing-error monitor states, respectively. A check that discards errors verifies only safe-prefix determination, a weaker condition that needs a separate safe-history premise. We use the full product, including errors. Failure of this plant-independent sufficient condition leaves RS unverified for the actual activation histories; it is not by itself a counterexample to the supplied plant's RS.

### S3.6 Lifecycle induction and residual meanings

At entry, each component is old, all update operations are pending, old monitor states are inherited from the reachable monitored endpoint, new monitors are inactive, and update-interval monitors receive the supplied  $\iota_r$ . This establishes the version/pending equivalence and  $t \in \text{ActiveM}(P) \iff \mu(t) \neq \perp$ .

An ordinary step changes the active product and advances exactly its active monitors, preserving tags and pending operations. A transfer changes one old tag to new, selects every supplied transfer outcome, consumes its pending operation, and advances all active monitors once. A stop removes its old monitor and pending operation while advancing the other active monitors; the stopped monitor does not consume its own stop. A start advances the previously active monitors and installs its new monitor from the residual after its activation boundary, so the new monitor does not consume that boundary twice. The pending set enforces one-shot execution. The non-preemption guard only makes an update bucket empty; it does not alter any successful transition. Induction therefore preserves both invariants and the declared observation intervals.

Old monitors inherit their history from a verified old endpoint. For an activated requirement  $r$ , let  $h \in H_r(\xi_{act})$  be its supplied activation history and let  $w$  be a subsequently observed monitored suffix. Policy safety ensures  $w \in \mathcal{L}_r(\iota_r(\xi_{act}))$ . The external RS premise gives

$$\mathcal{L}_r(\iota_r(\xi_{act})) \subseteq \bigcap_{h \in H_r(\xi_{act})} \{w : hw \in \text{SafePref}_r\}.$$

Thus the combined observed history satisfies the supplied requirement meaning. For A, this step quantifies over the admissible safe histories and establishes the chosen post-activation obligation; it does not certify a violating pre-activation trace under H. For E, it uses the fresh entry-scoped reference languages and requires the independently justified entry initializer. Conservative residuals may remove legal continuations and can make a supplied game lose; algorithmic completeness is relative to that supplied game. Type, domain, and non-error checks on  $\iota_r$  do not establish RS or the accuracy of  $\text{SafePref}_r$  as a description of intended application requirements.

At goal, every update operation has occurred, all new monitors are active, old monitors are inactive, and update monitors are safe. Equality of component and new-monitor projections with  $\kappa(q)$  carries the new monitored continuation into the verified endpoint. The entry/load boundary contributes no ordinary observation, and update-interval monitoring ends at goal. This establishes the model-level interval safety used by the synthesis theorem.

## S4 Supplementary validation and cost tables

To follow the main paper, use Section S4.1 for granularity comparisons (RQ1), Section S4.2 for independent checks (RQ2), and Sections S4.3 and S4.4 for fixed-budget costs and policy diagnoses (RQ3). Auxiliary campaigns, extended budgets and Legacy references follow in separate blocks. Constructed pairs, application contracts, checker jobs and single diagnostic runs have different purposes and denominators and are not pooled.

In saved filenames, historical RQ1 (correctness) and RQ2 (capability) both support current RQ2; historical RQ4 (scaling) supports current RQ3. Current RQ1 draws on the E1/E6 granularity campaigns.

### S4.1 Constructed operational mechanisms and controls

The E6 input generators construct finite abstractions motivated by documented staged-replacement practices; they are not deployment measurements or replications of production incidents. The final index contains 398 terminal trials (195 WIN, 197 LOSS, 6 TO), including 395 Mac and 3 Xeon trials. The main comparison population is 55 pairs; scale extensions, assumption controls, E4 references and follow-up budgets do not increase that denominator. Full model definitions, generated inputs, prior failed versions, independent checks and primary-source scope are retained under `experiments/witness_20260929/e6/`.

Rolling models a fixed number of replicas and a readiness floor. Updating a batch introduces a finite startup gap; the checked threshold concerns this abstraction, not arbitrary asynchronous deployment. Canary adds observed set-valued healthy/broken outcomes and recovery: controls with only healthy outcomes win even when merged, while removing recovery loses under both granularities.

Policy's opposing requirement-boundary orders separate individual from globally merged boundaries. DB-Rolling models secondary-first replacement with a retained availability requirement; the no-slack control loses under both granularities. Rolling+Audit has only one old-stop/new-start pair, so boundary merging remains WIN; transfer merging loses. Its displayed six-step path comes from the Full policy, not Lazy's returned maximum rank.

Threads models backpressure versus saturation; the former wins both ways and the latter loses both ways. These null controls prevent reading every staged update as a granularity separation.

**PC2-Rolling: measured and checked facts.** The fine contract returns WIN on the two-arm physical plant with a calibration phase. Its exported game has 6639 discovered states, 381 policy states and 134128 queries; the Xeon counters agree with the saved Mac result. A separate enumeration of the old closed-loop product checks that both raw transfer domains contain every reachable old state. Their product nevertheless puts both arms in calibration, immediately violating the interval availability requirement. Ordinary old-product moves preserve the enumerated invariant, and any completing joint-transfer policy must eventually take that violating transfer; this gives a structural impossibility argument independently of the unresolved solver. The fine Lazy Xeon trial completes in 99.024 total JVM seconds; the merged Lazy and fine Direct-Full trials both time out at 7200 seconds with a 200 GiB heap. Earlier Mac timeouts remain separate records. Raw exit codes and runtime class hashes are unavailable for these Xeon exports and are not reconstructed. The saved invariant check is not a claim that the larger game was independently enumerated in full.

**Inherited-contract granularity transformations (E1).** E1 merges all transfer events, all old-stop/new-start events, or both on each Base/R1 inherited contract; R2 is excluded because its interval monitors name individual commands. These are contract transformations rather than encodings of the DUCS objective. All 54 transformations finish with 0 changes in decision. The raw physical transfer relations are total. The inherited models omit the explicit post-transfer startup/calibration availability phase used in the constructed examples. These observations explain why those particular mechanisms are absent; they do not constitute a theorem that every total-transfer contract permits simultaneous updating. Requirements can still reject the joint successor, as PC2-Rolling above demonstrates.

## S4.2 Independent checks for the contract and returned policies

The following evidence supports current RQ2 at different levels: finite-game decisions and certificates, frontend/source-monitor agreement, and focused handover fixtures. Functional checks and source-level sufficient conditions do not add application performance samples. Each check retains its explicit exclusions.

Table S2: Confirmed-semantic checks. Denominators exclude inapplicable checks, not failed cases.

| Check                                      | Correctness | Capability |
|--------------------------------------------|-------------|------------|
| Planned jobs                               | 92          | 14         |
| WIN / LOSS / invalid                       | 36 / 48 / 8 | 10 / 4 / 0 |
| Agreement with raw-LTS expectation         | 92/92       | 14/14      |
| Internal certificate, eligible jobs        | 84/84       | 14/14      |
| Separate exhaustive checker, eligible jobs | 84/84       | 14/14      |
| Atomic Link check, WIN jobs                | 36/36       | 10/10      |

The archived correctness campaign, labeled RQ1 in the saved files, contains 92 jobs: 41 semantic models under two methods and five endpoint-contract fixtures under two methods. Four invalid fixtures reject each method; the remaining contract fixture wins. Related variants share source models, so job count is not a count of independent applications. The archived capability campaign, labeled RQ2, has 14 jobs over eight finite LTS files, including two permanent branch-action bans and their neutral controls. These historical campaign names differ from the current main-paper RQs. All failures and invalid outcomes are retained. The three quiescence probes are separate from these fixed denominators. Historical PRISM results use the previous goal rule and are not included in the confirmed counts.

Ten original semantic models have 31 variants that rename or reorder elements, add unreachable states, or permute components. The code passes 41 focused tests, including five boundary tests; the broader run passes 99 tests with one opt-in integration test unexecuted.

**Current interpretation evidence.** The current results are NEW: 138 A inclusions (nonempty safe-history domains unverified except the stated PC2 tuple), and UPD: 135 E justifications (46 R1 and 89 R2). Direct ACT policy arguments cover PC2 and GSM. Separately, all 900 non-error lookup cells have the exact source-invariant continuation language under their reference observers; actual activation key/observation/fallback conformance remains a premise (Technical Appendix E.1). The additional R1 evidence below strengthens its earlier sufficient condition; no historical failures are removed.

**Direct frontend lookup validation.** We materialized the normal frontend’s NEW lookup objects for all 27 application inputs using the same frozen JAR and byte-identical inputs. Each separate JVM ran once with a 2 GiB heap and 120 s limit; all 27 completed, with no retry or omitted target. Frontend compilation constructs the endpoint controllers, but the exporter neither runs FG synthesis nor expands physical closure or the activation domain. These are functional checks, not performance samples or recovered historical JVM objects.

The independent Python checker matches every exported table’s keys and values and verifies observer order, alphabets and transitions against the saved definitions; Technical Appendix E.1 gives the complete counts and their relation to source-invariant exactness. It reconstructs the full monitor/observer table over repeatable update events, includes the first ERROR frontier, applies ERROR priority and compares every key and value. A second reconstruction uses the earlier check’s alphabet and order: all 138 alphabet differences and 114 column permutations are reconciled explicitly. The resulting non-error values and ERROR counts match the earlier CSV.

Five mutations of copies of real exports are detected: a safe value changed to ERROR, the converse, a deleted key with adjusted count, an omitted observer transition, and permuted columns without matching key permutation. Two separate Python unit cases cover initially-true normalization and

ERROR priority. Actual data contain no initially-true columns, safe/ERROR key collisions or multiple safe states needing nontrivial language-equivalence checks, so these Java branches are not validated by the 27 exports. Physical-state decoding, full activation-domain construction and actual safe-history nonemptiness remain unverified. The complete inputs, commands, tables, comparison and negative checks are in `evidence/frontend_lookup_demo/`; these files are included in the public artifact.

**Additional entry-invariant evidence.** The separate static checker `verify_r1_entry_invariant.py` verifies the guarded-invariant condition for R1 in GSM (1 occurrence), Industry (3), MetaSocket (1), PowerPlant (3), PC Arms=1 (6), PC Arms=2 (12), Railcab (7), Surveillance (7), and Workflow (6): 46/46 in total. Its CSV records each source formula and guard line, saved export selector, original run's input validation and update labels, and the old  $E'$  flags without altering them. The seven negative mutations are separate checker tests. No synthesis decision, input formula, measurement or historical diagnostic was replaced. These check files are included in the public artifact.

#### S4.2.1 Quiescent-goal functional checks

Three separate one-run fixtures test a safe target match with an enabled uncontrollable event. One old state has a controllable idle loop and a transfer to new state  $n_0$ . There are no monitors or dependencies. All new states have controllable idle loops. The three cases are:

| Case       | New uncontrollable edges     | Targets        | Decision |
|------------|------------------------------|----------------|----------|
| Persistent | $n_0 \xrightarrow{spin} n_0$ | $\{n_0\}$      | LOSS     |
| Finite     | $n_0 \xrightarrow{step} n_1$ | $\{n_0, n_1\}$ | WIN      |
| Restricted | $n_0 \xrightarrow{step} n_1$ | $\{n_0\}$      | LOSS     |

A target match at  $n_0$  is never a goal while UC is enabled. The finite case reaches quiescent  $n_1$ , with certificate ranks 2, 1, 0 and load into the new controller's state 1. The restricted case cannot return to its only allowed target; its quiescent  $n_1$  is not a goal.

The Python checker independently parses restricted FSP and constructs endpoint products, the entire game, NP and goals, and a least fixed point before comparing with Java. It checks all saved game states and buckets, and for WIN all certificate states, ranks, policy edges and endpoint links. The LOSS result is corroborated by independent full-game closure and Java certificate/exhaustive PASS logs; the CLI does not serialize losing-region membership, so no membership comparison is claimed. The frozen campaign JAR runs once per case under a 512 MiB heap and 120 s cap. Inputs, commands, exports and verification output are retained in `evidence/quiescent_goal_demo/`. These checks are outside the original 92 jobs and every performance statistic.

#### S4.2.2 Load-target selector functional checks

Three separate one-run JVM invocations use the frozen campaign JAR and the same one-component input, changing only `loadable_new_states`. The old environment/controller has one idle state. The new environment/controller starts at  $n_0$ , permits a controllable move to  $n_1$ , admits idle at both states and has no return. The sole transfer maps the old state to  $n_1$ . Selecting both endpoint states or only  $n_1$  returns WIN; selecting only  $n_0$  returns LOSS. Although  $n_0$  is reachable in the fixed endpoint, it is unreachable from the update entry.

Both WIN certificates contain two states and one transfer edge, with maximum rank one, and load new controller state 1 at endpoint `new-00000001`. All three complete update-game exports have two reachable states; the LOSS game has no goal. Goal states are terminal in the update export; subsequent idle edges remain in the endpoint bundle. The LOSS CLI exit 6 denotes unrealizability, not an execution failure.

The saved `evidence/load_target_selector_demo/verify_saved.py` parses the fixture’s restricted FSP independently, constructs the endpoint products and a Python least fixed point, and compares all exported edges, goals, physical tuples, empty monitor maps, pending transfer, rank and handover. All comparisons pass, as does Java’s separate exhaustive construction. There are no monitors, initializer obligations, uncontrollable events or precedence edges. These checks do not validate those features or deployment behavior and are separate from the original 92-job suite and all performance statistics. The directory retains all three inputs, commands, complete game exports, WIN endpoint bundles and verification outputs; no attempt was repeated.

### S4.3 Fixed-budget comparison on the 27 adapted contracts

This is the application comparison for current RQ3. All 135 instance–method cells are retained, including the Legacy reference column; same-game comparisons concern the four FG constructions. The outcome grid and absolute solver times appear first. Technical Appendix M is the complete table of all 14 finished Lazy/Direct-Full exploration pairs. Preparation measurements follow here; complete metric panels and JVM-time details are in Section S4.4.

The generated tables report all RQ3 model/method cells and metrics. Timing summaries require five valid, decision-consistent completions; other groups retain their statuses. Mac pilot and Industry/R1 diagnostic measurements are excluded from Xeon timing comparisons. The controlled-family table reports first-trial structure only. The five Legacy N/M (capture cap) cells are interrupted before synthesis by the author-side `M9MtsSnapshot` instrumentation’s fixed 250,000-state cap. They give no evidence of Legacy capability or solver failure. Saved process statuses remain unchanged; N/M is an evidence-based display annotation.

#### Benchmark contract coverage

Each triple is base/R1/R2. Transfer fan-out and loadable endpoint counts come from preserved first-trial outputs; component/requirement counts and explicit precedence are cross-checked against the parsed definitions.

| Model        | Components | Multi-target | Initializers | Prec. | $ Z_{load} $ |
|--------------|------------|--------------|--------------|-------|--------------|
| GSM          | 1          | No           | 1/2/3        | 0     | 9            |
| Industry     | 3          | No           | 3/6/9        | 0     | 81           |
| MetaSocket   | 2          | No           | 1/2/3        | 0     | 4            |
| PowerPlant   | 2          | No           | 3/6/10       | 0     | 4            |
| PC Arms=1    | 1          | No           | 6/12/18      | 0     | 9            |
| PC Arms=2    | 2          | No/No/–      | 12/24/36     | 0     | 137          |
| Railcab      | 4          | Yes          | 7/14/18      | 0     | 71           |
| Surveillance | 2          | No           | 7/14/21      | 0     | 107          |
| Workflow     | 5          | No           | 6/12/17      | 0     | 21           |

Multi-target reports whether a transfer source has more than one outcome. Initializers count NEW plus UPD requirements; NEW table lookup and UPD constant boundary initialization are distinguished in the RS table. PC Arms=2/R2 multi-target status is unmeasured (–): preserved runs do not contain the final transfer census. Zero explicit precedence does not mean transition monitors impose no ordering constraints.

Table S3: Full main-campaign outcome grid for all 135 instance–method cells.

| Model        | FG lazy | Direct-Full        | Eager  | Update-first | Legacy |
|--------------|---------|--------------------|--------|--------------|--------|
| GSM          | W/W/W   | W/W/W              | W/W/W  | W/W/W        | W/W/W  |
| Industry     | W/L/W   | W/T/W              | W/T/W  | W/L/W        | N/N/N  |
| MetaSocket   | W/W/W   | W/W/W              | W/W/W  | W/W/W        | W/W/W  |
| PowerPlant   | W/W/W   | W/W/W              | W/W/W  | W/W/W        | W/W/W  |
| PC Arms=1    | W/W/W   | W/T/W              | W/W/W  | W/W/W        | N/O/N  |
| PC Arms=2    | W/W/T   | T/T/T              | T/T/T  | T/T/T        | O/O/O  |
| Railcab      | W/W/W   | T/T/W              | T/T/W  | W/W/W        | O/O/O  |
| Surveillance | W/W/W   | T/T/T              | T/T/T  | W/W/W        | O/O/O  |
| Workflow     | W/W/W   | T/T/T <sup>‡</sup> | W/T/W  | W/W/W        | O/O/O  |
| Decided / 27 | 26      | 14                 | 17     | 24           | 9      |
| WIN / LOSS   | 25/1    | 14/0               | 17/0   | 23/1         | 9/0    |
| TO / OOM / N | 1/0/0   | 13/0/0             | 10/0/0 | 3/0/0        | 0/13/5 |

Each triple is Base/R1/R2; all 135 cells are retained. W/L: WIN/LOSS; T: timeout; O: out of memory; N: not measured before synthesis (author-side capture cap). <sup>‡</sup>: same-setting supplementary timeout, with the interrupted original preserved. PC Arms=2/R2 also times out in a separate 3,600 s FG trial. Legacy uses its native objective.

Table S4: RQ3 solver medians in seconds: Lazy / Direct-Full for all 27 supplied contracts. Every numeric median uses five completed runs.

| Input        | Base            | R1              | R2                     |
|--------------|-----------------|-----------------|------------------------|
| GSM          | 0.0224 / 0.0416 | 0.0228 / 0.045  | 0.0201 / 0.0327        |
| Industry     | 0.19 / 154      | 562 L / TO      | 1.15 / 28              |
| MetaSocket   | 0.0149 / 0.0408 | 0.0136 / 0.0426 | 0.0152 / 0.024         |
| PowerPlant   | 0.0463 / 1.85   | 0.125 / 5.51    | 0.18 / 1.01            |
| PC Arms=1    | 0.0741 / 157    | 0.0793 / TO     | 6.16 / 77.8            |
| PC Arms=2    | 2.61 / TO       | 2.6 / TO        | TO / TO                |
| Railcab      | 0.169 / TO      | 3.18 / TO       | 15.5 / 283             |
| Surveillance | 2.01 / TO       | 18.6 / TO       | 37.1 / TO              |
| Workflow     | 0.111 / TO      | 0.325 / TO      | 16.1 / TO <sup>†</sup> |

Times include checking and exclude JVM startup, frontend and endpoint preparation; three significant digits. Numeric cells are WIN unless L (LOSS); TO is an unfinished first trial, not a median. <sup>†</sup> Workflow/R2 retains its interruption and separate same-setting timeout.



**Preparation before game exploration.** The FSP benchmark driver constructs the fixed endpoint inputs before calling the FG-DUCS solver. Two saved, nonoverlapping intervals expose this work. Definition preparation ( $D$ ) includes old/new controller synthesis, transfer/component preparation, monitor preparation and fluent-observer lookup construction. Adapter preparation ( $P$ , next page) imports these structures, constructs closed-loop endpoints and activation tables, and validates the input contract. The four FG methods share this code; their exploration choices follow it.

Table S5 reports  $D$  for every supplied contract and method. Both preparation tables use the original fixed-budget runs: 81 completed cells with five repetitions each (405 runs), including the Industry/R1 LOSS results. The other 27 cells retain their outcomes. No preparation value is imputed from another method or contract.

Table S5: Definition preparation  $D$ , in seconds: five-run median [minimum–maximum]. TO is the process outcome; it does not locate the timeout within preparation or search.

| Contract            | Lazy             | Eager            | Update-first     | Direct-Full      |
|---------------------|------------------|------------------|------------------|------------------|
| GSM / Base          | .231 [.229–.233] | .232 [.227–.237] | .231 [.229–.233] | .230 [.228–.231] |
| GSM / R1            | .246 [.245–.249] | .246 [.241–.247] | .246 [.245–.247] | .246 [.243–.249] |
| GSM / R2            | .257 [.256–.259] | .258 [.257–.259] | .258 [.257–.259] | .259 [.257–.260] |
| Industry / Base     | .367 [.356–.371] | .369 [.364–.374] | .367 [.363–.370] | .371 [.366–.380] |
| Industry / R1       | .511 [.504–.517] | TO               | .512 [.501–.518] | TO               |
| Industry / R2       | .469 [.452–.475] | .471 [.467–.480] | .470 [.454–.485] | .467 [.463–.474] |
| MetaSocket / Base   | .234 [.228–.237] | .235 [.232–.237] | .233 [.232–.239] | .235 [.231–.241] |
| MetaSocket / R1     | .251 [.250–.253] | .250 [.247–.251] | .250 [.245–.253] | .249 [.245–.254] |
| MetaSocket / R2     | .261 [.259–.268] | .262 [.259–.265] | .264 [.259–.266] | .263 [.261–.268] |
| PowerPlant / Base   | .280 [.276–.282] | .276 [.273–.281] | .277 [.276–.280] | .278 [.276–.282] |
| PowerPlant / R1     | .318 [.317–.322] | .321 [.316–.323] | .316 [.313–.321] | .317 [.312–.325] |
| PowerPlant / R2     | .372 [.368–.374] | .372 [.369–.377] | .369 [.365–.378] | .373 [.371–.374] |
| PC1 / Base          | .372 [.371–.375] | .373 [.372–.374] | .375 [.372–.378] | .377 [.370–.379] |
| PC1 / R1            | .466 [.463–.476] | .464 [.459–.470] | .468 [.463–.471] | TO               |
| PC1 / R2            | .538 [.537–.543] | .538 [.537–.540] | .543 [.538–.545] | .538 [.537–.541] |
| PC2 / Base          | .620 [.611–.628] | TO               | TO               | TO               |
| PC2 / R1            | .817 [.814–.834] | TO               | TO               | TO               |
| PC2 / R2            | TO               | TO               | TO               | TO               |
| Railcab / Base      | .368 [.365–.371] | TO               | .369 [.366–.373] | TO               |
| Railcab / R1        | .471 [.467–.477] | TO               | .472 [.469–.476] | TO               |
| Railcab / R2        | .554 [.550–.562] | .555 [.549–.564] | .560 [.557–.565] | .552 [.551–.560] |
| Surveillance / Base | .432 [.426–.439] | TO               | .433 [.432–.440] | TO               |
| Surveillance / R1   | .538 [.534–.547] | TO               | .536 [.528–.551] | TO               |
| Surveillance / R2   | .641 [.634–.647] | TO               | .639 [.631–.658] | TO               |
| Workflow / Base     | .384 [.382–.389] | .383 [.381–.384] | .383 [.380–.390] | TO               |
| Workflow / R1       | .481 [.476–.489] | TO               | .482 [.480–.486] | TO               |
| Workflow / R2       | .587 [.584–.624] | .596 [.584–.615] | .592 [.581–.595] | TO <sup>†</sup>  |

The saved fields are `definition_prepare_total_time` and `revised_preparation_time`, originally in milliseconds.  $D$  ends before input-size census collection. Internal phase fields are nested within these intervals; adding them again would double-count work. Neither  $D + P$  nor JVM minus solver time isolates all frontend work: parsing, census collection, startup, output, linking and other overhead have different scopes. <sup>†</sup> Workflow/R2 Direct-Full retains its original interruption and the later same-setting timeout; neither supplies a preparation summary. Other timeout cells likewise lack these structured phase values.

**Activation closure and adapter preparation.** Table S6 gives  $P$  and the size of the activation closure built before solver exploration. Starting from all reachable old-endpoint physical tuples, the frontend closes ordinary transitions and component transfers. A tuple includes component versions and fluent-observer states, but omits endpoint-controller states, requirement-monitor states and pending commands. This is neither the explored update game nor an initializer-entry count. For each contract with a completed run, the count is identical across all completed methods and repetitions; PC2/R2 has no recorded count.

Table S6: Adapter preparation  $P$ , in seconds: five-run median [minimum–maximum], and precomputed activation-closure tuples. All 27 contracts and four methods remain visible.

| Contract            | Closure | Lazy                   | Eager               | Update-first        | Direct-Full      |
|---------------------|---------|------------------------|---------------------|---------------------|------------------|
| GSM / Base          | 19      | .030 [.030–.031]       | .030 [.028–.031]    | .029 [.028–.030]    | .029 [.028–.029] |
| GSM / R1            | 19      | .031 [.030–.031]       | .030 [.028–.030]    | .031 [.029–.031]    | .030 [.029–.030] |
| GSM / R2            | 19      | .032 [.031–.033]       | .031 [.031–.032]    | .032 [.030–.033]    | .032 [.031–.033] |
| Industry / Base     | 7,432   | .557 [.551–.572]       | .556 [.542–.573]    | .550 [.545–.563]    | .551 [.543–.579] |
| Industry / R1       | 7,432   | .550 [.544–.579]       | TO                  | .568 [.557–.569]    | TO               |
| Industry / R2       | 7,432   | .561 [.555–.563]       | .555 [.543–.573]    | .560 [.556–.568]    | .562 [.557–.563] |
| MetaSocket / Base   | 18      | .030 [.029–.031]       | .030 [.029–.031]    | .030 [.029–.032]    | .030 [.029–.033] |
| MetaSocket / R1     | 18      | .031 [.030–.033]       | .032 [.029–.032]    | .031 [.030–.033]    | .031 [.031–.033] |
| MetaSocket / R2     | 18      | .032 [.031–.033]       | .032 [.031–.033]    | .032 [.030–.034]    | .032 [.032–.033] |
| PowerPlant / Base   | 68      | .051 [.045–.055]       | .051 [.048–.053]    | .049 [.047–.050]    | .050 [.045–.052] |
| PowerPlant / R1     | 68      | .056 [.054–.060]       | .058 [.050–.059]    | .056 [.054–.058]    | .058 [.052–.058] |
| PowerPlant / R2     | 68      | .052 [.051–.056]       | .053 [.051–.056]    | .054 [.052–.055]    | .052 [.050–.056] |
| PC1 / Base          | 74      | .116 [.114–.122]       | .113 [.106–.123]    | .114 [.103–.119]    | .116 [.106–.122] |
| PC1 / R1            | 74      | .110 [.108–.111]       | .108 [.100–.111]    | .108 [.106–.112]    | TO               |
| PC1 / R2            | 74      | .107 [.105–.125]       | .109 [.101–.119]    | .108 [.101–.109]    | .108 [.105–.110] |
| PC2 / Base          | 10,628  | 13.872 [13.696–14.177] | TO                  | TO                  | TO               |
| PC2 / R1            | 10,628  | 14.221 [13.974–14.411] | TO                  | TO                  | TO               |
| PC2 / R2            | –       | TO                     | TO                  | TO                  | TO               |
| Railcab / Base      | 7,467   | .919 [.907–.942]       | TO                  | .913 [.899–.927]    | TO               |
| Railcab / R1        | 7,467   | .914 [.908–.926]       | TO                  | .905 [.879–.923]    | TO               |
| Railcab / R2        | 7,467   | .907 [.902–.919]       | .913 [.900–.923]    | .922 [.909–.925]    | .919 [.910–.924] |
| Surveillance / Base | 8,761   | 1.001 [.991–1.004]     | TO                  | 1.001 [.984–1.008]  | TO               |
| Surveillance / R1   | 8,761   | .994 [.983–1.013]      | TO                  | .996 [.984–1.000]   | TO               |
| Surveillance / R2   | 8,761   | 1.006 [1.001–1.013]    | TO                  | 1.013 [1.009–1.028] | TO               |
| Workflow / Base     | 52,608  | 5.730 [5.660–5.737]    | 5.735 [5.659–5.769] | 5.736 [5.725–5.782] | TO               |
| Workflow / R1       | 52,608  | 5.507 [5.444–5.615]    | TO                  | 5.511 [5.457–5.643] | TO               |
| Workflow / R2       | 52,608  | 5.568 [5.544–5.659]    | 5.511 [5.440–5.562] | 5.603 [5.552–5.700] | TO <sup>†</sup>  |

$P$  includes endpoint products, the closure, global-tuple initializer construction and endpoint-contract validation; it is not a closure-only timer. The count is `revised_reachable_physical_states_for_activation`. TO and <sup>†</sup> have the preceding table’s meaning. Rounding is to milliseconds; the CSV preserves the recorded precision.

**Reading the cost claims.** Lazy defers update-game states and successor buckets after this preparation. For example, its  $P$  medians are 13.872/14.221 s for PC2 Base/R1 and 5.507–5.730 s for Workflow. Thus reduced search does not remove global preparation. Main-paper Table 6 separately reports total JVM ratios. The public `scripts/generate_rq3_preparation.py` regenerates these tables and `build/generated/rq3-preparation.csv` directly from the archived campaign metrics, without executing synthesis.

## S4.4 Detailed fixed-budget measurements and policy diagnoses

The metric panels complete the fixed-budget record from Section S4.3. The Industry and Railcab blocks then explain particular outcomes using separate saved diagnostic runs. Their measurement scopes remain distinct.

### S4.4.1 Complete metric panels and completion ranks

The output-state and output-transition panels below count the linked old–update–new controller (saved field `output_controller_states`), not the update certificate’s rank domain (`certificate_states`). Lazy’s 25 WIN contracts have 15–1,260 linked states and 8–1,071 certificate states. Linking includes reachable old/new closed-loop states and non-goal update states, identifying goals with their selected new states. Neither quantity is the explored game size.

**Windows Xeon campaign; all 27 instances and five methods retained.**

**Discovered states (first trial)**

| Model        | Req. | FG lazy | Eager   | Update-first | Direct-Full     | Legacy |
|--------------|------|---------|---------|--------------|-----------------|--------|
| GSM          | BASE | 31      | 75      | 32           | 75              | –      |
|              | R1   | 34      | 114     | 34           | 115             | –      |
|              | R2   | 34      | 68      | 46           | 68              | –      |
| Industry     | BASE | 272     | 1.1e+05 | 2470         | 2.8e+05         | N/M    |
|              | R1   | 3.9e+04 | TO      | 3.1e+04      | TO              | N/M    |
|              | R2   | 7717    | 2.8e+04 | 2875         | 5.1e+04         | N/M    |
| MetaSocket   | BASE | 8       | 71      | 11           | 82              | –      |
|              | R1   | 8       | 88      | 11           | 95              | –      |
|              | R2   | 16      | 31      | 17           | 34              | –      |
| PowerPlant   | BASE | 73      | 5634    | 101          | 8168            | –      |
|              | R1   | 759     | 2.3e+04 | 911          | 2.4e+04         | –      |
|              | R2   | 1330    | 2810    | 600          | 3594            | –      |
| PC Arms=1    | BASE | 83      | 4.1e+05 | 6018         | 4.1e+05         | N/M    |
|              | R1   | 94      | 3.3e+06 | 2e+04        | TO              | OOM    |
|              | R2   | 5.6e+04 | 1.6e+05 | 9304         | 1.6e+05         | N/M    |
| PC Arms=2    | BASE | 2502    | TO      | TO           | TO              | OOM    |
|              | R1   | 2433    | TO      | TO           | TO              | OOM    |
|              | R2   | TO      | TO      | TO           | TO              | OOM    |
| Railcab      | BASE | 171     | TO      | 1.4e+04      | TO              | OOM    |
|              | R1   | 1.3e+04 | TO      | 3.7e+04      | TO              | OOM    |
|              | R2   | 9.8e+04 | 3e+05   | 3.8e+04      | 3.2e+05         | OOM    |
| Surveillance | BASE | 5319    | TO      | 1.4e+04      | TO              | OOM    |
|              | R1   | 9.6e+04 | TO      | 6.7e+05      | TO              | OOM    |
|              | R2   | 1.8e+05 | TO      | 1.2e+05      | TO              | OOM    |
| Workflow     | BASE | 151     | 4.3e+06 | 173          | TO              | OOM    |
|              | R1   | 1098    | TO      | 519          | TO              | OOM    |
|              | R2   | 1.5e+05 | 2.9e+05 | 3.3e+04      | TO <sup>‡</sup> | OOM    |

### Successor queries (first trial)

| Model        | Req. | FG lazy | Eager   | Update-first | Direct-Full     | Legacy |
|--------------|------|---------|---------|--------------|-----------------|--------|
| GSM          | BASE | 27      | 658     | 31           | 658             | –      |
|              | R1   | 44      | 695     | 38           | 706             | –      |
|              | R2   | 78      | 352     | 44           | 352             | –      |
| Industry     | BASE | 3336    | 1.4e+06 | 3.1e+04      | 5.1e+06         | N/M    |
|              | R1   | 2.5e+05 | TO      | 2.1e+05      | TO              | N/M    |
|              | R2   | 5.3e+04 | 2.1e+05 | 1.8e+04      | 7.3e+05         | N/M    |
| MetaSocket   | BASE | 12      | 190     | 18           | 289             | –      |
|              | R1   | 12      | 201     | 18           | 257             | –      |
|              | R2   | 24      | 53      | 22           | 79              | –      |
| PowerPlant   | BASE | 165     | 2.7e+04 | 242          | 4.8e+04         | –      |
|              | R1   | 1927    | 6.4e+04 | 2619         | 6.6e+04         | –      |
|              | R2   | 3118    | 7528    | 1149         | 1.4e+04         | –      |
| PC Arms=1    | BASE | 840     | 2.5e+06 | 5.2e+04      | 3.8e+06         | N/M    |
|              | R1   | 852     | 1.2e+07 | 2e+05        | TO              | OOM    |
|              | R2   | 3.5e+05 | 8.5e+05 | 5.9e+04      | 1.4e+06         | N/M    |
| PC Arms=2    | BASE | 5e+04   | TO      | TO           | TO              | OOM    |
|              | R1   | 4.4e+04 | TO      | TO           | TO              | OOM    |
|              | R2   | TO      | TO      | TO           | TO              | OOM    |
| Railcab      | BASE | 1575    | TO      | 1.4e+05      | TO              | OOM    |
|              | R1   | 9.8e+04 | TO      | 3.7e+05      | TO              | OOM    |
|              | R2   | 4.4e+05 | 1.4e+06 | 1.6e+05      | 2.8e+06         | OOM    |
| Surveillance | BASE | 3.8e+04 | TO      | 1.1e+05      | TO              | OOM    |
|              | R1   | 4.8e+05 | TO      | 5.5e+06      | TO              | OOM    |
|              | R2   | 8.8e+05 | TO      | 5.8e+05      | TO              | OOM    |
| Workflow     | BASE | 1131    | 3e+07   | 1329         | TO              | OOM    |
|              | R1   | 7142    | TO      | 4499         | TO              | OOM    |
|              | R2   | 5.9e+05 | 1.1e+06 | 1.2e+05      | TO <sup>‡</sup> | OOM    |

### Transition outcomes (first trial)

| Model        | Req. | FG lazy | Eager   | Update-first | Direct-Full     | Legacy |
|--------------|------|---------|---------|--------------|-----------------|--------|
| GSM          | BASE | 24      | 162     | 25           | 162             | –      |
|              | R1   | 28      | 170     | 28           | 172             | –      |
|              | R2   | 35      | 98      | 39           | 98              | –      |
| Industry     | BASE | 417     | 3.2e+05 | 5762         | 9.3e+05         | N/M    |
|              | R1   | 7e+04   | TO      | 5.7e+04      | TO              | N/M    |
|              | R2   | 1.2e+04 | 5e+04   | 4292         | 1.3e+05         | N/M    |
| MetaSocket   | BASE | 7       | 156     | 10           | 184             | –      |
|              | R1   | 7       | 163     | 10           | 177             | –      |
|              | R2   | 19      | 44      | 16           | 50              | –      |
| PowerPlant   | BASE | 105     | 2.1e+04 | 141          | 2.8e+04         | –      |
|              | R1   | 1441    | 4.9e+04 | 1753         | 5e+04           | –      |
|              | R2   | 2519    | 6024    | 873          | 7910            | –      |
| PC Arms=1    | BASE | 131     | 9.6e+05 | 1.6e+04      | 9.6e+05         | N/M    |
|              | R1   | 134     | 5.7e+06 | 5e+04        | TO              | OOM    |
|              | R2   | 1.2e+05 | 3.4e+05 | 2e+04        | 3.4e+05         | N/M    |
| PC Arms=2    | BASE | 5615    | TO      | TO           | TO              | OOM    |
|              | R1   | 4987    | TO      | TO           | TO              | OOM    |
|              | R2   | TO      | TO      | TO           | TO              | OOM    |
| Railcab      | BASE | 216     | TO      | 3.5e+04      | TO              | OOM    |
|              | R1   | 2.7e+04 | TO      | 8.2e+04      | TO              | OOM    |
|              | R2   | 1.8e+05 | 5.7e+05 | 6.9e+04      | 6.2e+05         | OOM    |
| Surveillance | BASE | 1.2e+04 | TO      | 3.1e+04      | TO              | OOM    |
|              | R1   | 2.5e+05 | TO      | 1.9e+06      | TO              | OOM    |
|              | R2   | 4.5e+05 | TO      | 2.5e+05      | TO              | OOM    |
| Workflow     | BASE | 149     | 1.2e+07 | 185          | TO              | OOM    |
|              | R1   | 2197    | TO      | 999          | TO              | OOM    |
|              | R2   | 2.4e+05 | 4.6e+05 | 5.4e+04      | TO <sup>‡</sup> | OOM    |

# **Solver time (seconds; five complete trials)**

| Model        | Req. | FG lazy                      | Eager                        | Update-first                 | Direct-Full                  | Legacy                    |
|--------------|------|------------------------------|------------------------------|------------------------------|------------------------------|---------------------------|
| GSM          | BASE | W 0.0224<br>[0.0208, 0.0236] | W 0.0332<br>[0.0323, 0.0362] | W 0.0233<br>[0.0222, 0.0235] | W 0.0416<br>[0.0398, 0.0421] | W 0.241<br>[0.234, 0.252] |
|              | R1   | W 0.0228<br>[0.0216, 0.0232] | W 0.0348<br>[0.0315, 0.0357] | W 0.0243<br>[0.0233, 0.0249] | W 0.045<br>[0.0438, 0.0472]  | W 0.306<br>[0.303, 0.309] |
|              | R2   | W 0.0201<br>[0.0197, 0.0219] | W 0.0281<br>[0.0271, 0.0285] | W 0.0225<br>[0.0219, 0.0232] | W 0.0327<br>[0.0311, 0.0346] | W 0.25<br>[0.234, 0.252]  |
| Industry     | BASE | W 0.19<br>[0.187, 0.201]     | W 14.5<br>[14.2, 14.7]       | W 0.542<br>[0.528, 0.566]    | W 154<br>[152, 166]          | N/M                       |
|              | R1   | L 562<br>[534, 566]          | TO                           | L 461<br>[366, 512]          | TO                           | N/M                       |
|              | R2   | W 1.15<br>[1.13, 1.17]       | W 2.96<br>[2.93, 3.12]       | W 0.455<br>[0.446, 0.463]    | W 28<br>[27.7, 28.3]         | N/M                       |
| MetaSocket   | BASE | W 0.0149<br>[0.0144, 0.0165] | W 0.0302<br>[0.027, 0.0308]  | W 0.0165<br>[0.0154, 0.0183] | W 0.0408<br>[0.0392, 0.0423] | W 0.323<br>[0.323, 0.324] |
|              | R1   | W 0.0136<br>[0.013, 0.0158]  | W 0.0278<br>[0.0264, 0.0298] | W 0.0147<br>[0.0146, 0.0153] | W 0.0426<br>[0.0392, 0.0441] | W 0.443<br>[0.44, 0.45]   |
|              | R2   | W 0.0152<br>[0.0143, 0.0162] | W 0.018<br>[0.0169, 0.0183]  | W 0.0154<br>[0.0145, 0.0161] | W 0.024<br>[0.0234, 0.0291]  | W 0.29<br>[0.283, 0.29]   |
| PowerPlant   | BASE | W 0.0463<br>[0.0427, 0.0502] | W 0.487<br>[0.483, 0.494]    | W 0.052<br>[0.0506, 0.053]   | W 1.85<br>[1.82, 1.87]       | W 18<br>[17.1, 25.7]      |
|              | R1   | W 0.125<br>[0.12, 0.147]     | W 1.08<br>[1.04, 1.1]        | W 0.152<br>[0.146, 0.155]    | W 5.51<br>[5.43, 5.58]       | W 896<br>[805, 910]       |
|              | R2   | W 0.18<br>[0.178, 0.18]      | W 0.251<br>[0.246, 0.259]    | W 0.114<br>[0.111, 0.117]    | W 1.01<br>[0.991, 1.03]      | W 8.25<br>[7.88, 8.31]    |
| PC Arms=1    | BASE | W 0.0741<br>[0.0678, 0.0754] | W 26<br>[25.5, 27]           | W 0.886<br>[0.848, 0.912]    | W 157<br>[156, 159]          | N/M                       |
|              | R1   | W 0.0793<br>[0.0748, 0.0811] | W 188<br>[185, 190]          | W 3.15<br>[3.05, 3.23]       | TO                           | OOM                       |
|              | R2   | W 6.16<br>[6.06, 6.89]       | W 12.2<br>[12.1, 12.6]       | W 1.17<br>[1.16, 1.19]       | W 77.8<br>[77, 78.2]         | N/M                       |
| PC Arms=2    | BASE | W 2.61<br>[2.49, 2.95]       | TO                           | TO                           | TO                           | OOM                       |
|              | R1   | W 2.6<br>[2.52, 2.78]        | TO                           | TO                           | TO                           | OOM                       |
|              | R2   | TO                           | TO                           | TO                           | TO                           | OOM                       |
| Railcab      | BASE | W 0.169<br>[0.164, 0.17]     | TO                           | W 2.26<br>[2.25, 2.29]       | TO                           | OOM                       |
|              | R1   | W 3.18<br>[3.12, 3.19]       | TO                           | W 6.74<br>[6.64, 6.89]       | TO                           | OOM                       |
|              | R2   | W 15.5<br>[15.5, 15.6]       | W 38.7<br>[38.6, 38.8]       | W 4.16<br>[4.13, 4.19]       | W 283<br>[279, 284]          | OOM                       |
| Surveillance | BASE | W 2.01<br>[1.99, 2.03]       | TO                           | W 2.63<br>[2.41, 2.71]       | TO                           | OOM                       |
|              | R1   | W 18.6<br>[18.4, 18.8]       | TO                           | W 126<br>[125, 130]          | TO                           | OOM                       |
|              | R2   | W 37.1<br>[36.9, 37.5]       | TO                           | W 16.1<br>[15.3, 16.4]       | TO                           | OOM                       |
| Workflow     | BASE | W 0.111<br>[0.108, 0.114]    | W 508<br>[491, 519]          | W 0.116<br>[0.114, 0.119]    | TO                           | OOM                       |
|              | R1   | W 0.325<br>[0.323, 0.33]     | TO                           | W 0.215<br>[0.212, 0.221]    | TO                           | OOM                       |
|              | R2   | W 16.1<br>[15.7, 16.3]       | W 23.9<br>[23.4, 24.5]       | W 3.18<br>[3.11, 3.27]       | TO <sup>‡</sup>              | OOM                       |

### Sampled peak RSS (GiB; five complete trials)

| Model        | Req. | FG lazy                    | Eager                     | Update-first              | Direct-Full               | Legacy                    |
|--------------|------|----------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| GSM          | BASE | W 0.113<br>[0.112, 0.116]  | W 0.122<br>[0.115, 0.124] | W 0.112<br>[0.11, 0.113]  | W 0.12<br>[0.116, 0.126]  | W 0.144<br>[0.143, 0.152] |
|              | R1   | W 0.111<br>[0.108, 0.115]  | W 0.118<br>[0.115, 0.12]  | W 0.111<br>[0.11, 0.115]  | W 0.124<br>[0.121, 0.131] | W 0.147<br>[0.145, 0.154] |
|              | R2   | W 0.108<br>[0.107, 0.111]  | W 0.108<br>[0.107, 0.118] | W 0.11<br>[0.109, 0.111]  | W 0.12<br>[0.115, 0.123]  | W 0.148<br>[0.141, 0.149] |
| Industry     | BASE | W 0.509<br>[0.498, 0.529]  | W 2.94<br>[2.69, 3.1]     | W 0.703<br>[0.697, 0.79]  | W 5.77<br>[5.58, 6.4]     | N/M                       |
|              | R1   | L 1.48<br>[1.39, 1.93]     | TO                        | L 1.6<br>[1.54, 1.6]      | TO                        | N/M                       |
|              | R2   | W 0.881<br>[0.876, 0.906]  | W 1.47<br>[1.39, 1.6]     | W 0.649<br>[0.623, 0.659] | W 1.98<br>[1.86, 2.03]    | N/M                       |
| MetaSocket   | BASE | W 0.102<br>[0.0995, 0.107] | W 0.112<br>[0.107, 0.113] | W 0.103<br>[0.102, 0.105] | W 0.116<br>[0.112, 0.116] | W 0.153<br>[0.146, 0.159] |
|              | R1   | W 0.105<br>[0.102, 0.106]  | W 0.107<br>[0.107, 0.108] | W 0.104<br>[0.101, 0.105] | W 0.112<br>[0.111, 0.115] | W 0.195<br>[0.19, 0.202]  |
|              | R2   | W 0.105<br>[0.103, 0.105]  | W 0.104<br>[0.102, 0.106] | W 0.104<br>[0.101, 0.105] | W 0.106<br>[0.105, 0.108] | W 0.141<br>[0.14, 0.143]  |
| PowerPlant   | BASE | W 0.127<br>[0.124, 0.135]  | W 0.436<br>[0.426, 0.448] | W 0.132<br>[0.13, 0.137]  | W 1.13<br>[1.08, 1.17]    | W 3.25<br>[3.05, 3.28]    |
|              | R1   | W 0.181<br>[0.175, 0.182]  | W 0.643<br>[0.601, 0.654] | W 0.177<br>[0.171, 0.184] | W 1.57<br>[1.56, 1.6]     | W 8.07<br>[7.89, 12.5]    |
|              | R2   | W 0.188<br>[0.185, 0.193]  | W 0.23<br>[0.228, 0.234]  | W 0.177<br>[0.165, 0.186] | W 0.579<br>[0.572, 0.587] | W 2.18<br>[2.16, 2.56]    |
| PC Arms=1    | BASE | W 0.145<br>[0.143, 0.15]   | W 5.74<br>[5.19, 6.01]    | W 0.528<br>[0.474, 0.562] | W 5.37<br>[5.17, 5.51]    | N/M                       |
|              | R1   | W 0.146<br>[0.144, 0.151]  | W 22.6<br>[22, 23.7]      | W 1.07<br>[1.06, 1.14]    | TO                        | OOM                       |
|              | R2   | W 1.65<br>[1.52, 1.75]     | W 3.08<br>[2.91, 3.16]    | W 0.448<br>[0.445, 0.458] | W 3.27<br>[3.18, 3.34]    | N/M                       |
| PC Arms=2    | BASE | W 7.26<br>[7.2, 7.47]      | TO                        | TO                        | TO                        | OOM                       |
|              | R1   | W 7.36<br>[7.19, 7.54]     | TO                        | TO                        | TO                        | OOM                       |
|              | R2   | TO                         | TO                        | TO                        | TO                        | OOM                       |
| Railcab      | BASE | W 0.794<br>[0.778, 0.806]  | TO                        | W 0.932<br>[0.916, 0.942] | TO                        | OOM                       |
|              | R1   | W 1.47<br>[1.46, 1.5]      | TO                        | W 1.6<br>[1.5, 1.62]      | TO                        | OOM                       |
|              | R2   | W 2.86<br>[2.51, 2.95]     | W 5.26<br>[5.26, 5.29]    | W 1.65<br>[1.62, 1.68]    | W 5.47<br>[5.33, 5.58]    | OOM                       |
| Surveillance | BASE | W 0.933<br>[0.928, 0.951]  | TO                        | W 1.4<br>[1.13, 1.45]     | TO                        | OOM                       |
|              | R1   | W 3<br>[2.86, 3.02]        | TO                        | W 14.2<br>[14, 14.5]      | TO                        | OOM                       |
|              | R2   | W 4.4<br>[4.37, 4.4]       | TO                        | W 2.53<br>[2.49, 2.55]    | TO                        | OOM                       |
| Workflow     | BASE | W 1.54<br>[1.52, 1.63]     | W 42.3<br>[37.6, 45]      | W 1.53<br>[1.5, 1.61]     | TO                        | OOM                       |
|              | R1   | W 1.54<br>[1.54, 1.55]     | TO                        | W 1.57<br>[1.51, 1.62]    | TO                        | OOM                       |
|              | R2   | W 2.54<br>[2.54, 2.64]     | W 3.74<br>[3.71, 3.97]    | W 1.61<br>[1.58, 1.62]    | TO <sup>‡</sup>           | OOM                       |

### Linked controller states (first trial)

| Model        | Req. | FG lazy | Eager | Update-first | Direct-Full     | Legacy |
|--------------|------|---------|-------|--------------|-----------------|--------|
| GSM          | BASE | 41      | 42    | 42           | 41              | 86     |
|              | R1   | 39      | 41    | 41           | 41              | 92     |
|              | R2   | 34      | 41    | 39           | 40              | 77     |
| Industry     | BASE | 459     | 475   | 476          | 459             | N/M    |
|              | R1   | –       | TO    | –            | TO              | N/M    |
|              | R2   | 441     | 460   | 458          | 438             | N/M    |
| MetaSocket   | BASE | 15      | 18    | 18           | 15              | 83     |
|              | R1   | 15      | 18    | 18           | 15              | 79     |
|              | R2   | 15      | 16    | 16           | 15              | 61     |
| PowerPlant   | BASE | 59      | 59    | 70           | 59              | 2434   |
|              | R1   | 32      | 61    | 53           | 65              | 945    |
|              | R2   | 58      | 59    | 46           | 53              | 1117   |
| PC Arms=1    | BASE | 62      | 78    | 70           | 61              | N/M    |
|              | R1   | 62      | 78    | 70           | TO              | OOM    |
|              | R2   | 72      | 75    | 74           | 59              | N/M    |
| PC Arms=2    | BASE | 625     | TO    | TO           | TO              | OOM    |
|              | R1   | 625     | TO    | TO           | TO              | OOM    |
|              | R2   | TO      | TO    | TO           | TO              | OOM    |
| Railcab      | BASE | 206     | TO    | 196          | TO              | OOM    |
|              | R1   | 136     | TO    | 186          | TO              | OOM    |
|              | R2   | 203     | 221   | 212          | 204             | OOM    |
| Surveillance | BASE | 1260    | TO    | 1522         | TO              | OOM    |
|              | R1   | 376     | TO    | 976          | TO              | OOM    |
|              | R2   | 982     | TO    | 829          | TO              | OOM    |
| Workflow     | BASE | 193     | 199   | 199          | TO              | OOM    |
|              | R1   | 182     | TO    | 194          | TO              | OOM    |
|              | R2   | 164     | 198   | 147          | TO <sup>‡</sup> | OOM    |

### Linked controller transitions (first trial)

| Model        | Req. | FG lazy | Eager | Update-first | Direct-Full     | Legacy |
|--------------|------|---------|-------|--------------|-----------------|--------|
| GSM          | BASE | 49      | 50    | 50           | 49              | 119    |
|              | R1   | 47      | 49    | 49           | 49              | 117    |
|              | R2   | 42      | 49    | 47           | 48              | 102    |
| Industry     | BASE | 1021    | 1037  | 1038         | 1021            | N/M    |
|              | R1   | –       | TO    | –            | TO              | N/M    |
|              | R2   | 1008    | 1022  | 1020         | 1000            | N/M    |
| MetaSocket   | BASE | 21      | 24    | 24           | 21              | 156    |
|              | R1   | 21      | 24    | 24           | 21              | 143    |
|              | R2   | 21      | 22    | 22           | 21              | 107    |
| PowerPlant   | BASE | 78      | 78    | 89           | 78              | 7796   |
|              | R1   | 51      | 80    | 72           | 84              | 2303   |
|              | R2   | 77      | 78    | 65           | 72              | 3148   |
| PC Arms=1    | BASE | 80      | 96    | 88           | 79              | N/M    |
|              | R1   | 80      | 96    | 88           | TO              | OOM    |
|              | R2   | 90      | 93    | 92           | 77              | N/M    |
| PC Arms=2    | BASE | 1360    | TO    | TO           | TO              | OOM    |
|              | R1   | 1360    | TO    | TO           | TO              | OOM    |
|              | R2   | TO      | TO    | TO           | TO              | OOM    |
| Railcab      | BASE | 287     | TO    | 276          | TO              | OOM    |
|              | R1   | 215     | TO    | 266          | TO              | OOM    |
|              | R2   | 283     | 300   | 293          | 283             | OOM    |
| Surveillance | BASE | 1638    | TO    | 1900         | TO              | OOM    |
|              | R1   | 754     | TO    | 1354         | TO              | OOM    |
|              | R2   | 1360    | TO    | 1207         | TO              | OOM    |
| Workflow     | BASE | 243     | 249   | 249          | TO              | OOM    |
|              | R1   | 232     | TO    | 244          | TO              | OOM    |
|              | R2   | 214     | 248   | 197          | TO <sup>‡</sup> | OOM    |

W/L: winning/losing. Seconds: median [min, max] for five valid, consistent completions; ranges in S4. TO: 1,200 s timeout; OOM: out of memory. N/M (capture cap): author-side M9MtsSnapshot instrumentation stops before synthesis at its fixed 250,000-state cap; no evidence of Legacy capability or solver failure. Original operational interruptions are retained in S4. Legacy uses native semantics. Other incomplete groups show statuses, without medians.

### Censored JVM attempts and elapsed time

The 1,200 s cap covers the whole JVM invocation, including startup, frontend and endpoints, solving, checks and output. For each Direct-Full timeout with five consistent FG completions,  $b = 1200 / \text{median}(t_{\text{FG,elapsed}})$  is a lower bound on that censored attempt relative to the FG elapsed median, not a Direct-Full median or a solver-only speedup. The last column records the  $1200 / \text{median}(t_{\text{FG,solver}})$  quotient for transparency; it has no solver-time lower-bound interpretation because the cap includes non-solver work. Displayed bounds are rounded downward.

| Model        | Target | FG elapsed (s) | FG solver (s) | $b (\geq)$ | Cap/solver |
|--------------|--------|----------------|---------------|------------|------------|
| Industry     | r1     | 563.93         | 562.13        | 2.12       | 2.13       |
| PC Arms=1    | r1     | 1.32           | 0.08          | 910.06     | 15125.54   |
| PC Arms=2    | base   | 19.36          | 2.61          | 61.99      | 459.89     |
| PC Arms=2    | r1     | 19.86          | 2.60          | 60.43      | 461.22     |
| Railcab      | base   | 2.23           | 0.17          | 539.14     | 7090.31    |
| Railcab      | r1     | 5.25           | 3.18          | 228.62     | 377.56     |
| Surveillance | base   | 4.34           | 2.01          | 276.32     | 596.89     |
| Surveillance | r1     | 21.16          | 18.64         | 56.71      | 64.37      |
| Surveillance | r2     | 39.89          | 37.07         | 30.08      | 32.38      |
| Workflow     | base   | 7.06           | 0.11          | 169.92     | 10810.32   |
| Workflow     | r1     | 7.17           | 0.33          | 167.45     | 3688.87    |
| Workflow     | r2     | 23.05          | 16.06         | 52.05      | 74.70      |

Workflow/R2 uses the authorized supplementary attempt; the original operational interruption remains in the provenance table. Industry/R1 compares an unresolved attempt with a completed LOSS decision. The table measures time to decide the supplied game, not time to return a controller.



**Total JVM elapsed seconds: five-run median [min, max]**

| Model        | Req. | FG lazy                   | Eager                     | Update-first              | Direct-Full               | Legacy                 |
|--------------|------|---------------------------|---------------------------|---------------------------|---------------------------|------------------------|
| GSM          | BASE | W 0.913<br>[0.913, 0.913] | W 0.913<br>[0.913, 0.914] | W 0.913<br>[0.912, 0.915] | W 0.913<br>[0.913, 0.916] | W 1.22<br>[1.11, 1.22] |
|              | R1   | W 0.913<br>[0.912, 0.914] | W 0.913<br>[0.913, 0.914] | W 0.913<br>[0.913, 0.914] | W 0.913<br>[0.913, 1.01]  | W 1.22<br>[1.22, 1.22] |
|              | R2   | W 0.914<br>[0.912, 0.915] | W 0.914<br>[0.914, 0.914] | W 0.913<br>[0.913, 0.914] | W 0.914<br>[0.913, 0.914] | W 1.12<br>[1.11, 1.22] |
| Industry     | BASE | W 1.92<br>[1.92, 1.92]    | W 16.3<br>[16, 16.5]      | W 2.22<br>[2.22, 2.33]    | W 156<br>[154, 168]       | N/M                    |
|              | R1   | L 564<br>[536, 568]       | TO                        | L 462<br>[368, 514]       | TO                        | N/M                    |
|              | R2   | W 2.93<br>[2.93, 2.93]    | W 4.75<br>[4.74, 4.95]    | W 2.23<br>[2.22, 2.33]    | W 29.8<br>[29.5, 30.1]    | N/M                    |
| MetaSocket   | BASE | W 0.912<br>[0.813, 0.914] | W 0.914<br>[0.91, 0.915]  | W 0.914<br>[0.911, 0.914] | W 0.913<br>[0.91, 0.914]  | W 1.22<br>[1.21, 1.32] |
|              | R1   | W 0.913<br>[0.913, 0.913] | W 0.914<br>[0.914, 0.915] | W 0.914<br>[0.912, 0.914] | W 0.913<br>[0.912, 0.913] | W 1.42<br>[1.42, 1.43] |
|              | R2   | W 0.913<br>[0.913, 0.914] | W 0.913<br>[0.913, 0.914] | W 0.914<br>[0.912, 0.915] | W 0.914<br>[0.913, 0.915] | W 1.22<br>[1.22, 1.22] |
| PowerPlant   | BASE | W 1.01<br>[1.01, 1.02]    | W 1.62<br>[1.52, 1.62]    | W 1.01<br>[1.01, 1.02]    | W 2.83<br>[2.83, 2.83]    | W 19.2<br>[18.3, 27]   |
|              | R1   | W 1.12<br>[1.12, 1.12]    | W 2.13<br>[2.02, 2.13]    | W 1.22<br>[1.12, 1.22]    | W 6.56<br>[6.45, 6.66]    | W 898<br>[807, 912]    |
|              | R2   | W 1.22<br>[1.22, 1.22]    | W 1.32<br>[1.32, 1.32]    | W 1.22<br>[1.21, 1.22]    | W 2.12<br>[2.02, 2.13]    | W 9.29<br>[8.88, 9.38] |
| PC Arms=1    | BASE | W 1.22<br>[1.21, 1.22]    | W 27.6<br>[27.2, 28.5]    | W 2.03<br>[2.02, 2.12]    | W 159<br>[157, 160]       | N/M                    |
|              | R1   | W 1.32<br>[1.31, 1.32]    | W 191<br>[188, 193]       | W 4.43<br>[4.25, 4.55]    | TO                        | OOM                    |
|              | R2   | W 7.56<br>[7.46, 8.27]    | W 13.6<br>[13.6, 14.1]    | W 2.43<br>[2.42, 2.53]    | W 79.4<br>[78.5, 79.6]    | N/M                    |
| PC Arms=2    | BASE | W 19.4<br>[19.2, 19.9]    | TO                        | TO                        | TO                        | OOM                    |
|              | R1   | W 19.9<br>[19.7, 20.3]    | TO                        | TO                        | TO                        | OOM                    |
|              | R2   | TO                        | TO                        | TO                        | TO                        | OOM                    |
| Railcab      | BASE | W 2.23<br>[2.22, 2.23]    | TO                        | W 4.24<br>[4.24, 4.25]    | TO                        | OOM                    |
|              | R1   | W 5.25<br>[5.24, 5.25]    | TO                        | W 8.88<br>[8.76, 9.07]    | TO                        | OOM                    |
|              | R2   | W 17.9<br>[17.8, 17.9]    | W 41.2<br>[41.2, 41.4]    | W 6.36<br>[6.35, 6.46]    | W 285<br>[282, 287]       | OOM                    |
| Surveillance | BASE | W 4.34<br>[4.34, 4.44]    | TO                        | W 5.15<br>[4.85, 5.25]    | TO                        | OOM                    |
|              | R1   | W 21.2<br>[20.8, 21.3]    | TO                        | W 129<br>[129, 134]       | TO                        | OOM                    |
|              | R2   | W 39.9<br>[39.9, 40.4]    | TO                        | W 18.8<br>[18, 19.1]      | TO                        | OOM                    |
| Workflow     | BASE | W 7.06<br>[6.96, 7.06]    | W 518<br>[501, 528]       | W 7.06<br>[7.06, 7.16]    | TO                        | OOM                    |
|              | R1   | W 7.17<br>[7.06, 7.26]    | TO                        | W 7.06<br>[6.96, 7.17]    | TO                        | OOM                    |
|              | R2   | W 23.1<br>[22.8, 23.4]    | W 30.9<br>[30.4, 31.6]    | W 10.2<br>[10.1, 10.3]    | TO <sup>‡</sup>           | OOM                    |

Incomplete cells retain their original display statuses and receive no timing estimate. Elapsed time is measured by the harness monotonic clock; timeout termination overhead is excluded from the conservative 1,200 s numerator above.

**Completion-rank coverage.** All 25 winning Lazy contracts have five saved worst-completion-rank values, for 125 records. Human-readable and structured metric entries agree in every saved output, and each contract's rank is identical across its five repetitions. Maximum retained ranks range from 4 to 34, with median 19 across contracts; Railcab Base/R1 have 21/33. The raw summary preserves the other two contracts as LOSS/TO with no winning rank. These are upper bounds on observed game events under the stated execution-progress premises, not downtime measurements or minimal update lengths. The script `evidence/summarize_completion_ranks.py` and all 27 rows in `evidence/completion_ranks.csv` permit rechecking without rerunning synthesis.

#### S4.4.2 Industry/R1 losing region and a separate contract variant

**Industry/R1 diagnostic.** One Mac run of the frozen JAR enables the separate exhaustive checker and confirms UNREALIZABLE by its reachable-state fixed point (110,992 states, 873,132 successor queries, 211,940 outcomes). Its exported game stops at goals, giving 110,825 states; a Python fixed point on this export identifies 89 winning and 42 losing entries among 131 roots and checks every losing-region closure condition. These counts concern correctness, not performance.

The stored graph identifies a losing entry  $Q77863$ : TOR has already completed once, DSD1 has not, and pending inputs have been received. All nine update events can execute in stop-reconfigure-start order, reaching safe, uncontrollable-quiescent  $Q19580$  with no pending update event, but with a signature outside the new endpoint's compatible set. From there the environment can sustain

$$Q19580 \xrightarrow{\text{validatedDSD1}} Q29028 \xrightarrow{\text{dsd1Ok}.0} Q21628 \xrightarrow{\text{reviewDSD1}} Q19204 \xrightarrow{\text{receivedDSD1}} Q19580.$$

Validation and review are controllable; rejection and receipt are uncontrollable. At each safe state, alternative GF1/TOR validations violate an active monitor; declining validation at quiescent  $Q19580$  deadlocks. The four safe states and eight unsafe alternatives form a 12-state spoiling certificate. R1 disallows ordinary controllable events between stopping the old requirements and starting the new ones. The global losing-region check, rather than this illustrative update order alone, rules out alternative policies. The exported game, command, outputs, collector diagnostics, and reproducible graph analysis are in `raw/industry-r1-independent/`. Update-command preemption alone cannot remove a cycle with no pending update commands; whether a different command/handover contract avoids this region remains unverified.

**Separate validation-permitting contract variant.** Keep the original Industry components, endpoint controllers, transfer relations, roots, precedence and Post/goal rules. In each of the three R1 update-time properties, replace

$$\begin{aligned} & \Box(\text{AfterStopBeforeStart}_r \Rightarrow \neg \text{AnyAction}) \\ \text{by } & \Box(\text{AfterStopBeforeStart}_r \Rightarrow \neg(\text{AnyAction} \wedge \neg \text{validateGF1})). \end{aligned}$$

Thus the same one event, `validateGF1`, becomes permitted in those intervals; the other seven ordinary controllable labels stay forbidden. This assumes that gate-form validation is an acceptable operation during update. It changes the supplied contract, not the synthesis semantics, and its application-level acceptability is an external premise.

One Mac run with the frozen JAR returns WIN, passing the 214-state winning certificate, atomic Link and separate exhaustive checks (145,864 reachable states). A Python fixed point over the export confirms that the 131 roots match the original tuples and all win, including the original 42 losing entries. Its independently extracted rank certificate covers 273 states and 407 successor buckets; its different size reflects policy extraction, not a different decision.

At the illustrative losing root's counterpart, a winning path reconfigures the three components, stops the three old requirements, starts the new no-duplicate requirement, and validates GF1. Each of the uncontrollable outcomes `adjustGF1`, `approveGF1` and `cancelGF1` resets the relevant TOR/DSD1 history fluents; receipt of GF1 and activation of the remaining two requirements then reach a compatible goal. All outcomes are retained in the certificate. The exact three-line LTS diff, executable checks, source-fluent definitions and raw evidence are in `experiments/industry_r1_contract_variant/`. This decision-only diagnosis supplies no timing comparison and is excluded from the 27-instance table; the original R1 LOSS remains unchanged.

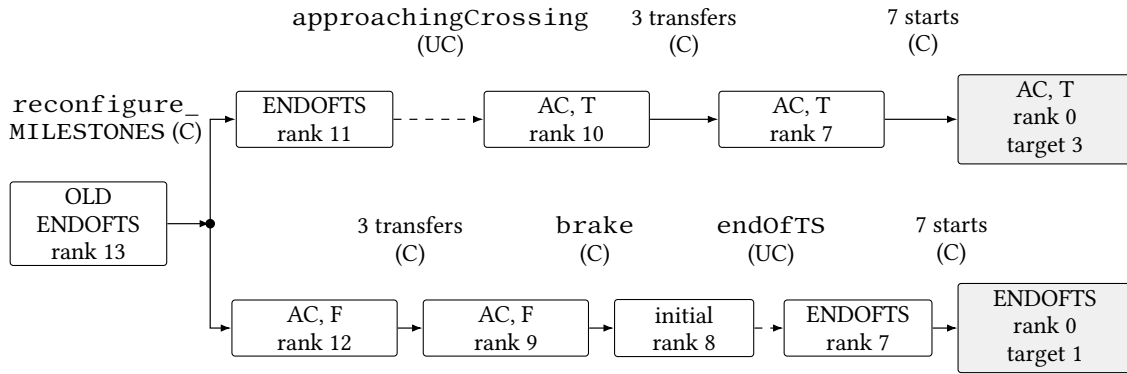
### S4.4.3 Qualitative Railcab policy exports

**Export scope.** The two qualitative illustrations are separate, single executions on macOS/ARM64 with OpenJDK 17.0.19, using the saved RQ3 campaign JAR, input model, solver properties and 64 GiB heap limit. Only the output options change: `--transition-output full` and `--deployment-output handoff-bundle.json` retain complete policy bodies. The scripts, exact commands, outputs and extraction checks are in `evidence/railcab_policy_demo` and `evidence/railcab_r1_policy_demo`. Their times are excluded from every performance table. Each was executed once; both return WIN with certificate and Link checks. No independent full-game enumeration or actual runtime experiment was added.

Both exports have 22 old entries and 71 new endpoint states. Their structural fields agree with the saved campaign. Full realized-controller transition counts differ from certificate counts and are not substituted for them.

**Base: both outcomes of one transfer.** Base exports 115 certificate states, 117 buckets, 118 outcomes, two reached goals and maximum rank 21.

Figure S1 shows the entire descendant fragment of Base state M69 (rank 13), reached from old entry O1 after five old stops. All four components are old; every old monitor is inactive, every new monitor is unstarted and Base has no interval monitor. Both outcomes of one transfer and their distinct endpoint continuations are retained. AC,T and AC,F have the same milestone location but different `ApproachingCrossingHappen` observer values; they must not be merged. The fragment has 26 states and 25 edges, with state identifiers and every expanded edge in `policy-fragment-edges.csv`. M69's outcomes are M15 (rank 11) and M19 (rank 12); their suffixes end at M20 and M7, loading controller states 3 and 1.



C: controllable; UC: uncontrollable (dashed). Shaded boxes: handover targets.

Figure S1: Both outcomes of one Railcab/Base transfer after five old stops and before any new start; Base has no interval monitor. AC is approaching-crossing; T/F records whether that event occurred since `endOfTS`. Boxes show milestone state and rank; targets identify supplied new-controller states. Only single-path sequences of three transfers or seven starts are contracted. Every branch of this 26-state, 25-edge fragment is retained.

Each contracted transfer block uses `ANOTHER_REQUEST`, `REQUEST_ATOMIC`, then `URNS`. Each start block activates, in order, `P_BRAKE_DURING_LASTBRAKE`, `P_BRAKE_IF_SOMETHING_BAD`, `P_CHECK_STATUS_AFTER_APPROACHING_CROSSING`, `P_CHECK_STATUS_BEFORE_LAST_EMERGENCY_BRAKE`, `P_EBRAKE_DURING_ELASTBRAKE`, `P_EBRAKE_IF_SOMETHING_BAD`, and `P_NEW_NOT_CRASH`. Every internal contracted state has exactly one bucket and one outcome. Extraction verifies all 118 rank decreases and both component/new-monitor matches with the fixed endpoint.

**R1: two uncontrollable events.** R1 exports 44 certificate states, 46 buckets, 46 outcomes, one goal and maximum rank 33. It retains seven UPD monitors throughout all 44 certificate states, including its goal. Their guards become true after any old stop until the associated new start. Together they forbid `requestEnter`, `brake`, `emergencyBrake`, `idle_c` and `checkCrossingStatus` during the relevant intervals; they do not forbid uncontrollables. The returned policy performs its ordinary controllable steps before its first stop. Milestone transfers occur at LB or NR and each has one outcome. R1 has *zero* multi-outcome buckets. Its branch at M30 instead has two distinct uncontrollable labels, `enterAllowed.0` and `enterAllowed.1`. For entry O1, all 32 descendant states/32 edges are retained; branches merge before the suffix of five stops, three remaining transfers and seven starts. All  $46 \times 7 = 322$  interval-monitor steps match the source DFA, and the goal matches the physical tuple and NEW monitors of new controller state 3. These are checks of the particular returned certificate, not a proof that every winning policy has this order.

**How interval guards change the stop/brake order.** The Base and R1 exports share old endpoint entry O1, controller state 1, with identical component and old-monitor states. Their update roots are Base M64 (rank 18) and R1 M28 (rank 27); identifiers are local to each export. Base takes five old stops to M69 (rank 13), then its MILESTONES transfer has two outcomes retained in the saved export and the paths below. On the lower branch, three further transfers reach M16 (rank 9), whose brake edge reaches M0 (rank 8).

The R1 prefix, without any old stop, consists of:

| Source   | Event                                                 | Target   |
|----------|-------------------------------------------------------|----------|
| M28 (27) | <code>requestEnter</code> (C)                         | M30 (26) |
| M30 (26) | <code>enterAllowed.0</code> (UC)                      | M31 (20) |
| M31 (20) | <code>lastBrake</code> (UC)                           | M37 (19) |
| M37 (19) | <code>reconfigure_MILESTONES</code> (C)               | M19 (18) |
| M19 (18) | <code>brake</code> (C)                                | M1 (17)  |
| M1 (17)  | <code>endOfTS</code> (UC)                             | M2 (16)  |
| M2 (16)  | <code>approachingCrossing</code> (UC)                 | M13 (15) |
| M13 (15) | <code>stopOldSpec_P_BRAKE_DURING_LASTBRAKE</code> (C) | M14 (14) |

Parentheses give ranks. The final four sources each have one successor. All five old monitors remain active until M13. The stop removes one and changes all seven R1 monitor guards from 0 to 1. Four further stops, three transfers and seven starts reach M3 (rank 0).

The separate *enterAllowed.1* branch completes from the same old entry without brake. This extraction does not establish that braking is necessary for every run. The file `railcab_stop_brake_paths.csv` preserves every edge of the selected Base/R1 entry-to-goal paths (18/22 edges) and all alternative outgoing edges from states on those selected paths. The accompanying extraction script checks path membership, ownership, ranks, common entry and saved UPD-monitor transitions without rerunning synthesis. Original campaign measurements remain unchanged.

## S4.5 Archived auxiliary campaigns and historical checks

This section retains older sufficient-condition diagnostics, the auxiliary requirement-boundary fixtures, construction/contract variants, and controlled scaling families. They supplement the current RQs without replacing fixed-budget failures or increasing the main constructed-pair denominator.

### S4.5.1 Historical initialization coverage

**Archived H/A/E' sufficient-condition results.** The historical table below is retained unchanged. For UPD, the current entry-scoped conclusion combines two sufficient conditions:

| UPD population              | Archived E' check | Current E justification |
|-----------------------------|-------------------|-------------------------|
| 89 stage-fluent occurrences | Exact             | Lemma S3.3              |
| 46 guarded R1 occurrences   | Unverified        | Exact by Lemma S3.4     |

Together they establish exact entry initialization for 135 UPD occurrences; application-level execution conformance remains unchecked.

### RS sufficient-condition coverage

Counts are requirement occurrences over base/R1/R2; repeated new requirements remain in the denominator. The independent free product includes absorbing ERROR and evolving fluent values after error. A failed sufficient check leaves RS untested and does not establish a reachable contract violation.

| Model        | New / upd | H checked | H FD | H exact | H untested | A sound <sup>a</sup> | E' exact <sup>b</sup> | A/E' unverified |
|--------------|-----------|-----------|------|---------|------------|----------------------|-----------------------|-----------------|
| GSM          | 3 / 3     | 6         | 0    | 0       | 6          | 3                    | 2                     | 1               |
| Industry     | 9 / 9     | 18        | 0    | 0       | 18         | 9                    | 6                     | 3               |
| MetaSocket   | 3 / 3     | 6         | 0    | 0       | 6          | 3                    | 2                     | 1               |
| PowerPlant   | 9 / 10    | 19        | 0    | 0       | 19         | 9                    | 7                     | 3               |
| PC Arms=1    | 18 / 18   | 36        | 0    | 0       | 36         | 18                   | 12                    | 6               |
| PC Arms=2    | 36 / 36   | 72        | 0    | 0       | 72         | 36                   | 24                    | 12              |
| Railcab      | 21 / 18   | 39        | 0    | 0       | 39         | 21                   | 11                    | 7               |
| Surveillance | 21 / 21   | 42        | 0    | 0       | 42         | 21                   | 14                    | 7               |
| Workflow     | 18 / 17   | 35        | 0    | 0       | 35         | 18                   | 11                    | 6               |

FD means that equal reachable fluent valuations determine the same monitor state, including ERROR. Exact additionally requires compatibility with the actual initializer: NEW uses its non-error lookup; UPD uses a constant monitor state after `hotSwapIn`. Per-requirement state counts, two conflicting histories, and update-boundary checks are in `rs-requirements.csv`. <sup>a</sup>All 138 NEW occurrences satisfy RS under A: the checked safe-prefix language condition gives equality when the actual safe-history set is nonempty; with an empty set, inclusion is automatic. Plant-level nonemptiness is not established, so exactness remains conditional. <sup>b</sup>The archived E' column uses the syntactic sufficient condition (Lemma E'): every referenced fluent is declared and affected only by update labels, which cannot occur before entry; the constant initializer must match the non-error state after one `hotSwapIn`. Remaining event/plant predicates retain unverified reference-language comparisons. Earlier E diagnostic columns and all H/A results are unchanged. Update labels occur at most once in the new prefix explorations; continuation-language equivalence is checked over all words, a stronger condition.

### S4.5.2 Auxiliary requirement-boundary inputs

The capability campaign contains 14 decisions. Its requirement-boundary comparison follows. Each input has one old entry. Endpoints allow only *idle*, with *align* additionally permitted at neutral's new endpoint. OLD/NEW testers have one safe state and an error sink; NEW activation requires a false forbidden-event observer. Entry sets *Aligned* false. In separation, *align* sets persistent *Aligned* and a separate observer of the last ordinary event. *idle* clears only the observer; update events preserve both.

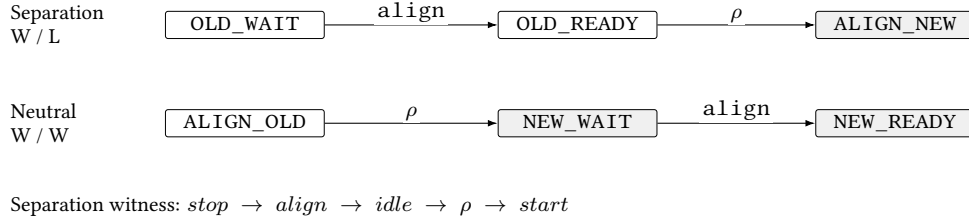


Figure S2: Measured requirement-boundary inputs. Row outcomes are free/block (W: WIN, L: LOSS). All events are controllable; shaded states are new. Each node has an *idle* loop; neutral also has *bad* loops.  $\rho$  is the sole transfer. Old/new monitors forbid *align* in separation and *bad* in neutral. An interval monitor permits *start* only after *align* since entry. A requirement block forbids intervening events/transfers between *stop* and *start*. Separation is WIN with free boundaries and LOSS with a block; neutral is WIN in both.

The source-derived sequence *stop*, *align*, *idle*,  $\rho$ , *start* matches the saved six-state certificate and maximum rank five. Old requirement prevents *align* before *stop*, so a contiguous block cannot complete. Neutral permits  $\rho$ , *align* before the block. Its block/free runs return ranks four/five; ranks need not be minimal. The saved counts do not identify the returned paths.

### S4.5.3 Granularity and construction ablations

The E1 transformations are described with the granularity evidence above. The remaining E4/E5 variants and E2 construction comparison are additional campaigns; the generated tables also preserve the full E1 and E6 series indices without expanding the main comparison populations.

E4-1 checks the transformations against saved finite fixtures. Transfer/both merging recovers the cell's losing bulk case. Merging the stop commands with each other and the start commands with each other does not forbid ordinary steps between stop and start: the transformed boundary fixture is therefore distinct from the hand-written contiguous-block restriction. E4-2's corrected  $\text{Cell}_n$  rings preserve one circulating workpiece; local and boundary-merged inputs win, transfer/both-merged inputs lose. The original series' input-validation errors are retained, not treated as synthesis losses. The series are stored in `experiments/witness_20260929/e4_1/` and `experiments/witness_20260929/e4_2_corrected/`.

E5 uniformly restricts each physical transfer domain to the declared initial local state. Of 18 fine/merged pairs, 0 separate, 6 lose both ways, 2 win both ways and 10 remain unresolved. The 25 attempted jobs and 29 deadline-excluded jobs remain separate. Railcab/Base's fine job timed out; its merged job and the other deadline-excluded jobs were not run. Enabled UC requests at the restricted states can prevent a safe quiescent transfer opportunity. Sources and domain checks are in `experiments/witness_20260929/e5/tables/`.

E2 changes graph construction alone: Full+UC omits controllable buckets at states with enabled UC events, then constructs the reachable graph before solving. The Mac series has an initial capped pass and a longer pass over its timeouts; its final decisions are 16 WIN, 1 LOSS and 10 unresolved. State counts, unlike runtime, can be compared across these saved deterministic constructions. On comparable cells, Full+UC is smaller than Direct-Full in 10 and equal in 6. It is equal in count to Eager in 14 and larger in 2 (GSM/R1 and PowerPlant/R1); equality of the state sets was not checked. OTF without UC pruning is not implemented, so this is an incomplete two-factor comparison. The separate Xeon series uses one fresh trial per cell; Industry/R1 is TO there despite Mac's LOSS. No Mac runtime is used in the paired Xeon table. Sources: `experiments/ablation_20260928/tables/` and `experiments/ablation_20260928/raw/e2_xeon/`.

Table S8: Construction choices: saved state counts (E2: Mac; other methods: fixed-budget Xeon) and separate Xeon timings. DF = Direct-Full; F+UC = full construction with UC pruning. Time is solve plus internal checking (seconds); F+UC is one trial, Eager a five-run median.

| Input         | Lazy  | Discovered states |        |        | Xeon seconds |        |
|---------------|-------|-------------------|--------|--------|--------------|--------|
|               |       | Eager             | DF     | F+UC   | F+UC         | Eager  |
| Industry/Base | 272   | 109726            | 282709 | 109726 | 50.455       | 14.516 |
| PC1/Base      | 83    | 407019            | 407019 | 407019 | 158.667      | 26.016 |
| PC1/R2        | 56443 | 158218            | 158218 | 158218 | 77.342       | 12.155 |
| Railcab/R2    | 97522 | 299700            | 317094 | 299700 | 268.563      | 38.669 |

Table S9: Selected E2 Xeon timing observations (seconds).

| Contract      | Full+UC states | Full+UC (1) | Eager (5 med.) | Ratio |
|---------------|----------------|-------------|----------------|-------|
| Industry/Base | 109,726        | 50.455      | 14.516         | 3.48  |
| PC1/Base      | 407,019        | 158.667     | 26.016         | 6.10  |
| PC1/R2        | 158,218        | 77.342      | 12.155         | 6.36  |
| Railcab/R2    | 299,700        | 268.563     | 38.669         | 6.95  |

Full+UC is one trial; Eager is the median of five valid completed trials. Both timing columns are Xeon observations using the LTS solve-plus-internal-check timer, 64 GiB heap and 1,200 s whole-JVM cap. The displayed Full+UC state counts are identical in the saved Mac and Xeon trials. The full E2 structural comparison uses Mac Full+UC counts and existing Xeon counts for the other methods, retaining each marked budget; it pools no times across hosts. Ratios are descriptive, without repetition-based uncertainty for Full+UC. Equal counts do not establish equal state sets. OTF without UC pruning remains unimplemented.

Table S10: Granularity checks on adapted contracts and controlled variants.

| Series              | Comparisons/trials | WIN | LOSS | TO/NR |
|---------------------|--------------------|-----|------|-------|
| E1 transformations  | 54                 | 51  | 3    | 0     |
| E4 fixtures         | 24                 | 19  | 5    | 0     |
| E4 corrected rings  | 45                 | 27  | 18   | 0     |
| E5 scheduled trials | 54                 | 4   | 12   | 9/29  |

E1 changes no decision in 54 comparisons; 44 discovered-state counts decrease and 10 are equal. The 22 original physical transfer relations are total, which alone does not prove safety or completion. E4 rings cover  $n = 2, \dots, 10$ ; the initial series with 30 input errors is retained separately. Boundary batching is not the handmade contiguous-block restriction. E5 has 18 pairs: no witness, six both-LOSS, two both-WIN, ten unresolved; NR denotes the deadline exclusion.

Table S11: E6 registered trials by saved series, with hosts and additional budgets kept separate.

| Series                         | Trials | WIN | LOSS | TO |
|--------------------------------|--------|-----|------|----|
| rolling/v1                     | 100    | 50  | 50   | 0  |
| canary/v1                      | 60     | 30  | 30   | 0  |
| policy/v2                      | 6      | 3   | 3    | 0  |
| db_rolling/v2                  | 8      | 2   | 6    | 0  |
| rolling_audit/v1               | 8      | 4   | 4    | 0  |
| threads/v1                     | 80     | 40  | 40   | 0  |
| rolling_scale/v1               | 80     | 40  | 40   | 0  |
| canary_controls/v1             | 48     | 24  | 24   | 0  |
| pc2_rolling/v2                 | 3      | 1   | 0    | 2  |
| pc2_rolling/extended_7200      | 2      | 0   | 0    | 2  |
| pc2_rolling/xeon_20260929_2344 | 3      | 1   | 0    | 2  |

The 55 main comparisons are 33 witnesses, 11 both-LOSS, 10 both-WIN and one unresolved PC2 comparison. Rolling scale (20 pairs), Canary controls (12 pairs), and E4 references (9 pairs) are separate. These deliberately constructed cases do not estimate prevalence. PC2 measured TO remains TO despite its separately checked safety obstruction. Audit boundary merging remains WIN; Threads supplies ten both-WIN and ten both-LOSS controls. All new timings are single trials, not medians.

#### S4.5.4 Controlled families, Travel, and hub scaling

These are additional scaling populations for current RQ3; RQ4 in generated captions is the archived campaign name. Their first-trial structure, completion eligibility, censored outcomes, and complete plotted ranges are retained below.

##### Controlled families: first-trial structure

All 132 jobs are retained. These single trials provide no timing medians.

| Model                             | FG lazy | Eager  | Update-first | Direct-Full |
|-----------------------------------|---------|--------|--------------|-------------|
| ctrl_fanout_01                    | W: 2    | W: 2   | W: 2         | W: 2        |
| ctrl_fanout_02                    | W: 2    | W: 2   | W: 2         | W: 2        |
| ctrl_fanout_04                    | W: 2    | W: 2   | W: 2         | W: 2        |
| ctrl_fanout_08                    | W: 2    | W: 2   | W: 2         | W: 2        |
| ctrl_fanout_16                    | W: 2    | W: 2   | W: 2         | W: 2        |
| transfer_outcomes_01              | W: 2    | W: 2   | W: 2         | W: 2        |
| transfer_outcomes_02              | W: 3    | W: 3   | W: 3         | W: 3        |
| transfer_outcomes_04              | W: 5    | W: 5   | W: 5         | W: 5        |
| transfer_outcomes_08              | W: 9    | W: 9   | W: 9         | W: 9        |
| transfer_outcomes_16              | W: 17   | W: 17  | W: 17        | W: 17       |
| uc_events_00                      | W: 2    | W: 2   | W: 2         | W: 2        |
| uc_events_01                      | W: 3    | W: 3   | W: 3         | W: 3        |
| uc_events_02                      | W: 5    | W: 5   | W: 5         | W: 5        |
| uc_events_04                      | W: 9    | W: 9   | W: 9         | W: 9        |
| uc_events_08                      | W: 17   | W: 17  | W: 17        | W: 17       |
| uc_outcomes_01                    | W: 3    | W: 3   | W: 3         | W: 3        |
| uc_outcomes_02                    | W: 5    | W: 5   | W: 5         | W: 5        |
| uc_outcomes_04                    | W: 9    | W: 9   | W: 9         | W: 9        |
| uc_outcomes_08                    | W: 17   | W: 17  | W: 17        | W: 17       |
| requirements_00                   | W: 2    | W: 2   | W: 2         | W: 2        |
| requirements_01                   | W: 4    | W: 10  | W: 4         | W: 15       |
| requirements_02                   | W: 6    | W: 26  | W: 6         | W: 94       |
| requirements_04                   | W: 10   | W: 82  | W: 10        | W: 2556     |
| requirements_08                   | W: 18   | W: 290 | W: 18        | W: 1179640  |
| precedence_closure_00             | W: 5    | W: 11  | W: 5         | W: 16       |
| precedence_closure_01             | W: 5    | W: 8   | W: 5         | W: 12       |
| precedence_closure_02             | W: 5    | W: 7   | W: 5         | W: 9        |
| precedence_closure_03             | W: 5    | W: 6   | W: 5         | W: 8        |
| precedence_closure_06             | W: 5    | W: 5   | W: 5         | W: 5        |
| decision_transfer_complete        | W: 3    | W: 3   | W: 3         | W: 3        |
| decision_transfer_partial         | L: 3    | L: 3   | L: 3         | L: 3        |
| decision_self_loop_controllable   | W: 2    | W: 2   | W: 2         | W: 2        |
| decision_self_loop_uncontrollable | L: 1    | L: 1   | L: 1         | L: 1        |

Cells show decision and discovered states. Queries and outcomes: `rq4-controlled.csv`.



### Travel and hub: completed returns

Travel retains all 48 instances; hub retains 30 seeds under each of three UC profiles. W/TO/OOM are first-trial counts. Five-run timing eligibility is shown separately.

| Family | Method       | W  | TO | OOM | Five-run cells |
|--------|--------------|----|----|-----|----------------|
| Travel | FG lazy      | 20 | 26 | 2   | 20             |
| Travel | Eager        | 20 | 26 | 2   | 20             |
| Travel | Update-first | 20 | 26 | 2   | 20             |
| Travel | Direct-Full  | 18 | 28 | 2   | 18             |
| Hub    | FG lazy      | 90 | 0  | 0   | 90             |
| Hub    | Eager        | 90 | 0  | 0   | 90             |
| Hub    | Update-first | 90 | 0  | 0   | 90             |
| Hub    | Direct-Full  | 90 | 0  | 0   | 90             |

All completed decisions agree. Travel has 960 planned repetition slots: 504 executed trials and 456 explicit resource-failure skips; hub completes all 1,800 trials. `rq4-points.csv` retains all plotted method/instance cells; the two `rq4-*comparisons.csv` files retain paired ratios and unresolved statuses. Travel lazy/Direct-Full state ratios range from 0.03435 to 0.46154 over 18 completed pairs. The upper endpoint is  $N = 1, K = 1$  (12/26 states); all eight  $N = 1$  ratios exceed one third. No case is excluded to tighten this range. Lazy and update-first match states, queries and outcomes on all 20 completed Travel instances. Unresolved instances have no measured structure and do not count as equal pairs. Hub timings are solve-only; Travel LTS timings include the internal checks. Ratios are paired within each family, never pooled across these scopes.

Table S13: All Travel first-trial outcomes. Entries are Lazy/Eager/Update-first/Direct-Full; W = WIN, T = timeout, O = out of memory.

| $N \backslash K$ | 1    | 2    | 4    | 6    | 8    | 12   | 16   | 20   |
|------------------|------|------|------|------|------|------|------|------|
| 1                | WWWW | WWWW | WWWW | WWWW | WWWW | WWWW | WWWW | WWWW |
| 2                | WWWW | WWWW | WWWW | WWWW | WWWW | WWWW | WWWW | WWWW |
| 4                | WWWW | WWWW | WWWT | WWWT | TTTT | TTTT | TTTT | TTTT |
| 6                | TTTT | TTTT | OOOO | OOOO | TTTT | TTTT | TTTT | TTTT |
| 8                | TTTT | TTTT | TTTT | TTTT | TTTT | TTTT | TTTT | TTTT |
| 12               | TTTT | TTTT | TTTT | TTTT | TTTT | TTTT | TTTT | TTTT |

$N$  counts amenities;  $K$  changes UC choices, path length and local state count together. Each outcome comes from the retained first trial under 1,200 s/64 GiB. Resource failures have no median.

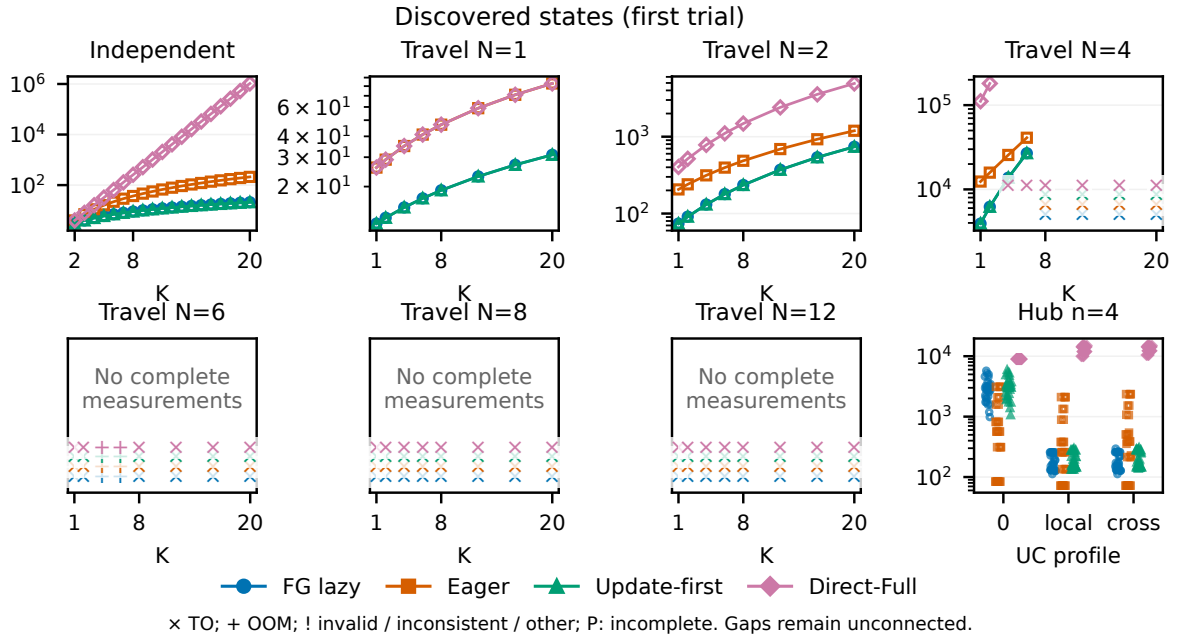


Figure S3: Full scaling-state panels, retaining every tested Travel concurrency level and every hub profile. RQ4 is the archived campaign name; these results appear under RQ3 (Scaling limits) in the main paper. The saved plots label the independent component count as  $K$ ; it is  $n$  in the main paper and S2, distinct from Travel's choice count  $K$ . Censored outcomes have status marks and no numeric ordinate.

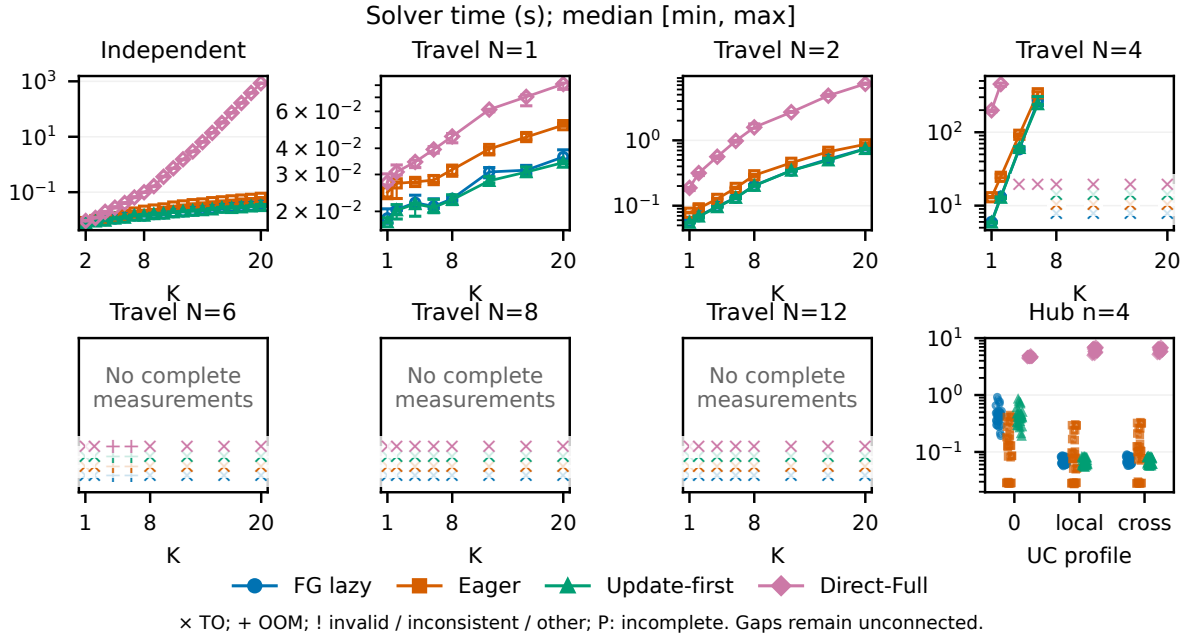


Figure S4: Full scaling-time panels from the archived RQ4 campaign. Medians and ranges require five valid, consistent completions; TO/OOM and other incomplete outcomes remain visible.

## S4.6 Extended-budget trials

These single trials extend the budget for previously unresolved conditions; the fixed-budget results remain unchanged. Missing raw is not a failed trial, and a capped single trial supplies no five-run median.

Table S14: Extended-budget single trials (separate from the fixed-budget campaign).

| Cell / campaign          | Method       | GiB | Cap (s) | Status | JVM (s)  | Solver (s) | RSS (GiB) | States  | JVM ratio |
|--------------------------|--------------|-----|---------|--------|----------|------------|-----------|---------|-----------|
| Travel(4,4)/base [ext1]  | DF           | 64  | 7200    | WIN    | 1.87e+03 | 1.81e+03   | 7.36      | 412688  | 15.7      |
| Travel(4,6)/base [ext1]  | DF           | 64  | 7200    | WIN    | 6.99e+03 | 6.77e+03   | 12.7      | 817362  | 13.4      |
| Travel(4,8)/base [ext1]  | Lazy         | 64  | 7200    | WIN    | 1.64e+03 | 756        | 15.9      | 47829   | –         |
| Travel(4,8)/base [ext1]  | Update-first | 64  | 7200    | WIN    | 1.66e+03 | 718        | 19.2      | 47829   | –         |
| Industry/r1 [ext2]       | DF           | 64  | 7200    | LOSS   | 1.7e+03  | 1.69e+03   | 2.79      | 110825  | 3.01      |
| PC1/r1 [ext2]            | DF           | 64  | 7200    | WIN    | 2.22e+03 | 2.22e+03   | 25.1      | 3252233 | 1.68e+03  |
| Travel(4,8)/base [ext3]  | DF           | 64  | 7200    | TO     | 7.2e+03  | –          | 17.2      | –       | –         |
| Travel(4,8)/base [ext3]  | Eager        | 64  | 7200    | WIN    | 2.04e+03 | 961        | 17.2      | 64248   | –         |
| Travel(4,12)/base [ext3] | Lazy         | 64  | 7200    | TO     | 7.2e+03  | –          | 32        | –       | –         |
| Travel(6,1)/base [ext3]  | Lazy         | 64  | 7200    | TO     | 7.2e+03  | –          | 47        | –       | –         |
| Travel(6,1)/base [ext3]  | DF           | 64  | 7200    | TO     | 7.2e+03  | –          | 48.6      | –       | –         |
| Railcab/base [ext4]      | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 190       | –       | –         |
| Surveillance/base [ext4] | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 213       | –       | –         |
| Workflow/base [ext4]     | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 191       | –       | –         |
| Workflow/r1 [ext4]       | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 192       | –       | –         |
| Railcab/r1 [ext4]        | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 212       | –       | –         |
| PC2/base [ext4]          | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 210       | –       | –         |
| PC2/r1 [ext4]            | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 209       | –       | –         |
| Surveillance/r1 [ext4]   | DF           | 200 | 7200    | OOM    | 5.27e+03 | –          | 211       | –       | –         |
| Workflow/r2 [ext4]       | DF           | 200 | 7200    | WIN    | 1.39e+03 | 1.38e+03   | 23.6      | 1774711 | 60.1      |
| Surveillance/r2 [ext4]   | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 159       | –       | –         |
| PC2/r2 [ext4]            | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 209       | –       | –         |
| PC2/r2 [ext4]            | Lazy         | 200 | 7200    | TO     | 7.21e+03 | –          | 187       | –       | –         |
| Travel(6,1)/base [ext5]  | Lazy         | 200 | 7200    | TO     | 7.21e+03 | –          | 78.6      | –       | –         |
| Travel(6,1)/base [ext5]  | DF           | 200 | 7200    | TO     | 7.21e+03 | –          | 78.9      | –       | –         |
| Travel(6,4)/base [ext5]  | Lazy         | 200 | 7200    | SKIP   | –        | –          | –         | –       | –         |
| Travel(6,4)/base [ext5]  | DF           | 200 | 7200    | SKIP   | –        | –          | –         | –       | –         |

Each row has one planned trial. WIN/LOSS display SUCCESS/UNREALIZABLE; TO is timeout at the stated cap. NOT\_RUN/INCOMPLETE means no completed return; – means unmeasured or ineligible, never zero. SKIP displays SKIPPED\_MONOTONE, a scheduling exclusion, not a measured resource failure or an inferred losing decision; reasons are in the CSV. ! marks a provenance/check failure; the raw status is retained. The ratio uses total JVM elapsed time divided by the same cell’s fixed-budget, five-valid-trial Lazy median (1,200 s, 64 GiB), only for matching checked decisions. It differs from the main paper’s solver-time ratios. The CSV separately reports solver-time ratios; timeouts have no ratio. An extended trial is never treated as a five-run median. RSS measures process memory, not heap occupancy. The 5 comparable extended decisions all match fixed-budget Lazy: Direct-Full on Travel (4, 4) and (4, 6), Industry/R1 (LOSS), PC1/R1 and Workflow/R2 (WIN). The 3 Travel (4, 8) decisions agree with one another, but fixed-budget Lazy timed out there; they are not baseline decision matches. JVM elapsed time includes preprocessing and other work outside the solver. For Travel (4, 8), Lazy, Update-first and Eager have JVM/solver times 1,636/756, 1,655/718 and 2,036/961 seconds. Table S14’s JVM ratios compare a single extended trial with a fixed-budget Lazy five-run median; the main paper’s solver ratios use solver time. Separate solver ratios are retained in the CSV.

Industry/R1’s 110,825 states here count the game with goal states terminal. The larger independent-checker count in the Industry/R1 diagnostic above includes reachable states before this truncation, so the enumeration domains differ. Surveillance/R1 has an explicit Java heap space failure. Timeout logs contain no heap-occupancy or GC measurements, so large process RSS alone does not prove a full heap or identify the bottleneck. The 2 Travel (6, 4) skips record the preregistered unmet success prerequisite for (6, 1) under both budgets; they are not observed timeouts or losing decisions.

**Additional timeout evidence.** The generated provenance below retains both the original interrupted attempt and its separate same-setting follow-up. The main PC Arms=2/R2 cell continues to

report its original 1,200 s TO; the single 3,600 s result is separate evidence.

### Authorized RQ3 follow-ups: original and returned records

Workflow/R2 Direct-Full uses the pre-authorized same-setting replacement for its table status (TO<sup>‡</sup>). The original operational interruption remains below. PC Arms=2/R2 retains its original 1,200 s TO; the 3,600 s single trial is a separate sensitivity check. Neither follow-up supplies a median or performance ratio.

| Field                | Recorded value                                                      |
|----------------------|---------------------------------------------------------------------|
| Record               | PC Arms=2/R2; fg_ducs_otf; original                                 |
| Campaign / config    | rq3 / config.json                                                   |
| Start / end (UTC)    | 2026-09-15T09:55:34.006646+00:00 / 2026-09-15T10:15:38.375540+00:00 |
| Cap / heap / status  | 1200 s / 64g / TIMEOUT                                              |
| Elapsed / peak RSS   | 1202.9343 s / 31.2834 GiB (process working set)                     |
| Original association | rq3/runs/productioncell_arms2_r2_rep01_fg_ducs_otf/meta.json        |
| Record               | PC Arms=2/R2; fg_ducs_otf; supplementary                            |
| Campaign / config    | rq3_supplement_pc_arms2_r2_fg_3600 / config.json                    |
| Start / end (UTC)    | 2026-09-18T06:58:43.395208+00:00 / 2026-09-18T07:58:50.033019+00:00 |
| Cap / heap / status  | 3600 s / 64g / TIMEOUT                                              |
| Elapsed / peak RSS   | 3605.1630 s / 67.1467 GiB (process working set)                     |
| Original association | rq3/runs/productioncell_arms2_r2_rep01_fg_ducs_otf/meta.json        |
| Record               | Workflow/R2; direct_full; original                                  |
| Campaign / config    | rq3 / config.json                                                   |
| Start / end (UTC)    | 2026-09-15T00:12:34.878356+00:00 / not recorded                     |
| Cap / heap / status  | 1200 s / 64g / INTERRUPTED_NOT_RETRIED                              |
| Elapsed / peak RSS   | not recorded / not recorded                                         |
| Original association | rq3/runs/workflow_r2_rep01_direct_full/meta.json                    |
| Record               | Workflow/R2; direct_full; supplementary                             |
| Campaign / config    | rq3_supplement_workflow_r2_direct_full / config.json                |
| Start / end (UTC)    | 2026-09-18T06:38:27.266316+00:00 / 2026-09-18T06:58:30.415640+00:00 |
| Cap / heap / status  | 1200 s / 64g / TIMEOUT                                              |
| Elapsed / peak RSS   | 1201.6564 s / 13.8951 GiB (process working set)                     |
| Original association | rq3/runs/workflow_r2_rep01_direct_full/meta.json                    |

After repetition 1 times out, Workflow repetitions 2–5 have status `SKIPPED_AFTER_RESOURCE_FAILURE`; all five planned slots are retained in the generated provenance CSV. The PC follow-up planned one trial. Returned artifact digests and saved configuration/plan identities are verified. Input bytes, frozen JAR identity, JVM arguments, and stable host/runtime facts match the original records. The PC follow-up changes only the time cap and repetition count. All use the Xeon W-2265, 24 logical CPUs, Windows 10 build 19045, Temurin 17.0.20.1, and a 64 GiB heap cap. Peak RSS is sampled Windows process working-set memory. Heap occupancy was not recorded; the limiting resource remains unidentified. The recorded elapsed duration includes timeout termination overhead. The interrupted original has no recorded end time or RSS.

## S4.7 Separate Eager enlarged-heap trials

Each of the 10 conditions has one Eager trial on the same Xeon/JAR as the fixed-budget campaign, with a 200 GiB Java heap and a 7,200 s process-time cap; these are single observations, not timing medians, and no Mac times enter this comparison. All 4 decisions match fixed-budget Lazy; the state ratio is Eager's `states_discovered` divided by Lazy's `states_discovered` (structure repetition 1, equal over all five valid baseline repetitions), and is reported only when both decide. The 5 timeouts and 1 Java-heap OOM have no completed state/query/solver-time record or ratio; their JVM times include termination overhead, and sampled working-set RSS is process memory rather than heap occupancy. Counts describe the states discovered by the UC-pruned Eager search, which treats unsafe/goal states as terminal; they do not equate its state set with E2 or the unpruned Direct-Full game.

Table S16: Eager enlarged-heap trials (ext7). W = WIN; L = LOSS; TO = timeout. Times are seconds and RSS is GiB; each timing is one trial.

| Condition    | Status | States     | Queries     | Solver  | JVM     | RSS    | Lazy states | Ratio  | Src. |
|--------------|--------|------------|-------------|---------|---------|--------|-------------|--------|------|
| Industry/R1  | L      | 94,046     | 635,100     | 1,331.5 | 1,333.4 | 2.51   | 38,942      | 2.4    | a    |
| Railcab/Base | W      | 11,659,723 | 115,706,744 | 2,557.3 | 2,570.3 | 125.54 | 171         | 68,186 | b    |
| Workflow/R1  | W      | 26,594,216 | 111,677,312 | 2,406.6 | 2,428.9 | 184.94 | 1,098       | 24,221 | c    |
| Surv./Base   | TO     | –          | –           | –       | 7,205.1 | 211.94 | 5,319       | –      | d    |
| Railcab/R1   | TO     | –          | –           | –       | 7,205.2 | 209.62 | 13,339      | –      | e    |
| Surv./R1     | TO     | –          | –           | –       | 7,205.1 | 212.13 | 96,301      | –      | f    |
| Surv./R2     | W      | 10,841,519 | 55,401,437  | 1,575.3 | 1,591.7 | 134.43 | 181,116     | 59.9   | g    |
| PC2/Base     | TO     | –          | –           | –       | 7,205.2 | 209.66 | 2,502       | –      | h    |
| PC2/R1       | OOM    | –          | –           | –       | 2,981.3 | 209.06 | 2,433       | –      | i    |
| PC2/R2       | TO     | –          | –           | –       | 7,205.2 | 208.83 | –           | –      | j    |

Solver time includes internal checking; JVM includes preparation and process shutdown. The status/count/timing fields are cross-checked against each job's `meta.json`, `output.txt` and `stderr.txt`. Source directories below are relative to `experiments/rq3_xeon/raw/ext7_eager_heap200/`; baseline counts come from `paper/build/generated/rq3-cells.csv` and agree with all five corresponding raw repetitions.

- a `runs/industry__r1__rep01__fg_ducs_otf_eager_controllable/`
- b `runs/railcab__base__rep01__fg_ducs_otf_eager_controllable/`
- c `runs/workflow__r1__rep01__fg_ducs_otf_eager_controllable/`
- d `runs/surveillance__base__rep01__fg_ducs_otf_eager_controllable/`
- e `runs/railcab__r1__rep01__fg_ducs_otf_eager_controllable/`
- f `runs/surveillance__r1__rep01__fg_ducs_otf_eager_controllable/`
- g `runs/surveillance__r2__rep01__fg_ducs_otf_eager_controllable/`
- h `runs/productioncell_arms2__base__rep01__fg_ducs_otf_eager_controllable/`
- i `runs/productioncell_arms2__r1__rep01__fg_ducs_otf_eager_controllable/`
- j `runs/productioncell_arms2__r2__rep01__fg_ducs_otf_eager_controllable/`

## S4.8 Legacy reference, archived OTF path and fidelity checks

These original-source/fork comparisons use monolithic conditions and the reference tools' own continued-control objectives. They are decision and fidelity evidence for those contracts, separate from the four same-game FG constructions and from their timing ratios.

**Monolithic cross-check: original-source lab build versus this fork.** The input is the monolithic model from which the fine-grained adaptation was derived. Its bytes match the original-source lab build's bundled model after selecting  $T_{\text{Empty}}$  instead of  $T$ ; identity with a final published benchmark release remains unconfirmed. Only the two transition-selection lines change. Their definitions remain

$$T = (\text{StopOldSpec} \wedge \text{TORDone}) \Rightarrow (\neg \text{cancelGF1} \wedge \neg \text{approveGF1} \wedge \neg \text{startNewSpec}),$$

$$T_{\text{Empty}} = (\text{StopOldSpec} \wedge \neg \text{StartNewSpec}) \Rightarrow \neg \text{AnyAction}.$$

Both saved logs process the same named old/new safety formulas and  $T_{\text{Empty}}$ , report a 64-state old controller, and use deterministic GR synthesis after safety pruning. They disagree on the update result: the reference build returns a 2,312-state controller, while the fork returns no controller. The fork also reports a 240-state mapping product. Construction counts and the source-level cause appear below. The reference executable was built from original DUCS source; it is not an identified official TSE release. Full logs and unchanged input files are in `experiments/industry_r1_legacy_monolithic_check/`.

These decision-only results remain separate from RQ3. They rule out interpreting the FG LOSS as general unrealizability of R1: FG requires universal completion from all 131 entries into a compatible verified endpoint at quiescence. Published DUCS synthesizes continued control enforcing the new specification after the switch and takes no fixed new-controller endpoint as input.

**Source explanation of the cross-check.** We compare the fork's synthesis, environment construction and GR-goal code with the official MTSA source at commit d10ba71f. The inspected official revision is separate from the supplied lab build (source commit e1bf040); the latter JAR is the one used for the recorded GUI result. Both constructions retain the old closed loop and allow update entry throughout its reachable states; a GR solver's single initial-state check covers that entire pre-update behavior. Both traditional paths synthesize continued control under old/transition/new safety and a GR progress objective. A persistent only-turn-on fluent for the entry event is the GR assumption; persistent stop/start/reconfigure fluents give the three guarantees. Thus the input keyword `nonblocking` does not make this an existential-only completion check. Neither traditional path checks a fixed new-controller connection or quiescent handover. The supplied fork additionally has a separate OTF branch using `DirectedControllerSynthesisDUC` and an endpoint connection map; the inspected official synthesis file at d10ba71f has no such branch. The fork's plain OTF path synthesizes a new controller from the new environment and goal, matches component and new-safety states, and copies the matched controller's transitions into the result. This endpoint connection is therefore not specific to FG-DUCS. It is not used for the Traditional runs here, and inspection of this fork does not establish identity with a historical OTF release.

The upstream formula builder uses the model's `StopOldSpec/StartNewSpec`. The fork substitutes internal progress fluents (prefix `__DUC__`) when reconstructing the violation formulas. Its seven old-safety fluents are `TORDone`, `DSD1Done`, `Finished`, the three validation-event fluents, and internal `StopOldSpec`. The seven new-safety fluents replace `TORDone` by `NewTORDone` and internal `StopOldSpec` by internal `StartNewSpec`: their union has nine.  $T_{\text{Empty}}$  adds the model's `StopOldSpec/StartNewSpec` and eight controllable-event fluents, sharing three validation fluents. Hence the logged total is  $9 + 10 - 3 = 16$ ; these are not 16 additional endpoint obligations.

The fork's update-entry constant is `hotSwapIn`. The model's two progress fluents still reset on `beginUpdate`. `removeTopStates` composes fluent automata even when their labels lie outside the base environment alphabet. Consequently `beginUpdate` becomes an independent uncontrollable event, resetting the user fluents without resetting the internal ones. With the upstream entry

label `beginUpdate`, that reset instead synchronizes with entry. A finite Python reconstruction of this specific input, without modifying or executing either Java tool, gives:

| Product construction                                | States | Transitions |
|-----------------------------------------------------|--------|-------------|
| Old controller                                      | 64     | 148         |
| Mapping environment                                 | 240    | 1,108       |
| Fork entry/reset mismatch                           | 33,093 | 257,314     |
| Synchronized entry/reset                            | 14,763 | 99,676      |
| Fork after safety and at-most-once monitors         | 15,177 | 90,250      |
| Synchronized after safety and at-most-once monitors | 3,134  | 11,232      |

Both meta-product state/transition pairs exactly match the respective saved logs, as do the 15,177 versus 3,134 final safety-state counts. The synchronized final transition count is reconstructed, not printed by the lab-build log. The mapping count is reconstructed from the supplied model and printed by the fork; the lab-build log does not print that intermediate count. The reconstruction and its assertions are provided as `check_legacy_product.py` beside the two inputs and logs. It checks a concrete encoding discrepancy, not a new result for the frozen 27-instance campaign. The mismatch also explains the fork's negative decision: immediately after `hotSwapIn`, the safe `beginUpdate` self-loop can repeat forever without consuming any stop/start/reconfigure event. The GR entry assumption stays true while all three progress guarantees stay false. This uncontrollable trace survives safety and at-most-once filtering. It is a label-compatibility discrepancy, not a new-controller handover requirement. No input, configuration, JAR or synthesis source was changed to obtain this explanation.

The source model's `AnyAction` includes eight controllable events; its filename and `T_Empty` label alone do not establish identity with the published benchmark version.

The fixed RQ3 Legacy reference instead targets `UPDATE_CONTROLLER_FG` and its R1/R2 variants in the nine adapted input files. Those files use `hotSwapIn` and contain no `beginUpdate` label. The specific entry/reset mismatch above therefore belongs to the separate monolithic source-input cross-check; this distinction does not establish general source/fork equivalence.

**Aligned-entry execution and reference-build identity.** The reference was built from original DUCS source at commit `e1bf040`, with the executable and source identity retained in `experiments/legacy_fidelity/`. The eight monolithic inputs agree byte-for-byte across that source revision, compiled resources and executable. This identifies the reference build without asserting an official published-release identity.

A separate fork input changes only the three fluent reset lines for `StopOldSpec`, `StartNewSpec` and `Reconfigure`, replacing `beginUpdate` by `hotSwapIn`. One Mac execution of the frozen fork JAR through its Traditional path returns a 2,312-state controller, agreeing with the reference result; the meta-product and final safety input also match at 14,763 and 3,134 states. The earlier mismatched-input failure remains preserved. This is a decision-only fidelity check, excluded from RQ3 and its performance measurements. Input diff, command and original outputs are retained under `experiments/legacy_fidelity/`; the returned monolithic reference campaign is reported separately below.

The separate adapter pilot checks Industry under two monolithic conditions (states/transitions below); all four cells return WIN. The aligned fork's  $T_{\text{Empty}}$  cell reuses that one execution.

| Input              | Original-source lab build | Aligned fork  |
|--------------------|---------------------------|---------------|
| Supplied $T$       | 3,050 / 9,496             | 3,050 / 9,496 |
| $T_{\text{Empty}}$ | 2,312 / 6,615             | 2,312 / 6,615 |

Two initial adapter failures are preserved: synthesis produced a controller, but an incorrect Java type check rejected the outer composite. Correcting only that adapter yielded the two successful reference-tool cells. The final adapter additionally classifies an explicit no-controller result wrapped in a composition exception; four classification assertions check that branch. These Mac checks supply no performance evidence. The separate Xeon campaign has 21 conditions per tool (15 supplied targets and six named empty-transition alternatives); its returned results remain outside the main RQ3 table.

**Returned legacy-fidelity reference experiment.** These Windows campaigns compare the lab build and fork on mechanically generated monolithic conditions with all three entry-fluent resets aligned. Their contracts differ from FG; the results remain outside RQ3. Table S17 retains summaries, per-trial records and fork exceptions. Independently regenerated summaries match the returned CSVs byte-for-byte; raw statuses and reproduction outputs remain in the artifact.

Table S17: Separate monolithic reference experiment: all 21 conditions. S denotes SUCCESS (controller returned); sizes are first-trial states/transitions. Each completed cell has five consistent SUCCESS decisions; sizes may vary across repetitions (below). N/M labels annotate raw CRASH records, without changing them.

| Input / condition          | Original-source build | lab Fork legacy path     | Comparison    |
|----------------------------|-----------------------|--------------------------|---------------|
| GSM supplied               | S 86/119              | N/M<br>(fork parser)     | N/M           |
| GSM empty                  | S 92/117              | N/M<br>(fork parser)     | N/M           |
| Industry supplied          | S 3,050/9,496         | S 3,050/9,496            | MATCH         |
| Industry empty             | S 2,312/6,615         | S 2,312/6,615            | MATCH         |
| MetaSocket UPD64_128       | S 36/59               | S 36/59                  | MATCH         |
| MetaSocket UPD64_64COM     | S 36/59               | S 36/59                  | MATCH         |
| MetaSocket UPD128_64       | S 36/59               | S 36/59                  | MATCH         |
| MetaSocket UPD128_128COM   | S 36/59               | S 36/59                  | MATCH         |
| MetaSocket UPD64COM_64     | S 36/59               | S 36/59                  | MATCH         |
| MetaSocket UPD64COM_128COM | S 36/59               | S 36/59                  | MATCH         |
| MetaSocket UPD128COM_128   | S 36/59               | S 36/59                  | MATCH         |
| MetaSocket UPD128COM_64COM | S 36/59               | S 36/59                  | MATCH         |
| PowerPlant supplied        | S 79/141              | N/M<br>(fork parser)     | N/M           |
| ProductionCell supplied    | S 168/267             | S 168/267                | MATCH         |
| ProductionCell empty       | S 107/158             | S 107/158                | MATCH         |
| Railcab supplied           | S 448/803             | S 443/796                | SIZE-MISMATCH |
| Railcab empty              | S 471/708             | S 471/708                | MATCH         |
| Surveillance supplied      | S 1,394/2,959         | N/M<br>(instrumentation) | N/M           |
| Surveillance empty         | S 665/1,414           | N/M<br>(instrumentation) | N/M           |
| Workflow supplied          | S 679/1,144           | S 679/1,144              | MATCH         |
| Workflow empty             | S 193/287             | S 193/287                | MATCH         |



The two campaigns retain 210 planned slots: 190 executed trials and 20 explicit skips following the five fork exceptions. Those exceptions occur before a legacy GR decision and give no evidence of legacy synthesis capability or failure.

**Saved exception text: fork parser.** `ltsa.lts.LTSEException:`  
`name already defined LeavesPumpOn`  
`ltsa.lts.LTSEException:`  
`name already defined T_NO_UPDATE_WHILE_SEND`

**Saved exception text: instrumentation.** `java.lang.IllegalArgumentException:`  
`Derived old-action authority is inconsistent: weather.old`

| Tool      | Condition              | Rep. 1  | Rep. 2  | Rep. 3  | Rep. 4  | Rep. 5  |
|-----------|------------------------|---------|---------|---------|---------|---------|
| Fork      | ProductionCell / empty | 107/158 | 102/150 | 107/158 | 102/150 | 107/158 |
| Lab build | Railcab / supplied     | 448/803 | 448/809 | 449/805 | 443/796 | 442/797 |
| Fork      | Railcab / supplied     | 443/796 | 442/794 | 449/805 | 451/811 | 449/805 |
| Lab build | Railcab / empty        | 471/708 | 471/708 | 471/708 | 429/651 | 471/708 |

These are all 4 tool/condition cells whose returned controller sizes vary across the five successful trials; other completed cells retain their first-trial sizes. The fifteen size matches and five-state Railcab difference in Table S17 compare first trials only. In particular, the Railcab supplied lab-build trial 4 returns the same size as fork trial 1; a persistent tool-specific size difference or controller equivalence is not established.

**Causes of unmeasured fork conditions.** GSM reuses `T_NO_UPDATE_WHILE_SEND` for a property and an assertion; PowerPlant similarly reuses `LeavesPumpOn`. The original parser checks assertion definitions for name uniqueness, whereas the fork also checks constraints, rejecting these inputs during parsing. This accounts for the three N/M (fork parser) entries, not a synthesis LOSS.

Surveillance declares `weather` only in `NewControllableActions`, not in its old/new alphabets or environment transitions. The fork derives `weather.old` from the declaration and inserts that derived label into the alphabet. Author-side `M9TraditionalSafetySemanticsSnapshot` requires both the derived label and its base action in the alphabet; the absent base `weather` triggers the saved exception. This capture follows construction of the old controller and meta-environment, but precedes update safety pruning and the legacy GR decision. The two N/M (instrumentation) entries therefore provide no evidence of legacy synthesis capability or failure. No model, tool or JAR was changed and no trial was repeated to repair these outcomes.

**Railcab size variation and unresolved cause.** For supplied Railcab, both logs report 1,763 states before safety pruning and 1,582 afterward, then use nondeterministic controller synthesis. The first-trial difference is already present when the strategy is converted to an MTS and unreachable states are removed, before final `CompactState` conversion or outer composition. The inspected subset construction, knowledge-game solver and strategy-to-MTS paths share their operative code; preprocessing, fluent-valuation handling and enumeration still differ. The saved evidence does not establish corresponding safety graphs, valuations or strategies. Moreover, controller sizes vary within each tool across repetitions, with some sizes shared across tools. The cause of that variation and the first-trial difference remains unresolved after bounded static inspection; neither a persistent fork effect nor controller equivalence is inferred.

Table S18: Reference measurements only. End-to-end JVM wall seconds and sampled process peak RSS (GiB): median [min,max] only for five complete, consistent runs. No cross-tool ratios are computed.

| Input / condition          | Lab wall (s)         | Fork wall (s)     | Lab RSS (GiB)           | Fork RSS (GiB)       |
|----------------------------|----------------------|-------------------|-------------------------|----------------------|
| GSM supplied               | 0.61 [0.609, 0.61]   | N/M               | 0.102 [0.0995, 0.102]   | N/M                  |
| GSM empty                  | 0.711 [0.71, 0.712]  | N/M               | 0.101 [0.0997, 0.105]   | N/M                  |
| Industry supplied          | 12.7 [12.6, 12.7]    | 16.7 [16.3, 16.8] | 1.91 [1.6, 2.05]        | 2.02 [1.97, 2.11]    |
| Industry empty             | 4.64 [4.44, 4.74]    | 7.66 [7.46, 7.77] | 0.814 [0.738, 0.821]    | 1.81 [1.73, 1.89]    |
| MetaSocket UPD64_128       | 0.611 [0.609, 0.612] | 1.21 [1.11, 1.21] | 0.0996 [0.0971, 0.103]  | 0.14 [0.137, 0.147]  |
| MetaSocket UPD64_64COM     | 0.609 [0.608, 0.61]  | 1.21 [1.21, 1.22] | 0.0996 [0.0995, 0.0998] | 0.141 [0.139, 0.144] |
| MetaSocket UPD128_64       | 0.61 [0.609, 0.611]  | 1.21 [1.21, 1.21] | 0.0995 [0.0981, 0.0998] | 0.142 [0.14, 0.143]  |
| MetaSocket UPD128_128COM   | 0.61 [0.609, 0.61]   | 1.21 [1.11, 1.22] | 0.0997 [0.0971, 0.103]  | 0.141 [0.14, 0.142]  |
| MetaSocket UPD64COM_64     | 0.61 [0.609, 0.611]  | 1.21 [1.11, 1.21] | 0.1 [0.098, 0.103]      | 0.144 [0.137, 0.145] |
| MetaSocket UPD64COM_128COM | 0.609 [0.609, 0.611] | 1.21 [1.21, 1.22] | 0.1 [0.0995, 0.103]     | 0.143 [0.139, 0.15]  |
| MetaSocket UPD128COM_128   | 0.61 [0.608, 0.612]  | 1.21 [1.11, 1.21] | 0.1 [0.0997, 0.103]     | 0.145 [0.141, 0.147] |
| MetaSocket UPD128COM_64COM | 0.61 [0.608, 0.61]   | 1.21 [1.21, 1.22] | 0.102 [0.1, 0.103]      | 0.144 [0.141, 0.144] |
| PowerPlant supplied        | 0.711 [0.71, 0.712]  | N/M               | 0.105 [0.101, 0.109]    | N/M                  |
| ProductionCell supplied    | 1.11 [1.11, 1.12]    | 1.92 [1.92, 1.92] | 0.199 [0.191, 0.206]    | 0.303 [0.3, 0.308]   |
| ProductionCell empty       | 1.12 [1.11, 1.12]    | 1.92 [1.82, 1.92] | 0.196 [0.182, 0.204]    | 0.309 [0.297, 0.322] |
| Railcab supplied           | 1.62 [1.62, 1.72]    | 2.63 [2.62, 2.63] | 0.482 [0.472, 0.496]    | 0.703 [0.696, 0.711] |
| Railcab empty              | 1.82 [1.82, 1.92]    | 3.03 [2.93, 3.03] | 0.558 [0.537, 0.566]    | 0.682 [0.671, 0.689] |
| Surveillance supplied      | 6.35 [5.84, 6.45]    | N/M               | 0.862 [0.852, 0.867]    | N/M                  |
| Surveillance empty         | 10.7 [10.4, 10.8]    | N/M               | 0.929 [0.877, 0.95]     | N/M                  |
| Workflow supplied          | 9.37 [9.17, 9.48]    | 12.3 [12.1, 12.4] | 1.03 [0.974, 1.1]       | 1.98 [1.96, 2.04]    |
| Workflow empty             | 4.53 [4.43, 4.64]    | 5.65 [5.54, 5.94] | 0.664 [0.631, 0.667]    | 1.8 [1.79, 1.94]     |

Internal times have different scopes: the lab-build adapter includes compilation, continued compilation and composition, whereas the fork exposes its solve-control-problem interval. The table instead uses end-to-end JVM wall time as reference information; these tools do not solve identical FG games and their times are not ratioed. The fork exceptions have one actual trial followed by four explicit skips; no median is reported. `experiments/legacy_fidelity/` holds the inputs, diffs, collector, renderer and raw evidence; the public package omits shaded JARs and the private delta's `lib/`.

## S4.9 Separate Legacy enlarged-heap reference

Table S19: Separate Legacy extended-budget resource observations: one Xeon trial per condition, 200 GiB heap and 3,600 s cap.

| Contract          | Display       | Peak states | Peak edges | JVM s  | RSS GiB | Source |
|-------------------|---------------|-------------|------------|--------|---------|--------|
| Railcab/BASE      | N/M (profile) | 5,320,658   | 80,379,005 | 1511.9 | 193.94  | a      |
| Workflow/BASE     | OOM           | –           | –          | 3541.4 | 207.59  | b      |
| Surveillance/BASE | OOM           | –           | –          | 2668.6 | 207.71  | c      |
| PC2/BASE          | OOM           | –           | –          | 1550.6 | 207.80  | d      |
| Railcab/R1        | OOM           | –           | –          | 3209.7 | 207.71  | e      |
| Workflow/R1       | OOM           | –           | –          | 3396.3 | 207.64  | f      |
| Surveillance/R1   | OOM           | –           | –          | 1888.1 | 207.73  | g      |
| PC2/R1            | OOM           | –           | –          | 1696.5 | 207.75  | h      |
| Railcab/R2        | N/M (profile) | 5,320,658   | 80,379,005 | 1646.1 | 200.00  | i      |
| Workflow/R2       | OOM           | –           | –          | 2593.9 | 207.72  | j      |
| Surveillance/R2   | OOM           | –           | –          | 2067.3 | 207.79  | k      |
| PC2/R2            | OOM           | –           | –          | 774.1  | 207.76  | l      |
| PC1/R1            | OOM           | –           | –          | 2240.0 | 209.41  | m      |
| PC1/BASE          | N/M (profile) | 1,814,373   | 25,589,560 | 417.2  | 72.76   | n      |
| PC1/R2            | N/M (profile) | 1,814,373   | 25,589,560 | 436.2  | 80.83   | o      |
| Industry/BASE     | N/M (profile) | 683,129     | 8,136,485  | 141.5  | 24.81   | p      |
| Industry/R1       | N/M (profile) | 2,888,907   | 34,072,018 | 713.7  | 93.66   | q      |
| Industry/R2       | N/M (profile) | 683,129     | 8,136,485  | 145.6  | 25.95   | r      |

N/M (profile) preserves raw CRASH: all seven records report an `IllegalArgumentException` at `M9MtsSnapshot.requireResourceCensus` while capturing the safety meta-environment, before final controller synthesis. The snapshot profile limits include 250,000 states; this is separate from the heap/time budget. State/edge figures are the recorded peak intermediate state space, not a controller or completed solution. All eleven OOM records report Java heap space. RSS is sampled Windows working set, not heap occupancy. Failure runtimes are whole-JVM observations, not successful-solver timings or medians. No decision or general tool limitation is inferred. Text macros round recorded state counts (millions) and OOM RSS (GiB) to one decimal. Sources below are relative to `experiments/rq3_xeon/raw/ext6_legacy_heap200/`; each directory supplies `meta.json`, `output.txt` and `stderr.txt`.

```

a runs/railcab__base__rep01__legacy_ducs/
b runs/workflow__base__rep01__legacy_ducs/
c runs/surveillance__base__rep01__legacy_ducs/
d runs/productioncell_arms2__base__rep01__legacy_ducs/
e runs/railcab__r1__rep01__legacy_ducs/
f runs/workflow__r1__rep01__legacy_ducs/
g runs/surveillance__r1__rep01__legacy_ducs/
h runs/productioncell_arms2__r1__rep01__legacy_ducs/
i runs/railcab__r2__rep01__legacy_ducs/
j runs/workflow__r2__rep01__legacy_ducs/
k runs/surveillance__r2__rep01__legacy_ducs/
l runs/productioncell_arms2__r2__rep01__legacy_ducs/
m runs/productioncell_arms1__r1__rep01__legacy_ducs/
n runs/productioncell_arms1__base__rep01__legacy_ducs/
o runs/productioncell_arms1__r2__rep01__legacy_ducs/
p runs/industry__base__rep01__legacy_ducs/
q runs/industry__r1__rep01__legacy_ducs/
r runs/industry__r2__rep01__legacy_ducs/

```

## Artifact reproduction status

The public anonymous repository contains the full derived source tree, saved RQ1/RQ2 evidence, compiled-monitor exports and Python RS checker, and generated tables/figures. The earlier clean HTTPS clone, empty-cache source build and RQ1/RQ2 derive-only smoke checks passed; rederived fields matched the saved expectations after path normalization. Tests were compiled but not executed. Their logs are now under `reproduce/previous-validation/clean-clone/`. Third-party dependencies are fetched from official upstream sources; no shaded JAR is distributed. The expanded raw-output package is included in three lossless archive parts within the repository. The current `reproduce/check.py` verifies and restores these files, then checks saved correctness/supplement evidence and E-series/ext1–7 results without rerunning benchmarks. The README documents this command and its scope. The implementation and experimental evidence snapshot is identified by tag `fgducs-invariants-20261002`; the paper-matched PDFs contain the current formal statements and presentation. Its primary reading pair is the main paper and integrated supplementary material, with complete sources and both component PDFs retained for rebuilding and compatibility. The mathematical arguments and presentation use the unchanged experimental observations.

## S5 Expanded related-work comparison

Main-paper Table 2 summarizes three update interfaces. The tables here expand their coverage and progress conditions, then compare synthesis and switching foundations. Publication details appear in the main bibliography. Section and definition numbers identify the inspected native contracts; differences do not establish an inability to encode one approach in another.

### S5.1 Update and adaptation interfaces

Table S20: Native inputs, progress conditions and outputs of update and adaptation approaches.

| Work                            | Inputs and change mechanism                                                                                                                               | Coverage and progress                                                                                                       | Constructed result and scope                                                                                                        |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| DUCS (2016, 2020)               | One global old-stop/new-start pair, with distinct events; partial, set-valued global $R$ ; optional stages in $E'$ ; $T$ constrains order (Secs. 4.2–4.3) | All old states via total $H$ ; all outcomes; eventual update events after hotSwap; legal and deadlock-free (Defs. 4.1–4.3)  | Update controller; FLTL control reduction and MTSA synthesis                                                                        |
| Environment partitioning (2020) | Pre/post environment LTSs and global state-preservation edges labeled <code>reconfigure</code> ( $T_{upd}$ )                                              | The environment construction joins supplied pre/post states; subsequent control uses the existing update-synthesis method   | Generated $E_{upd}$ from $E_{bef}$ , $E_{aft}$ and $T_{upd}$ (Sec. V, Algorithm 1)                                                  |
| GR(1) updating (2022)           | Two specifications and switching assertion; old safety guarantees before switch, new specification afterward                                              | All prefixes meeting new environment safety; uniformly bounded switch (Def. 3.1); subsequent GR(1) liveness assumes justice | New controller $C_2$ and symbolic reachability bridge; switching bookkeeping; winning set $W$ or current-state early stop (Sec. 4)  |
| Live Synthesis (2022)           | Old/new LTL and trace or transition system; residual $\Psi(\eta, \varphi)$ ; old obligations finitely satisfied                                           | Universal variant: all old finite traces and new traces; reduction assumes one eventual update (Thm. 6.4)                   | Obligation monitor and LTL synthesis; LiveLTL language inclusion                                                                    |
| Jiao et al. (2013)              | Existing workcell supervisor and added machine; event relabeling and communication transitions produce coordinated supervisors (Sec. III.B)               | Old states map to new supervisor-state pairs; instant switching is assumed faster than UC events (Sec. III.C)               | Reuses and transforms the controller; the example checks equivalence to monolithic synthesis; no staged all-outcome handover search |
| Dynamic Feature (2024)          | Feature come/go, component reinitialization, simultaneous changes; behavioral/configuration constraints                                                   | Mixed controllability; UC preserved; every reachable controlled state has a jointly marked continuation (Sec. 7)            | CIF BDD synthesis; maximal safe controllable nonblocking supervisor                                                                 |
| Zhang et al. (2005)             | Configurations, dependency invariants, adaptive actions, implementations and costs; generated safe adaptation graph (Secs. 3.1, 4.1–4.2)                  | Supplied source/target request; minimum-cost path; failed steps may roll back or await intervention (Sec. 4.4)              | Dijkstra path and manager/agent protocol; action-specific safe states, atomic isolated adaptation (Secs. 3.1, 4.3)                  |
| Network updates (2015)          | Per-switch replacement; searched update order; packet-flush waits                                                                                         | All single-packet traces satisfy LTL; failure freedom, fair forwarding, bounded processing (Sec. 3.1)                       | Sequence; DFS and incremental checking; simple/careful completeness                                                                 |

## S5.2 Synthesis and supervisory switching

Existing synthesis and switching methods supply foundations for FG-DUCS. In particular, state correspondence and switching between existing supervisors are established ideas. FG-DUCS combines supplied endpoints with staged component transfers, individual requirement lifetimes and finite completion under all permitted outcomes.

Table S21: Synthesis and switching contracts. An available completing continuation differs from completion of every maximal update run.

| Work                                | Inputs and change mechanism                                                                                     | Coverage and progress                                                                                                                           | Constructed result and scope                                                                                                                            |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCS (2016)                          | Deterministic component LTSs; safety/co-safety event goals                                                      | Initial product state; safety and eventual goal occurrence on every closed-loop trace; states enabling controllable and UC events treated as UC | Controller or failure; heuristic OTF composition, deferred control alternatives and backward propagation (Secs. I, III–IV)                              |
| OTF-DCS (2023)                      | Component automata, marked/illegal states and control partition                                                 | All UC enabled; designated disturbances; eager director (Def. 3); every controlled finite trace has a marked extension (Def. 4)                 | Safe nonblocking director; heuristic OTF composition                                                                                                    |
| Tripakis–Altisen (1999)             | Finite game graph, controllable/UC edges, safe/target sets                                                      | UC-successor closure; controllable path to target from each node (Sec. 2.1); environment may move first                                         | Strategy subgraph; fully OTF for untimed graphs                                                                                                         |
| Strong planning (1998)              | Action/fluent domain, nondeterministic effects and goal formula                                                 | All initial states and outcomes; strong completion; fair iterative extension separate (pp. 40–43)                                               | OBDD state/action plan; universal outcome preimages; initial-set early stop                                                                             |
| DES state attraction (2015)         | Common deterministic plant; request/startup sets; task language $K$ and recognizer                              | All request states; UC preserved; weakly invariant startup set; bounded first completion (Def. 2, Thm. 1)                                       | Attractor supervisor; plant/recognizer product and state-attraction algorithm                                                                           |
| Switched supervisory control (2020) | One plant and two safe/live supervisors; switching and state-reinitialization functions (Sec. 3.3, Defs. 4–6)   | Safe, deadlock-free switching; every finite trace has an extension reaching a switching opportunity (Problem 1); repeated switching             | Also synthesizes the supervisor pair; supplied-pair construction retains compatible plant/supervisor states, not a one-time finite completion objective |
| Switched DES invariants (2017)      | Given operational modes and transitions, uncontrollable mode switches and a safety predicate per mode (Sec. II) | Invariants under every permitted transition; all mode switches retained; finite convergence concerns computation, not handover time             | Maximally permissive supervisor; distributed modal computation reaches the same solution (Thms. 1–2)                                                    |